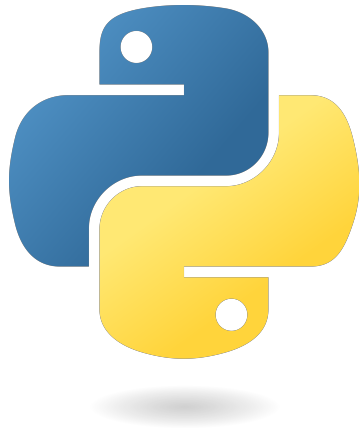


FINANCIAL RISK PYTHON IMPLEMENTATION



Financial risk means potential loss in the world of finance. If you invest, risk is everywhere and sometimes is considered even more important than financial return. The flavor of financial risk ranges from day-to-day fluctuations in stock prices, also called volatility, to broader, more unpredictable shocks caused by global events. Common types of financial risk include market risk (price fluctuations), credit risk (borrower defaults), and liquidity risk (difficulty in asset liquidation). These risks are not just abstract ideas; they bring about real challenges that can impact investment portfolios, financial institutions, and even the entire financial system as a whole.

For example, a recent study by Marani et al. (2021) analyzed disease outbreaks over the past four centuries and revealed that extreme pandemics are more frequent than previously assumed. The research estimates an annual probability of approximately 2% for a pandemic with an impact similar to COVID-19, suggesting that an individual born in the year 2000 would have about a 38% chance of experiencing such an event by now. Furthermore, the study indicates that a pandemic of comparable scale could be expected within the next 59 years. Such an extreme pandemic can disrupt markets, causing volatility, liquidity shortages, and systemic failures. This unpredictability makes it essential that financial professionals have a solid understanding of risk dynamics.

Pandemics such as COVID-19 have had undeniably profound effects on our daily lives, reshaping economies, healthcare systems, and social norms, transforming practices like remote work from niche to mainstream. For instance, the S&P 500 plummeted by approximately 34% from its peak in February 2020 to its trough in March 2020 (see Figure 1-1), and the world economy decreased by 3.5% in 2020. In light of these far-reaching impacts, countries and companies are increasingly prioritizing supply chain resilience, emphasizing the importance of diversifying suppliers and establishing alternative sources to mitigate the risk of primary supply disruptions. To properly manage supply chain risk in case of another pandemic, the overall supply chain system needs to have both adaptability and redundancy, which seems to move in the opposite direction against lean management.

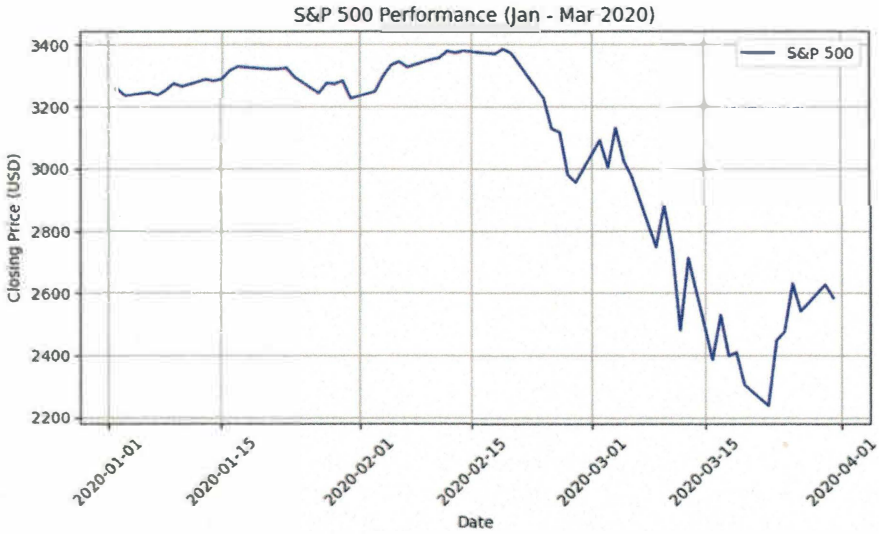


Figure 1-1 The S&P 500 plummeted by over 30% from its peak in February 2020 to its trough in March 2020

See Listing 1-1 used to generate Figure 1-1.

```

1 import yfinance as yf
2 import matplotlib.pyplot as plt
3
4 # Define the ticker symbol for S&P 500 ETF
5 ticker = '^GSPC'
6
7 # Fetch data from January 1, 2020, to April 1, 2020
8 sp500 = yf.download(ticker, start='2020-01-01', end='2020-04-01')
9
10 # Plot the closing prices
11 plt.figure(figsize=(8, 5))
12 plt.plot(sp500.index, sp500['Close'], label='S&P 500', color='
    blue')
13 plt.title('S&P 500 Performance (Jan - Mar 2020)')
14 plt.xlabel('Date')
15 plt.ylabel('Closing Price (USD)')
16 plt.legend()
17 plt.grid(True)
18 plt.xticks(rotation=45)
19 plt.tight_layout()
20 plt.show()

```

Listing 1-1 S&P 500 price curve

But why does risk exist in the first place? Given the inherent uncertainty in the world around us, risk exists because the future is unpredictable and driven by complex interactions between human behavior, environmental changes, and

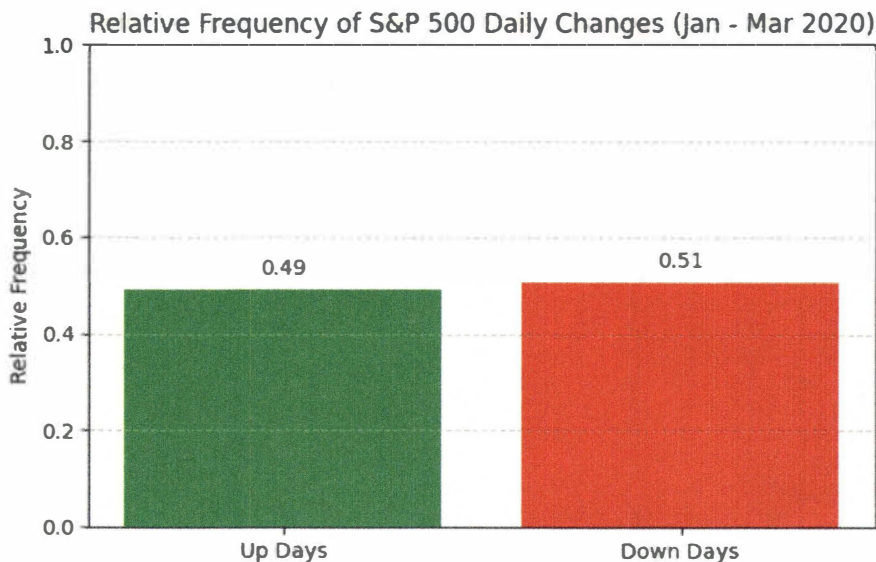


Figure 1-2 Calculating the relative frequency as the empirical probability of S&P trending up or down in this period

biological evolution. This uncertainty means that we cannot accurately forecast when or where a pandemic might occur, its severity, or its specific impacts. And, as a result, the outcome is a random event. Therefore, quantifying and managing such uncertainty and randomness is the central theme of risk management.

The most effective tool for characterizing the randomness of uncertain events is the probability distribution, which captures all possible outcomes and assigns a probability to each. For example, when predicting the likelihood of rain tomorrow, we might assign a probability of 70% to the event of rain. This implies a 30% chance that it will not rain, as the probabilities of all possible outcomes must sum to one. Referring to our previous S&P 500 daily price curve in Figure 1-1, we can calculate the relative frequency of the next day's price moving up or down by counting the occurrences of each outcome and turning the absolute count into relative frequency. These frequencies allow us to approximate the probability that the market is trending upward or downward the following day.

As illustrated in Figure 1-2, the relative frequency, which serves as an empirical probability measure, indicates a 51% likelihood that the S&P 500 closing price would decrease during the period from January to March 2020. Although it is impossible to predict with absolute certainty whether the index will rise or fall on any given day, this approach provides a quantified perspective on the most probable outcome among all possibilities.

See Listing 1-2 used to generate Figure 1-2.

```

1 # Calculate daily percentage change
2 sp500['Daily Change'] = sp500['Close'].diff()
3
4 # Calculate relative frequency of going up or down
5 total_days = len(sp500['Daily Change'].dropna())
6 up_days = len(sp500[sp500['Daily Change'] > 0])
7 down_days = len(sp500[sp500['Daily Change'] < 0])
8
9 up_frequency = up_days / total_days
10 down_frequency = down_days / total_days
11
12 # Prepare data for the bar chart
13 categories = ['Up Days', 'Down Days']
14 frequencies = [up_frequency, down_frequency]
15
16 # Plot the bar chart
17 plt.figure(figsize=(6, 4))
18 bars = plt.bar(categories, frequencies, color=['green', 'red'])
19 plt.title('Relative Frequency of S&P 500 Daily Changes (Jan - Mar
20         2020)')
21 plt.ylabel('Relative Frequency')
22 plt.ylim(0, 1)
23 plt.grid(axis='y', linestyle='--', alpha=0.7)
24
25 # Add exact values as legend
26 for bar, freq in zip(bars, frequencies):
27     plt.text(bar.get_x() + bar.get_width() / 2, bar.get_height()
28             + 0.02,
29             f"{freq:.2f}", ha='center', va='bottom', fontsize
30             =10)
31
32 plt.tight_layout()
33 plt.show()

```

Listing 1-2 Calculating the relative frequency of going up or down

This book, *Quantitative Risk Management Using Python*, provides a clear and structured guide to the diverse risks that are analyzed and managed in modern financial markets. We will attempt to quantify and manage these risks from multiple perspectives, including financial data, statistical techniques, and mathematical models, ultimately supporting more informed and risk-aware decision-making. At times, effectively managing risk can be even more critical than pursuing high financial returns, especially when dealing with large-scale portfolios. Beyond conceptual and theoretical discussions, we also place a strong emphasis on practical implementation using Python, hoping that readers gain both a robust theoretical framework and the hands-on skills needed to navigate real-world challenges.

In the following sections of this chapter, we will first discover key categories of financial risk, including market, credit, and liquidity risk, and then learn common financial instruments and their use in risk management. We aim to build a good

foundational understanding of the risk landscape and a practical set of quantitative tools to help you navigate the world of financial risk management with confidence.

1.1 Understanding Different Types of Risk in Financial Markets

To start, let us take a closer look at what financial markets are about and the different types of risk involved. In financial markets, retail investors, wholesale institutions, and even central governments engage in financial transactions such as buying and selling certain financial instruments, such as stocks, bonds, commodities, and derivatives. These activities promote price discovery, in that they reflect the aggregate beliefs of market participants and provide a way to approximate each instrument's "fair value" through the dynamics of supply and demand toward market equilibrium.

Although some traders seek arbitrage profits by exploiting temporary deviations from the fair value of an asset (which could be due to sudden shocks), others prioritize maximizing returns, minimizing risk, or a combination of both objectives. For example, retail investors can hope to achieve high returns by investing in long-term growth stocks, whereas large institutions often employ low-risk strategies to preserve capital with steady returns and minimal volatility. Consequently, trading activities can vary substantially based on an investor's goals, resources, and market outlook. In general, market participants have different intentions and employ varying strategies within financial markets.

These activities also come with various risks that can pose significant challenges if not properly managed. For example, **market risk** captures the potential losses due to fluctuations in market prices. Examples include a sudden drop in stock prices after the market observes a negative earnings report or a massive sell-off triggered following an announcement of a political policy (think about how the real estimate market reacts when the government introduces a new policy). If you are not prepared, market risk can lead to substantial losses in your investment portfolio since it will likely fluctuate on the downside (everyone likes upside fluctuation). Besides, there is **credit risk**, which comes into play when a borrower fails to repay the debt. This risk is particularly significant for banks and bond investors, who are often regulated to report and manage risk exposure should a default event occur. Lastly, **liquidity risk** says that an investor might not be able to buy or sell an asset quickly as they hope, or even if it is possible, such a fast trade comes with a significant price change (think about selling a property at a fairly low price when in urgent need of cash), potentially leading to losses or missed opportunities.

Beyond these, there are other risks, such as **operational risk**, which deals with failures in internal processes or systems in companies; **legal risk**, which involves potential losses from lawsuits or changes in regulations; **systemic risk**, which is the risk of a collapse in the entire financial system; and **model risk**, which presents the potential loss due to misspecification of the model assumption or incorrect model estimation. The challenges of managing these risks are further compounded by the

volatile nature of financial markets. For example, a predictive model trained and deployed today may not be as valid two months later due to changes in market conditions. In this section, we will introduce these various types of risk and explore how they can impact financial assets.

Consider an investment firm that operates both in the corporate bond market and in the stock market, where volatility in the price of the shares introduces market risk. One morning, the firm discovers that a major bond issuer has defaulted on its debt, exposing the institution to considerable credit risk: there is now a real threat that they may not fully recover their investment. At the same time, the firm faces unexpected withdrawal requests from clients, which require them to sell real estate assets rapidly at a substantial discount. This action demonstrates liquidity risk, as the firm must quickly convert assets into cash at the expense of their market value. Meanwhile, the firm's forecasting algorithm, which is based on a black-box model, does not account for sudden changes in market dynamics, thus resulting in greater potential losses than anticipated; this issue exemplifies the model risk.

In response to these challenges, regulators introduce new measures (reminiscent of post-2008 financial reforms), placing greater compliance demands on the firm and amplifying its legal risk. In addition, a technical glitch on a critical trading day completely halts order executions, illustrating just how impactful operational risk can be when internal systems fail. Finally, the firm's global reach makes it susceptible to external shocks: a significant geopolitical event, such as an armed conflict, triggers a worldwide sell-off that jeopardizes the stability of the entire financial system, presenting significant systemic risk.

Let us look at each specific type of financial risk in detail.

1.1.1 Market Risk

Market risk is arguably the most common form of risk faced by individual investors in financial markets. It involves potential losses arising from unforeseen fluctuations in asset prices, which can affect everything from stocks and bonds to foreign exchange rates and commodities such as oil and gold. Various factors, including economic news, political events, natural disasters, and changes in investor sentiment, can trigger these price swings. Moreover, because price discovery is an ongoing process, supply and demand constantly shape asset values, making any apparent market equilibrium only temporary.

We can classify market risk according to the underlying market of the financial instrument, such as equity, interest rate, currency, or commodity markets. Let us examine these distinct risks in more detail.

Equity risk refers to the potential for changes in stock prices to affect the value of your investments. When you hold shares in a company, you effectively share in its successes and setbacks. For example, if a company underperforms and releases a poor earnings report, its stock price can drop, causing losses for shareholders. Equity risk also includes broader market fluctuations: a sudden economic downturn can spark a widespread sell-off, impacting virtually all stocks. Because stock prices

are inherently unpredictable, owning equities carries the possibility of a decline in value (assuming you hold a long position). Of course, various protective measures, such as purchasing a put option, can help limit losses, and we will explore these hedging techniques later in the book.

Interest rate risk refers to the uncertainty in potential changes in the interest rate, which can have a pronounced effect on the value of fixed-income securities such as bonds. As an example, imagine that you own a bond that pays an annual coupon at 2%. If interest rates on the market later increase to 3%, newly issued bonds will likely offer that higher rate. Consequently, investors will be less willing to pay the full price for a bond yielding only 2%, causing its market value to drop. This inverse relationship between bond prices and interest rates stems from the fact that the fixed coupon of a bond becomes more (or less) attractive depending on the prevailing interest rate environment.

Currency risk, also known as **foreign exchange (or forex) risk**, arises whenever you invest in assets denominated in a currency other than your home currency. For example, suppose that you are a US-based investor who purchases shares of a European company. Now, you are exposed to two types of risk: equity risk, as introduced before, and currency risk, which comes from a depreciation of the euro against the dollar. Such depreciation would lead to a lower investment return when you convert your investment back to dollars. Conversely, if the euro appreciates against the dollar, your returns would increase accordingly. Currency risk is especially important for multinational companies and international portfolio managers with investment portfolio across multiple countries.

Commodity risk involves fluctuating prices of raw materials, such as those of oil, gold, and agricultural products. These prices can be highly volatile due to supply-demand imbalances in the market, policy instability, or natural disasters around the world. These price shocks can affect both producers on the supply side and consumers on the demand side. For example, when political conflicts disrupt oil production in the Middle East, oil prices can rapidly rise, affecting not only energy companies but also associated industries dependent on oil. Again, investors and firms exposed to commodities often adopt hedging strategies (such as purchasing futures and options) to protect themselves against sudden price swings.

Market risk is an essential consideration in almost every type of financial investment. It arises because prices in financial markets are perpetually changing, driven by a complex blend of economic indicators, political developments, changing investor sentiment, and unforeseen global events. Although it is impossible to completely eliminate market risk—given that no one can fully control these external influences—we can still employ several strategies to mitigate and manage its effects.

One of the most widely used approaches to managing market risk is **diversification**, which involves spreading investments across different asset classes, sectors, or geographic regions. As Harry Markowitz famously observed, “Diversification is the only free lunch in Finance.” By avoiding “putting all your eggs in one basket,” the underperformance of one asset can be balanced by the outperformance of another, thereby reducing overall portfolio volatility. When equity markets become volatile, holding bonds or commodities can often help stabilize the portfolio. As an example,

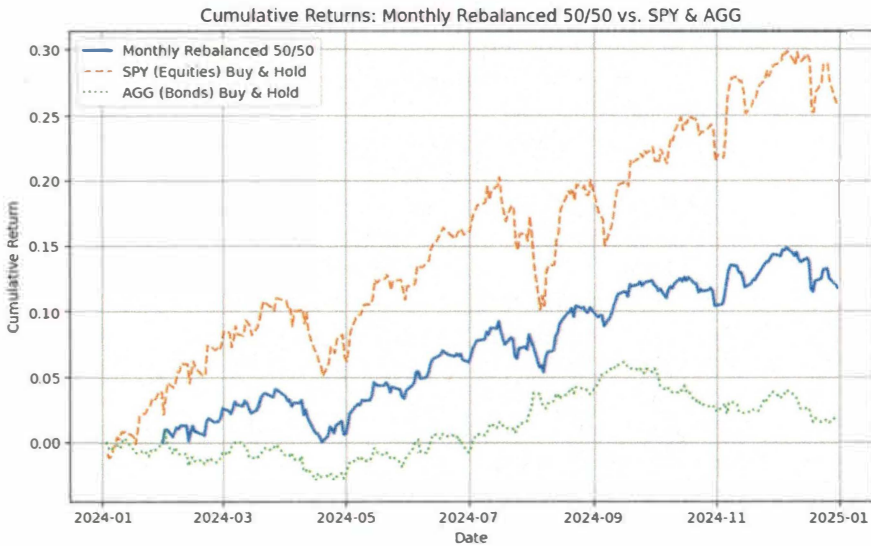


Figure 1-3 Comparing the wealth curve of three strategies in 2024

let us maintain a balanced 50/50 portfolio of equities (SPY) and bonds (AGG) in the year of 2024. We adopt an equal-weightage strategy with monthly rebalancing so that the risk profile of the portfolio remains relatively stable even if one asset class outperforms the other over a given month. This is the central idea of diversification: rather than relying exclusively on volatile equities or conservative bonds, the portfolio spreads its exposure, thereby lowering overall volatility compared to a single-asset strategy.

As shown in Figure 1-3, the monthly rebalanced portfolio of 50/50 shows a significantly smoother return path than the 100% buy-and-hold strategy (SPY), with an annualized volatility of approximately 7.27% compared to SPY 12.58%. Although this balanced approach does not match the upside potential of SPY, it provides a middle ground that delivers a higher return than bonds alone (AGG) but far less volatile than stocks. The result illustrates a key benefit of diversification: by combining two relatively uncorrelated assets and periodically rebalancing, the portfolio avoids the extreme fluctuations of equities while still capturing a portion of stock market gains.

See Listing 1-3 used to generate Figure 1-3.

```
1 import yfinance as yf
2 import pandas as pd
3 import numpy as np
4 import matplotlib.pyplot as plt
5
6 # -----
7 # 1. Download Historical Data
8 # -----
```

```

11 tickers = ["SPY", "AGG"]
12 data = yf.download(tickers, start="2024-01-01", end="2025-01-01")
13     ["Adj Close"]
14
15 # Drop any rows with missing values for simplicity
16 data.dropna(inplace=True)
17
18 # If the index has time zone info, remove it:
19 if data.index.tz is not None:
20     data.index = data.index.tz_localize(None)
21
22 # Calculate daily returns
23 daily_returns = data.pct_change().dropna()
24
25 # -----
26 # 2. Define Monthly Rebalance Dates
27 # -----
28 # We'll use the last calendar day of each month, but then we'll
29 # map that to the nearest *actual* trading date via asof().
30 # This ensures we don't hit missing index errors.
31
32 calendar_month_ends = pd.date_range(
33     start=daily_returns.index[0],
34     end=daily_returns.index[-1],
35     freq='M'
36 )
37
38 # Convert these month-ends to actual trading days (or the most
39 # recent trading day before them):
40 monthly_rebalance_dates = []
41 for date in calendar_month_ends:
42     # .asof(date) gives the last valid index on or before 'date'
43     # If no valid date is found, it returns NaT, so we skip it
44     trading_date = daily_returns.index.asof(date)
45     if pd.notnull(trading_date):
46         monthly_rebalance_dates.append(trading_date)
47
48 monthly_rebalance_dates = pd.DatetimeIndex(
49     monthly_rebalance_dates)
50
51 # -----
52 # 3. Simulate Monthly Rebalancing
53 # -----
54
55 initial_capital = 100_000.0
56 target_weights = np.array([0.5, 0.5]) # 50% in SPY, 50% in AGG
57
58 # Series to hold portfolio values for plotting
59 portfolio_values = pd.Series(dtype=float)
60
61 # Initialize positions on the first rebalance date
62 start_date = monthly_rebalance_dates[0]
63 spy_price = data.loc[start_date, "SPY"]
64 agg_price = data.loc[start_date, "AGG"]

```

```

59 spy_shares = (initial_capital * target_weights[0]) / spy_price
60 agg_shares = (initial_capital * target_weights[1]) / agg_price
61
62 # Record the initial portfolio value
63 portfolio_values.loc[start_date] = spy_shares * spy_price +
    agg_shares * agg_price
64
65 # Go through each subsequent monthly period
66 for i in range(len(monthly_rebalance_dates) - 1):
67     period_start = monthly_rebalance_dates[i]
68     period_end = monthly_rebalance_dates[i + 1]
69
70     # Get all trading days in [period_start, period_end]
71     period_days = daily_returns.loc[period_start:period_end].
    index
72
73     # Track day-to-day value within the month
74     for day in period_days:
75         # Update the portfolio value based on unchanged share
76         counts
77         spy_price = data.loc[day, "SPY"]
78         agg_price = data.loc[day, "AGG"]
79         daily_value = spy_shares * spy_price + agg_shares *
    agg_price
80         portfolio_values.loc[day] = daily_value
81
82     # At period_end, we rebalance to 50/50 (unless period_end is
    the final date)
83     if i < len(monthly_rebalance_dates) - 1:
84         # Use .asof() again for the final day price if necessary
85         final_day_price_spy = data["SPY"].asof(period_end)
86         final_day_price_agg = data["AGG"].asof(period_end)
87
88         final_value = spy_shares * final_day_price_spy +
    agg_shares * final_day_price_agg
89
90         # Re-allocate to 50/50
91         spy_shares = (final_value * target_weights[0]) /
    final_day_price_spy
92         agg_shares = (final_value * target_weights[1]) /
    final_day_price_agg
93
94 # -----
95 # 4. Calculate Portfolio Returns and Cumulative Returns
96 # -----
97 portfolio_returns = portfolio_values.pct_change().fillna(0)
98 cumulative_portfolio = (1 + portfolio_returns).cumprod() - 1
99
100 # -----
101 # 5. Compare to SPY and AGG Buy-and-Hold (No Rebalancing)
102 # -----
103 cumulative_spy = (1 + daily_returns["SPY"]).cumprod() - 1

```

```

104 cumulative_agg = (1 + daily_returns["AGG"]).cumprod() - 1
105
106 # Align indices for plotting (in case of any missing dates at the
    ends)
107 cumulative_portfolio = cumulative_portfolio.reindex(daily_returns
    .index, method='ffill')
108
109 # -----
110 # 6. Calculate Annualized Volatility
111 # -----
112 trading_days_per_year = 252
113 # The daily std. dev. of the portfolio returns * sqrt(252)
114 vol_portfolio = portfolio_returns.std() * np.sqrt(
    trading_days_per_year)
115 vol_spy = daily_returns["SPY"].std() * np.sqrt(
    trading_days_per_year)
116 vol_agg = daily_returns["AGG"].std() * np.sqrt(
    trading_days_per_year)
117
118 print("Annualized Volatility:")
119 print(f"Monthly Rebalanced 50/50 Portfolio: {vol_portfolio:.2%}")
120 print(f"SPY (Equities) Buy & Hold: {vol_spy:.2%}")
121 print(f"AGG (Bonds) Buy & Hold: {vol_agg:.2%}")
122
123 # -----
124 # 7. Plot the Cumulative Returns
125 # -----
126 plt.figure(figsize=(10, 6))
127 plt.plot(cumulative_portfolio, label='Monthly Rebalanced 50/50',
    linewidth=2)
128 plt.plot(cumulative_spy, label='SPY (Equities) Buy & Hold',
    linestyle='--')
129 plt.plot(cumulative_agg, label='AGG (Bonds) Buy & Hold',
    linestyle=':')
130 plt.title("Cumulative Returns: Monthly Rebalanced 50/50 vs. SPY &
    AGG")
131 plt.xlabel("Date")
132 plt.ylabel("Cumulative Return")
133 plt.legend()
134 plt.grid(True)
135 plt.show()

```

Listing 1-3 Comparing the wealth curve of three strategies in 2024

As shown in Figure 1-4, applying the same equal-weight strategy in 2020, when the COVID-19 pandemic initially emerged, resulted in smaller drawdowns than a purely equity-focused approach, illustrating how spreading exposure across asset classes can mitigate losses under adverse conditions. Observing the annualized volatilities, the monthly rebalanced 50/50 portfolio has a volatility of about 18.08%, substantially lower than the 33.45% registered by an all-equity (SPY) portfolio. In contrast, the all-bond (AGG) allocation remains the least volatile at 8.37%. Notably, diversification is especially beneficial during market downturns.

Note that in real-world settings, such rebalancing is often done at intervals (e.g., monthly or quarterly) to manage both market risk and transaction costs. We have seen that this combination of assets leads to reduced portfolio swings. Although returns on a balanced portfolio may be somewhat lower in booming equity markets (as shown in Figure 1-3), the reward comes during downturns, when the bond component tends to cushion losses (see Figure 1-4). We can continue to add even more asset classes (commodities, international equities, alternative investments, etc.) so as to further diversify the market risk. However, we must be aware that diversification only reduces unsystematic risk specific to a company or industry, not eliminate systematic risk that affects the entire market.

Another approach to managing market risk is hedging, which involves taking off-setting positions in different financial instruments to limit potential losses. Hedging often involves the use of derivatives, such as options, futures, and swaps. For example, an investor with a substantial equity portfolio might purchase put options that could generate profit (when exercised) if the underlying stocks decline, thereby cushioning against losses. Likewise, currency forwards or futures can lock in the transaction price in advance and help mitigate exchange rate fluctuations in international investments. However, these hedging activities often come with costs, such as premiums for options. Thus, a comprehensive risk management strategy needs to consider these additional transaction costs as well, as they could also diminish potential upside gains.

Beyond diversification and hedging, we can also employ a variety of other strategies to manage market risk. For example, we can perform tactical asset allocation to adjust portfolio exposures periodically in response to evolving market

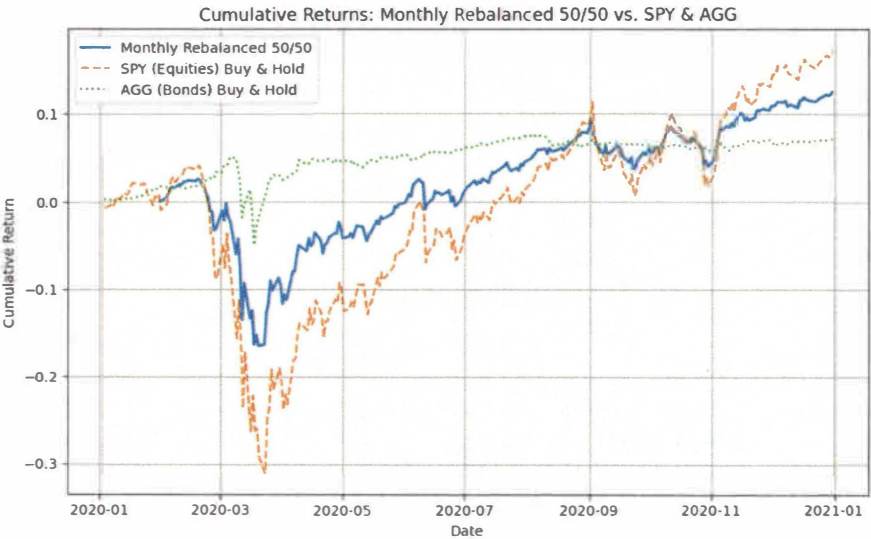


Figure 1-4 Comparing the wealth curve of three strategies in 2020

conditions. We have seen this in the previous equal weightage strategy, but such periodic rebalancing may also be done to cater to investor's specific risk-reward preference. Some investors also implement stop-loss orders, which automatically liquidate positions once they fall below a predetermined price, thereby preventing small losses from spiraling into large ones. As another example, risk parity strategies allocate capital based on the risk contribution of each asset rather than traditional fixed percentage allocations, aiming to balance volatility between different asset classes.

Managing market risk begins with recognizing the uncertainty inherent in market prices and adopting strategies to mitigate or even minimize potential losses. Because higher returns typically come with higher risk, reducing volatility often involves sacrificing some portion of potential gains, but this trade-off can prove to be well worth it if it prevents significant drawdowns during market downturns. In addition, effective risk management calls for a proactive approach: rather than simply reacting to sudden changes in market conditions, proactive risk management recommends that we continuously monitor trends and recalibrate portfolios (such as monthly rebalancing) as needed to maintain an acceptable risk-return balance.

We now provide a mathematical view on why mixing stocks and bonds in a portfolio typically reduces overall risk. Assume that the correlation between stock and bond is less than 1. In our two-asset portfolio setup, let R_S be the random variable for stock returns, and R_B be the random variable for bond returns. We form a portfolio P by allocating a fraction w to the stock index (such as SPY) and $(1 - w)$ to the bond index (such as AGG). Then the total portfolio return R_P can be expressed as

$$R_P = w R_S + (1 - w) R_B.$$

Similarly, the portfolio variance can be expressed as

$$\sigma_P^2 = \text{Var}(R_P) = \text{Var}(w R_S + (1 - w) R_B).$$

By standard properties of variance, we have

$$\sigma_P^2 = w^2 \text{Var}(R_S) + (1 - w)^2 \text{Var}(R_B) + 2 w (1 - w) \text{Cov}(R_S, R_B).$$

Now denote

$$\sigma_S^2 = \text{Var}(R_S), \quad \sigma_B^2 = \text{Var}(R_B), \quad \rho_{S,B} = \text{Corr}(R_S, R_B).$$

Then $\text{Cov}(R_S, R_B) = \rho_{S,B} \sigma_S \sigma_B$.

Therefore, we have

$$\sigma_P^2 = w^2 \sigma_S^2 + (1 - w)^2 \sigma_B^2 + 2 w (1 - w) \rho_{S,B} \sigma_S \sigma_B.$$

Since $\rho_{S,B} < 1$ (it is very unlikely that bonds move in perfect sync with stocks), the product $\rho_{S,B} \sigma_S \sigma_B$ is less than $\sigma_S \sigma_B$. Hence, the cross-term

$$2 w (1 - w) \rho_{S,B} \sigma_S \sigma_B$$

is smaller than the corresponding term if stocks and bonds perfectly moved together ($\rho_{S,B} = 1$). This implies that

$$\sigma_P^2 < w^2 \sigma_S^2 + (1 - w)^2 \sigma_B^2 + 2 w (1 - w) \sigma_S \sigma_B$$

(where we have used the fact that $\rho_{S,B} < 1$)

Simplifying, we have

$$\sigma_P < w \sigma_S + (1 - w) \sigma_B$$

This means that the standard deviation of the portfolio is strictly lower than the weighted sum of the individual standard deviations when the correlation is less than 1. Therefore, as long as the returns of stocks and bonds are not perfectly positively correlated ($\rho_{S,B} \neq 1$), the combined variance is reduced compared to owning only stocks or only bonds, in any weight allocation. This lower variance (or volatility) is precisely the mathematical reason why diversifying between stocks and bonds helps reduce risk in a portfolio.

Now that we have shown that diversification helps reduce total portfolio risk, the next question is to what extent we can reduce the risk. That is, how do we allocate the weights across these two assets so as to achieve the minimal risk of our portfolio? To answer this question, we need to derive the global minimum-variance (GMV) portfolio weights for our two-asset model, assuming that there are no constraints such as short-sale bans or leverage limits.

Recall the definition of portfolio variance that we would like to minimize:

$$\sigma_P^2 = w^2 \sigma_S^2 + (1 - w)^2 \sigma_B^2 + 2 w (1 - w) \rho_{S,B} \sigma_S \sigma_B.$$

Since this is a quadratic function of w , we can take the derivative $\frac{d}{dw} \sigma_P^2$ and set it to zero to obtain the closed-form optimality condition:

$$\frac{d\sigma_P^2}{dw} = 2w \sigma_S^2 - 2(1 - w) \sigma_B^2 + 2 \rho_{S,B} \sigma_S \sigma_B [1 - 2w] = 0.$$

After a bit of algebra, we can solve for w^* :

$$w^* = \frac{\sigma_B^2 - \rho_{S,B} \sigma_S \sigma_B}{\sigma_S^2 + \sigma_B^2 - 2 \rho_{S,B} \sigma_S \sigma_B}.$$

This is the weight of the stock index (SPY) that produces the minimum variance in the two-asset framework. Consequently, the weight for the bond index (AGG) is

$$1 - w^* = 1 - \frac{\sigma_B^2 - \rho_{S,B} \sigma_S \sigma_B}{\sigma_S^2 + \sigma_B^2 - 2 \rho_{S,B} \sigma_S \sigma_B}.$$

Of course, when our goal is to maximize the portfolio return (recall that $R_P = w R_S + (1 - w) R_B$), we can simply invest all our budget in the higher-return asset, which is the stock index.

Now, let us look at credit risk in the following section.

1.1.2 Credit Risk

Credit risk evaluates the ability of a borrower, also known as a counterparty, to repay the principal and the interest agreed on in a loan. Typical lenders include banks, specialized lending firms, and bond investors, all of whom expect to recoup their invested principal plus any associated interest. However, there is no guarantee of repayment; borrowers may default on part or all of their debt. Thus, credit risk captures the possibility that a borrower or counterparty will fail to meet the obligations stipulated by a loan or debt agreement. In the following section, we will explore several common types of credit risk, including default risk, credit spread risk, and downgrade risk.

Default risk is often viewed as the most common type of credit risk, which captures both the probability that a borrower will fail to repay a loan and the potential loss incurred if such a default occurs. In the worst case, lenders or bondholders could face a complete loss of their principal. To manage this risk, financial institutions typically employ three interrelated models: the probability of default (evaluating the likelihood that the borrower will default), the loss given default (estimating how much might be lost when default occurs), and the exposure at default (estimating the total amount at risk at the time of default). When taken together, these models provide an overall expected credit loss calculation, indicating how much a lender might expect to lose should the borrower eventually fail to meet the debt obligations. Banks and financial institutions commonly use this framework to set credit limits, price loans, and maintain sufficient capital reserves.

Credit spread risk arises when the creditworthiness of a borrower deteriorates, even if this change does not result in a real default. The primary mechanism at work is the “credit spread,” defined as the yield differential between a risk-free bond (such as a US Treasury) and a bond exposed to credit risk. If investors suddenly perceive a company to be more likely to default, perhaps due to unfavorable financial news, adverse market conditions, or other negative signals, they will demand a higher yield to compensate for the increased uncertainty. Because coupon payments and face value of a bond are generally fixed, the only way for yield to increase is for the market price of the bond to drop accordingly. Consequently, bondholders can incur

losses based solely on a market-driven reassessment of the borrower's credit profile, regardless of any actual default.

Downgrade risk refers to the possibility that a borrower's credit rating may be lowered by agencies such as Moody's or Standard & Poor's, signaling increased credit risk. A recent example is Intel's removal from the Dow Jones Industrial Average on November 8, 2024, after a 25-year tenure, reflecting its declining market position and financial performance. Such downgrades typically lead to a decrease in the value of the issuer's securities. For bondholders, a downgrade results in a higher yield premium, causing the bond market price to fall. Consequently, if an investor holds a bond from a downgraded entity, the bond's market value diminishes, leading to a potential loss upon sale. This underscores the importance of monitoring credit ratings, as they directly influence bond valuations and investment returns.

As an example, suppose that an investor holds a corporate bond from a company with a face value of \$1,000, a fixed annual coupon of 5% (\$50 per year), a current yield to maturity (YTM) of 6%, and a risk-free yield of 3%. Following a credit rating downgrade from A to BBB due to deteriorating financials, the market now demands an 8% yield to compensate for the increased perceived risk. Although there is no immediate expectation of default, the bond will continue to pay its fixed coupon. To align with the market's demand for a higher yield, the price of the bond must decrease, since its fixed coupon and face value remain unchanged. This price decline, driven by the increase in the credit spread, reflects the risk of credit spread rather than an actual default loss, since the downgrade impacts the market value of the bond rather than its cash flow reliability.

The probability of default is one of the most commonly monitored risks in financial risk management, as it reflects the likelihood that a company will not meet its obligations. As a quick exercise, consider a firm with a 2% chance of default in the first year and a 3% chance in the second year, conditional on surviving the first year. To determine the overall probability of default at the end of Year 2 in a single step, note that the firm can default immediately in the first year with probability 0.02. Alternatively, if it survives the first year (which happens with probability 0.98), then it faces a 0.03 chance of default in the second year. Consequently, the total probability of default by the end of Year 2 is

$$0.02 + (0.98 \times 0.03) = 0.02 + 0.0294 = 0.0494 = 4.94\%.$$

As an extension to the credit risk framework, the credit migration matrix is a widely used tool to characterize the transition probabilities among different states. In particular, it systematically captures the probabilities of transitioning between different credit ratings over a given time period. This transition probability provides us with a measure of the likelihood of upgrades, downgrades, and defaults among different credit grades. For example, we can adopt a frequency-based approach to estimate these transition probabilities using historical data at different time periods, thus describing the evolving credit quality of borrowers. This matrix also measures intermediate changes in creditworthiness that can lead to changes in credit spreads, offering deeper insights into potential credit deterioration paths.

Next, we look at the liquidity risk.

1.1.3 Liquidity Risk

Liquidity risk measures the ease with which an asset can be bought or sold without causing a significant impact on its price. Highly traded assets, such as popular stocks, are typically very liquid because of consistent supply and demand throughout trading hours. This allows transactions to occur seamlessly, either at the prevailing market price via a market order or at a preset price using a limit order. In contrast, assets such as real estate can be highly illiquid, particularly when the property is in a less desirable location or condition. Selling such assets often requires drastically lowering the price to attract buyers. In markets with limited participants, this lack of liquidity can force sellers to accept prices below the intrinsic value of an asset or force buyers to pay more than the asset is worth. This highlights the challenges and potential costs associated with trading in less liquid markets.

There are two primary types of liquidity risk: asset liquidity risk and funding liquidity risk. Asset liquidity risk arises when an asset cannot be sold at the desired price due to a lack of willing buyers. In such cases, the seller faces two options: wait for a buyer willing to pay the desired price or reduce the asking price to attract more buyers. This risk highlights the challenges of selling less liquid assets promptly without incurring a loss. In contrast, funding liquidity risk refers to the inability to meet short-term financial obligations, such as covering day-to-day operational costs or making scheduled debt payments such as housing mortgages. Addressing this risk requires maintaining sufficient cash flow to meet these consistent and periodic commitments, ensuring that operations and debt servicing remain uninterrupted. Both types of liquidity risk underscore the importance of liquidity management in financial decision-making.

1.1.4 Operational Risk

Operational risk refers to internal factors that can disrupt an organization's day-to-day functions, compared to external forces such as market fluctuations or macroeconomic events. These disruptions often come from breakdowns in internal processes, hardware or software systems, and human activities, either inadvertent or deliberate. For example, a sudden power outage can stop trading floors or payment processing unless backup generators are in place, and a rogue trader may bypass internal oversight, incurring substantial unauthorized losses. Even seemingly minor errors, such as incorrect data entry or flawed calculations, can entail material, financial, or reputational damage. To mitigate these risks, firms commonly implement robust internal controls, formalized procedures, and carefully designed system redundancies. Automation can also reduce human error, while comprehensive guidelines ensure that staff are well-informed about best practices and compliance requirements. By recognizing potential vulnerabilities within oper-

ations and proactively countering them, an organization can better protect itself against disruptions that might otherwise lead to significant losses.

1.1.5 Model Risk

Model risk arises when the mathematical models used to guide financial decision-making, whether highly transparent or so-called “black-box” models, differ significantly from actual market behavior. As the adage goes, “All models are wrong, but some are useful.” In practice, the usefulness of a model in modern settings, particularly when employing deep learning or other advanced neural network architectures, hinges on three interlinked elements: the quality and representativeness of the underlying data, the choice of model architecture, and the available computational resources.

With respect to data, financial markets directly shape the information inputs that feed into these models. Predictive accuracy is highly dependent on having current and comprehensive data, capturing the genuine statistical signals needed for reliable forecasting. Outdated data may not reflect future phenomena, incomplete data could fail to capture critical aspects of the market, and flawed or misspecified data would inevitably lead to incorrect outputs or instability in model predictions. Ensuring representativeness in time, for example, by carefully selecting historical data that match as closely as possible the conditions likely to prevail in the future, is just as vital as preserving an appropriate signal-to-noise ratio.

The model assumptions also require close scrutiny. These assumptions, in effect, define how the chosen architecture transforms the data into the final predictions. Deep neural networks, for instance, impose relatively few explicit assumptions on functional form and can approximate highly complex relationships in the data. However, they often face the risk of overfitting, where a model that excels in historical or training data may not be able to generalize when faced with new information. In contrast, simpler models, such as linear regression, are built on stronger structural assumptions, usually linearity in relationships, and thus run the risk of underfitting. Balancing these trade-offs to obtain robust, generalizable models is a delicate but essential part of the model design process.

Model validation and ongoing monitoring further mitigate the risk of relying on a model that may have become outdated or simply failed to account for emerging realities. Once a model has been developed, often using historical data, it lacks immediate awareness of real-time events in the external environment. Consequently, a model might still issue optimistic forecasts for a company that, in reality, has suffered sudden reputational damage and may soon face a decline in share price. Regular re-evaluation, combined with updated data feeds, stress testing, and comparative benchmarks, can help detect such discrepancies before they lead to potential financial losses.

In a later chapter of the book, we will take a deeper look at model risk in finance, emphasizing the importance of robust validation techniques and dynamic adaptation. Next, we turn to legal and regulatory risks.

1.1.6 Legal and Regulatory Risk

Legal and regulatory risk refers to the prospect of financial or reputational damage resulting from changes in laws, regulations, or government directives. Such changes often require businesses to adapt their operational procedures and risk management strategies rapidly. For example, in the banking sector, dedicated teams are not uncommon, whose primary role is to interpret and implement policies issued by regulatory authorities, including the Prudential Regulation Authority (PRA) in the United Kingdom and the Monetary Authority of Singapore (MAS) in Singapore, to ensure continuous compliance.

One prominent example is the Basel regulatory framework, which governs the capital adequacy requirements for banks. This framework has undergone multiple refinements, including Basel I, Basel II, Basel III, and beyond, to keep up with ever-changing market dynamics and financial innovations. A similarly influential standard is the International Financial Reporting Standard 9 (IFRS 9), which imposes nuanced requirements to estimate expected losses across various portfolios. Noncompliance with these obligations can lead to severe penalties, including substantial fines and potential reputational setbacks for the institutions involved. These realities illustrate why proactive measures, including rigorous monitoring of legal and regulatory developments and strong internal controls, are vital to mitigating legal and regulatory risk in modern financial institutions.

1.1.7 Systemic Risk

Systemic risk refers to a major threat that endangers the entire financial system, often originating with the failure of a single institution or market and then triggering a broader chain reaction. Regulators and policymakers focus on systemic risk precisely because it reaches beyond isolated firms and markets, carrying the potential to ignite a full-blown financial crisis and produce severe economic downturns. One well-known example occurred in the late 1990s, when the hedge fund Long-Term Capital Management (LTCM) experienced severe distress. To avoid widespread repercussions, the Federal Reserve convened an emergency meeting with major banks in an effort to engineer a rescue plan. Such interventions aim to restore confidence because if market sentiment collapses, even a localized disruption could escalate into a global crisis.

When an entity is considered “too big to fail,” government agencies often intervene—potentially using taxpayer funds—to conduct a bailout. However, this assistance is not guaranteed, as the sovereign debt itself can default under extreme conditions. A major driver of systemic risk is the high degree of interconnectivity among financial institutions. Banks, insurance companies, and other financial entities often hold each other’s debt or rely on similar funding channels; consequently, the failure of a key player can have a ripple effect. The collapse of Lehman Brothers in 2008, for example, led to a huge impact on the global financial system due to

the firm's extensive network of loans, derivatives, and credit arrangements. These interdependencies indicate how quickly a localized problem can spread, posing a systemic threat that demands careful monitoring, robust regulation, and coordinated policy responses.

1.1.8 Environmental, Social, and Governance (ESG) Risk

ESG risk involves understanding the potential financial impact of environmental, social, and governance factors on investments. In recent years, ESG risk has become an important topic of financial analysis as investors and companies recognize their importance in managing real risks that can affect long-term performance. ESG provides different perspectives that allow us to assess how well a company is prepared to face the challenges of the modern world, from climate change to social responsibility to ethical leadership. Let's break down what ESG risk encompasses and discuss why it is becoming a big deal in the investing world.

Environmental risk contains a wide spectrum of concerns, including climate change, depletion of natural resources, waste management, and pollution. Businesses that rely heavily on fossil fuels or do not mitigate their environmental impact can face substantial legal, financial, and reputational challenges. Heightened regulatory measures, such as those designed to reduce carbon emissions, can significantly increase operational costs for companies not prepared to adapt. In parallel, organizations linked to environmental harm or ecological disasters risk losing public trust and enduring long-term damage to their brand. These realities underscore why investors and stakeholders increasingly focus on the environmental footprint of a company when evaluating its overall viability.

Social risk focuses on how companies manage relationships with employees, suppliers, customers, and the broader communities where they operate. Issues such as labor standards, diversity and inclusion, human rights, and consumer protection have become central to ensuring sustainability and fostering stakeholder confidence. As an example, an unfavorable review from a former employee on Glassdoor can put a bad look on the public image of an organization, potentially deterring both prospective talent and investors. In today's interconnected economy, companies demonstrating proactive and responsible social practices are often more resilient, with a stronger capacity to avoid costly controversies.

Governance risk, meanwhile, refers to the internal policies, leadership structures, and oversight mechanisms that guide corporate decision-making. It involves factors such as board diversity, executive compensation schemes, shareholder rights, and transparency in financial disclosures. Many companies, for example, strive to ensure that their boards meet certain benchmarks in terms of gender, professional background, or other diversity measures, recognizing that broader representation often correlates with more balanced and effective leadership. By maintaining high standards in environmental, social, and governance (ESG) practices, organizations can not only mitigate risk but also strengthen their reputations and long-term competitiveness.

Table 1-1 Overview of risks in financial markets

Type of Risk	Description
Market Risk	Risk of losses due to changes in market prices of financial instruments like stocks, bonds, foreign exchange rates, and commodities. Influenced by economic news, political events, natural disasters, and investor sentiment. Includes equity risk, interest rate risk, currency risk, and commodity risk.
Credit Risk	Risk that a borrower will not repay a loan or fulfill financial obligations. Key for banks, lenders, and bond investors. Includes default risk, credit spread risk, and downgrade risk.
Liquidity Risk	Risk that an investor cannot buy or sell an asset quickly without significantly impacting its price. Includes asset liquidity risk (lack of buyers or sellers) and funding liquidity risk (inability to access funds).
Operational Risk	Risk of loss due to failures in internal processes, systems, people, or external events. Examples are process failures, system outages, human errors, and natural disasters.
Model Risk	Risk arising from inaccuracies in the mathematical models used for financial decision-making. Can result from wrong assumptions, poor-quality data, or errors in implementation.
Legal and Regulatory Risk	Risk of financial loss due to changes in laws, regulations, or government policies. Includes regulatory changes, compliance costs, and potential legal actions.
Systemic Risk	Risk of a breakdown in the entire financial system, often triggered by the failure of a major institution. Can cause a chain reaction or contagion effect, leading to a global crisis.
ESG Risk	Risk related to environmental, social, and governance factors affecting investments. Includes environmental risks (e.g., climate change), social risks (e.g., labor practices), and governance risks (e.g., corporate policies).

1.1.9 A Summary of Common Risk Types

Table 1-1 provides a high-level overview of the various types of financial risk prevalent in financial markets. Each type of risk, be it market, credit, liquidity, operational, model, legal and regulatory, systemic, or ESG, presents unique challenges and requires specific strategies for effective management. By recognizing the characteristics and implications of each risk type, we can make more informed decisions, adapt to changing conditions, and better protect our assets against potential financial disruptions.

In the final section, we explore commonly used financial products and their varying risk profiles. These instruments serve as fundamental risk management tools in modern financial markets, enabling many practitioners to hedge, diversify, and control the myriad of uncertainties inherent in investment and lending activities in the financial markets.

1.2 Common Financial Instruments

Having explored the various types of financial risks, we now look at the common financial instruments as tools used to navigate these risks in the financial markets. These tools include stocks, bonds, derivatives, and more, which are often used by investors and institutions to build and manage portfolios. Not all financial instruments are created equal when it comes to risk, so we will also rank them in terms of their riskiness to help you better understand where they fit within a broader investment strategy.

For example, bonds are generally considered less risky than stocks. Bonds are debt products that generally offer an interest payment (called a coupon), which may be fixed or floating, over a set period, and return the principal at maturity. They are typically less volatile, making them a safer choice for more conservative investors. However, while stocks offer the potential for higher returns, they come with greater volatility and risk. Then, there are other instruments like derivatives, which can be used to hedge risks or, conversely, to take on additional risk (called leverage) in pursuit of higher returns. Understanding the risk profile of each instrument is crucial because it helps us determine how best to incorporate them into the investment strategy based on the specific risk tolerance and investment objective.

In this section, we will attempt to group common financial instruments by the typical risk levels, ranging from low-risk options that offer more stability to high-risk choices that come with the potential for higher returns, but also greater volatility. We start with low-risk assets.

1.2.1 Low-Risk Assets

Low-risk assets are generally known for their stability and are less prone to significant price swings compared to higher-risk investments. They are the bedrock of conservative investment strategies, appealing primarily to those who prioritize capital preservation and steady income over pursuing higher returns. These are assets that are used to protect the principal amount invested since they provide a relatively predictable, albeit modest, return. Although they may not deliver spectacular gains, such reliability makes them attractive, especially in uncertain or volatile market conditions. Let us take a closer look at some common low-risk assets and their risk-return characteristics.

- **Cash and Cash Equivalents:** These are considered the safest assets around. Cash equivalents include instruments such as money market funds, Treasury bills, and savings accounts. The primary appeal of these assets is that the principal is almost always guaranteed, which means you are highly unlikely to lose money. However, this safety comes at a cost—they offer very low returns, often barely keeping up with inflation. For example, a savings account might provide a return of 1–2% per year, which is minimal but stable. These are

ideal for risk-averse investors who prioritize liquidity, which means they can easily access their money as needed, without worrying about market fluctuations. They're also a great option for those who need to park their funds temporarily while deciding on more strategic and longer-term investments.

- **Government Bonds:** Government bonds, especially those issued by stable governments, such as US Treasury bonds, are another popular low-risk asset. They are considered low-risk because they are backed by the government's ability to tax its citizens and print money if needed. This means that government bonds are highly unlikely to default, although such defaults have happened before. A notable example is the 2012 Greek debt crisis, when Greece restructured its sovereign bonds under severe economic pressure. Government bonds provide regular interest payments, known as coupons, which can offer a reliable income stream to investors. They are less volatile than stocks, meaning that their prices do not fluctuate as much in response to market news or economic data. However, the trade-off is that the returns are typically lower than the riskier assets. For example, a ten-year US Treasury bond could yield around 2–3% annually, which is better than cash but still relatively modest. These are a good option for conservative investors or retirees who want to ensure a steady income without taking on much risk.
- **Investment-Grade Corporate Bonds:** These are bonds issued by companies with strong credit ratings, such as those rated BBB or higher by agencies like Standard & Poor's or Moody's. Although they are slightly riskier than government bonds (due to the small probability of default), they are still considered relatively low risk. To account for such default risk, the returns, or yields, on investment-grade corporate bonds are typically higher than those on government bonds, offering a bit more income for a bit more risk. For example, an investment-grade corporate bond might yield 3–5% per year, which is attractive to investors looking for a balance between risk and return. These bonds are suitable for those who want a higher yield than government bonds without taking too much risk in high-yield or junk bonds.

In summary, low-risk assets offer safety and stability. They're perfect for investors who want to protect their principal and enjoy a predictable income stream without worrying about market turbulence. Although the returns on these assets are lower than most higher-risk investments, the trade-off is still safety and stability, especially in times of economic uncertainty or for those nearing retirement who need to safeguard their savings.

1.2.2 Moderate-Risk Assets

Moderate-risk assets seek a sweet spot between risk and return. These assets are good investment vehicles for investors who are looking to take on a bit more risk, but are not yet ready to dive into high-risk investments. They offer a nice balance, providing the potential for higher returns than low-risk assets while still maintaining

a reasonable level of stability. Let us take a closer look at some common types of moderate-risk assets.

- **Dividend-Paying Stocks:** Dividend-paying stocks represent shares of listed companies that provide dividends to shareholders. These stocks are considered moderately risky because they provide a dual benefit: the potential for both capital appreciation (as the stock price increases) and a steady income stream through dividends (should this be provided). Companies that pay dividends are usually more mature and have stable cash flows, and issuing such dividends also has a promoting effect, thus leading to lower volatility in their stock prices compared to those of high-growth companies. However, a certain level of risk is involved. If the market goes down or the company's financial status worsens, the stock price will drop, and the dividend payout may be reduced or even avoided. Despite these risks, these dividends remain an important consideration for investors seeking long-term growth and short-term income, particularly during periods of market uncertainty.
- **Real Estate Investment Trusts (REITs):** REITs allow investors to participate in the real estate market without having to purchase and own properties directly. The purchase is made through a company that owns and operates the real estate to generate rental income. These real estate properties could include multiple types, including office buildings, shopping malls, or apartments. Investors then enjoy a percentage of the rental income based on the total ownership of shares. Like stocks, real estate property could increase in value, along with dividend payout, which amounts to a steady stream of income. REITs are subject to market risk that causes housing prices to move downward, as well as interest rate risk that influences property value given rising interest rates.
- **High-Yield Bonds (Junk Bonds):** High-yield bonds, often referred to as junk bonds, are issued by relatively risky companies with lower credit ratings. These companies are considered more likely to default on their debt than creditworthy companies, so bond yields are set higher to attract investors to take up the additional risk. This means that if the issuing company fails to meet its debt obligations, bondholders could lose part or all of their investment. The higher return of high-yield bonds thus comes from the risk premium taken by bond investors. In addition, these bonds also offer diversification benefits, as their returns are often less correlated with traditional stocks and (low-yield) bonds.

In summary, moderate-risk assets provide a middle ground for those who prefer more than just the safety of low-risk options but aren't quite ready to dive into high-risk territory. These assets combine the potential for higher returns with a level of stability that can help cushion against market volatility.

1.2.3 High-Risk Assets

High-risk assets carry the potential for substantial returns, but also bear the equivalent possibility of significant losses. These assets are geared toward investors who exhibit higher risk tolerance, seek notable portfolio growth, and are comfortable with the increased volatility that may ensue. In other words, the prices of high-risk assets can deliver considerable rewards but also fluctuate dramatically in response to market conditions, economic factors, or even company-specific news. Let us now look at several common categories of such high-risk assets.

Stocks (equities) issued by smaller companies or startups are commonly regarded as high-risk assets due to their increased volatility. Unlike dividend-paying stocks that can provide a steady income stream and often exhibit more moderate price fluctuations, shares of newly established or smaller-scale businesses face increased uncertainty. Their limited size makes them more susceptible to market dynamics, sometimes resulting in rapid price movements. Although this volatility can yield substantial returns in a bullish environment, it also creates the possibility of significant losses if the market declines or the company fails to meet performance expectations. To mitigate this elevated risk, investors often diversify their portfolios, for instance, by holding stocks from multiple sectors or asset classes.

Cryptocurrencies, such as Bitcoin and Ethereum, are among the most volatile assets in modern financial markets. Their rapid price fluctuations can generate sizable returns, yet this potential is also accompanied by a similarly high degree of risk. Prices are influenced by a multitude of factors, including market sentiment, technological innovation, regulatory developments, and broader macroeconomic conditions. In contrast to traditional assets, cryptocurrencies typically lack intrinsic value and government backing, which further amplifies their risk profile. In addition, ongoing regulatory uncertainties and the susceptibility of digital exchanges or wallets to security breaches add additional complexity to their investment landscape. Consequently, while cryptocurrencies can yield impressive gains, we must also recognize potential significant losses and exercise due diligence before allocating capital to this emerging asset class.

- **Commodities**, such as oil, gold, silver, and various agricultural products, can be highly speculative investments due to their wide exposure to factors such as geopolitical events (for instance, unrest in oil-producing regions), weather patterns (which affect crop yields), and global economic cycles (shaping industrial demand for metals). These factors can result in sudden and substantial price movements, leading to increased risk for investors. At the same time, commodities may offer valuable diversification benefits and provide a hedge against inflation, since they often appreciate in price during periods of rising inflation.
- **Private equity and venture capital** involve the acquisition of stakes in privately held companies, often startups or businesses in the early stages of development.

Although such investments can generate substantial returns, often exceeding those in public equity markets, this potential is also accompanied by increased risks. A venture may also face liquidity risk because private shares cannot be readily bought or sold on the open market. Those investing in private equity or venture often need to wait several years for the company to go public or be acquired, which typically brings a significant amount of return. Furthermore, the lack of regular financial disclosures and the high failure rate among early-stage companies amplify the risk profile. However, for those equipped to bear these uncertainties, private equity and venture capital can offer a capital boost for high-growth sectors. This requires a long-term investment horizon and the patience to tolerate elevated levels of volatility.

In general, high-risk assets can provide significant growth opportunities, but also require a careful approach to risk management.

Next, we look at another class of risky assets, called derivatives.

1.2.4 Derivatives

Derivatives are a different breed of financial instruments. Unlike traditional investments in which you own an asset directly, derivatives are contracts that derive their value from the performance of something else, such as a stock, bond, commodity, or even an interest rate. This indirect relationship gives derivatives their unique versatility: they can be used to hedge against risk, speculate on future price movements, or even take advantage of arbitrage opportunities when markets experience temporary frictions. Let us look at some common types of derivatives:

- **Options** serve as financial contracts that give the right, but not the obligation, to buy or sell an underlying asset at a predetermined strike price on or before a specified expiration date. This structure (assuming a call option) gives us a unique advantage in that we can reap the benefit of unlimited upside potential, along with limited downside risk, without committing to its full purchase or sale. Options can also be used for hedging purposes, which can mitigate downside risk and protect against adverse price movements. When used for speculative purposes, they can offer outsized returns relative to the initial premium (paid to purchase the option). However, options exhibit a varied risk profile. On the one hand, option buyers face limited risk, which is capped at the premium paid for the contract. However, option writers (sellers) may incur substantial losses if the market moves against them since their potential liability can far exceed the premium received. This combination of versatility and asymmetrical risk makes options a powerful but potentially dangerous tool.
- **Futures and Forwards** are derivative contracts that obligate one to buy or sell an underlying asset at a predetermined price on a specified future date. Futures are standardized instruments traded on regulated exchanges, which enjoy a high level of liquidity and transparency, while forwards are privately negotiated over-

the-counter (OTC) contracts that offer more customization, but lack sufficient oversight and market visibility. Hedgers such as farmers and oil producers are frequent users of futures and forwards as they are interested in locking the future prices to be transacted. After locking the prices, these farmers and producers are no longer subject to unfavorable market fluctuations, although they also give up opportunities for the upside volatility and resulting return. Similarly, speculators would bet on price movements for profit. Being derivative contracts, they provide a leverage effect, meaning even a small change in the price of the underlying asset could lead to substantial gains or losses.

- **Swaps** are derivative contracts that allow two parties to exchange specified cash flows or assets, often to manage or transfer risk. The most commonly encountered variants are interest rate swaps, wherein one party exchanges a fixed interest rate for a floating interest rate (or vice versa), and currency swaps, through which counterparties swap principal and interest payments denominated in different currencies. A typical use case involves converting a floating rate loan to a fixed rate obligation, thereby stabilizing payments and insulating the borrower from interest rate fluctuations. However, participating in a swap introduces counterparty risk, as each party depends on the other's capacity to meet ongoing payment obligations. If the counterparty's creditworthiness deteriorates, or if the counterparty defaults, the non-defaulting party can face financial losses and find itself without the anticipated hedge.

Derivatives represent a category of highly adaptable instruments that can be used to achieve various objectives, including hedging strategies and speculative positions. Although subsequent chapters will illustrate specific hedging examples, it is equally important to recognize the substantial losses that can arise from the improper or excessive use of these products. Furthermore, derivatives do not simply align with conventional low-, medium-, or high-risk classifications, given their structural complexities and the diverse ways in which they can be utilized. Consequently, it is more appropriate to treat them as a separate asset category in any comprehensive risk assessment.

1.2.5 A Summary of Financial Instruments by Risk Level

Table 1-2 categorizes different financial instruments by their typical risk levels, providing a unified overview of the spectrum of investment options available in financial markets. Low-risk assets, such as cash equivalents, government bonds, and investment-grade corporate bonds, are ideal for investors who prioritize stability and capital preservation. Moderate-risk assets, such as dividend-paying stocks, REITs, and high-yield bonds, offer a balanced approach, providing the potential for higher returns while maintaining a reasonable level of risk. On the other hand, high-risk assets, including stocks of smaller companies, cryptocurrencies, commodities, and private equity, cater to investors with a higher risk appetite, offering the possibility of significant returns but with greater volatility. Finally, derivatives are a unique class

Table 1-2 Classification of financial instruments by risk level

Risk level	Asset type	Examples
Low-Risk Assets	Stability and predictable income; ideal for conservative investors.	Cash Equivalents (e.g., savings accounts, money market funds), Government Bonds, Investment-Grade Corporate Bonds
Moderate-Risk Assets	Balance of risk and return, with moderate volatility.	Dividend-Paying Stocks, REITs, High-Yield Bonds
High-Risk Assets	High returns and high volatility; suited for high-risk tolerance.	Stocks (e.g., smaller companies or startups), Cryptocurrencies, Commodities (e.g., oil, gold), Private Equity, Venture Capital
Derivatives	Instruments that derive value from underlying assets; used for hedging or speculation.	Options, Futures, Forwards, Swaps

of instruments that can range from low to high risk depending on their usage; they are highly versatile tools that are used for both hedging and speculative strategies.

1.3 Summary

This chapter introduced the multifaceted nature of financial risk by examining how unexpected events, ranging from daily market fluctuations to historic pandemics, can severely impact investment portfolios and the broader financial system. We introduced standard categories of risk, including market, credit, liquidity, operational, model, legal and regulatory, systemic, and ESG considerations, each highlighting distinct challenges. Through real-world examples, such as the 2020 market sell-off and the ongoing focus on climate-driven regulations, we saw how these risk dimensions can manifest abruptly, underscoring the need for robust risk assessment and mitigation strategies.

We then explored how probability distributions serve as indispensable tools in capturing the randomness of market-driven outcomes, offering quantitative insights into the likelihood of favorable and adverse scenarios. Building on this probabilistic foundation, we demonstrated practical risk management techniques such as diversification, hedging, and tactical asset allocation. By presenting a balanced 50/50 equity-bond portfolio vs. a purely equity-oriented investment, we illustrated how mixing uncorrelated assets can reduce volatility and drawdowns, particularly during periods of market turbulence.

Finally, we surveyed an array of financial instruments and classified them by typical risk levels, from low-risk government bonds and investment-grade securities to moderate-risk assets such as dividend-paying stocks and REITs, all the way up

to high-risk ventures such as early-stage equities and cryptocurrencies. Derivatives, which function as contracts derived from underlying assets, were also discussed for their dual capacity to hedge against unwanted risk or amplify speculative positions. Together, these discussions serve as a gateway to the quantitative methods and analytical frameworks detailed in the following chapters.

The world of finance revolves around two fundamental concepts: risk and return. When we invest, the goal is often to maximize return, minimize risk, or reach an optimal trade-off between the two. However, a higher level of risk often accompanies higher potential returns, and these two quantities are often positively correlated. For example, the previous chapter introduced low-risk and high-risk products, which also correspond to (relatively stable) low-return and (likely volatile) high-return assets. This risk manifests itself in the uncertainty of the investment outcome, meaning that we are not certain about whether the investment will be profitable or not, and, if profitable, how much. Thus, a risk-averse investor tends to prefer low-risk products that deliver more or less guaranteed returns.

Understanding how risk and return are intertwined is crucial for making informed investment decisions. The trade-off is simple: to aim for higher returns, we need to face higher risk. This additional risk that investors take on is what generates the “risk premium,” an additional layer of return that compensates for the burden of this uncertainty. In essence, risk-seeking investors are not just gambling for higher rewards; they’re earning a premium for venturing into more volatile territories. This concept is similar to starting a business; if it is successful, the return could be tremendous, and the corresponding risk of losing market competition or not gaining the right market share is also high. Grasping this risk-return dynamic is therefore at the heart of constructing and managing a successful investment portfolio.

2.1 Understanding Return

So what is the return of a financial asset? Return is the financial reward we receive for putting our money to work in the financial market instead of not investing at all (even saving in the bank gives us fixed interest). Return measures how much our

investment has grown or decreased over some period of time. That is, we only need to use the price of our current portfolio P_t to subtract the previous price P_{t-1} and obtain the difference $P_t - P_{t-1}$. To avoid the impact of the unit, we often convert this difference in a relative term, giving $\frac{P_t - P_{t-1}}{P_{t-1}}$, which represents how much percentage change the portfolio experiences from period $t - 1$ to t . This is to facilitate the comparison across different assets and time periods.

Returns can come from a variety of sources, depending on the specific type of instrument invested. In practice, all components are included in the calculation of the return. For example, if our portfolio is a stock that pays dividends D_t in the time period t , then our previous return calculation will be $\frac{P_t - P_{t-1} + D_t}{P_{t-1}}$. Common sources of return include

- **Capital Appreciation:** This is the increase in the value of the asset itself, such as a stock price rising over time.
- **Dividends:** Dividends are cash payments made to shareholders, typically distributed from a company's profits.
- **Interest Payments:** Bonds and other fixed-income securities generate returns through regular interest payments (coupon payment).

These are three common streams of income: capital gains, dividends, and interest. However, we can also define custom derivative products with desired payoff structures to satisfy investor needs with either return maximization or risk minimization.

In addition, it is a common practice to annualize returns, which means converting the original calculated returns from the previous period (which could be daily, monthly, or quarterly) into a standardized annual basis. This gives a clearer picture of how an investment performs on a yearly basis and makes it easier to compare different investment opportunities, regardless of their time frames. However, converting to an annual basis is not a straightforward process, such as multiplying a quantity by 12 to convert the figure from monthly to yearly. In finance, money invested in the financial market (such as saving in a bank) gives interest, and different ways to calculate the interest could make a notable difference, due to the process of compounding.

For example, if a stock return is 2% in a single month, the annualized return would not simply be 12 times that monthly return. Due to the effect of compounding, the annualized return would be slightly higher, capturing the cumulative impact of returns reinvested over time. This practice of annualization provides a consistent and accurate measure of the performance of an investment over different periods.

Note that there are also different ways to report this return. For example, we can use the absolute return to measure the total amount of money gained or lost or the percentage return to represent the gain or loss as a proportion of the initial investment. We can use cumulative return to generate the wealth curve and indicate

the total wealth across each period of the investment horizon or use a single-period return to report the individual percentage return in each period. Additionally, we can use risk-adjusted return to consider the risk taken to achieve the return or use the ratio of the two (return divided by risk) to give the return per unit risk (similar to the idea of Sharpe ratio introduced later in this chapter).

2.2 Understanding Risk

Now, let us flip to the other side of the coin: risk. While return represents the financial reward of an investment, risk is the uncertainty surrounding how much of that return we will actually receive (if it is indeed a positive return). In finance, risk is most often associated with volatility, which measures the fluctuations in an investment's value over time. The greater the volatility, the more the value can swing from day to day, week to week, or year to year, indicating a higher level of uncertainty or risk. When drawn on the graph, a high-volatility asset would look more wiggly than a low-volatility one.

So far, we have used volatility as a measure of risk. However, such volatility means both upside and downside volatility. While everyone loves the upside jump (assuming that you have the asset), we are generally more interested in (and concerned with) the downside. In addition, we have learned that the risk itself can take many forms, such as market risk, credit risk, liquidity risk, and operational risk (see Chapter 1). These different types of risk do not exist in isolation. Often, a single investment may be subject to multiple risks simultaneously, each influencing the overall risk profile in complex ways (which are determined by their correlation). Although volatility provides a lens for understanding some of the risks, it is only part of the bigger picture in assessing and mitigating the uncertainty that comes with investing.

Let us look at one simulated example to compare the wealth curve of stable vs. volatile stocks and analyze how varying volatility levels can affect an investor's wealth trajectory over the course of a year. We first assign different drift (average return) and volatility parameters to two stocks—one labeled “stable” and the other “volatile”—to see how daily random fluctuations accumulate into different wealth paths. The stable stock shows less severe price changes but similarly limited upside potential, whereas the volatile stock experiences more dramatic gains or losses day by day. As seen in Figure 2-1, although both stocks begin with the same initial investment and share an identical average daily return, increased volatility in the second stock can lead to substantial divergence in their final outcomes, making its wealth curve more extreme compared to the first stock.



Figure 2-1 Comparing the wealth curve of stable and volatile stocks over one year

See Listing 2-1 used to generate Figure 2-1.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Number of days in a simulated year
5 num_days = 365
6
7 # Simulation parameters
8 initial_wealth = 100.0 # starting wealth for both stocks
9
10 # Stable Stock parameters
11 stable_daily_drift = 0.0005 # e.g., 0.05% mean daily return
12 stable_daily_vol = 0.005 # e.g., 0.5% standard deviation
13
14 # Volatile Stock parameters
15 volatile_daily_drift = 0.0005 # same drift for easier comparison
16 volatile_daily_vol = 0.015 # e.g., 1.5% standard deviation
17
18 # Arrays to hold wealth values over the simulation
19 stable_wealth = np.zeros(num_days)
20 volatile_wealth = np.zeros(num_days)
21
22 # Initialize
23 stable_wealth[0] = initial_wealth
24 volatile_wealth[0] = initial_wealth
25
26 # Seed for reproducibility; remove or change if desired
27 np.random.seed(42)
28
29 # Simulation loop for each day
```

```

40 for i in range(1, num_days):
41     # Random daily return for the stable stock
42     stable_shock = np.random.normal(stable_daily_drift,
43                                     stable_daily_vol)
44     stable_wealth[i] = stable_wealth[i-1] * (1 + stable_shock)
45
46     # Random daily return for the volatile stock
47     volatile_shock = np.random.normal(volatile_daily_drift,
48                                       volatile_daily_vol)
49     volatile_wealth[i] = volatile_wealth[i-1] * (1 +
50                                                 volatile_shock)
51
52 # Create a day index for plotting
53 days = np.arange(num_days)
54
55 # Plot both wealth curves
56 plt.figure(figsize=(10, 6))
57 plt.plot(days, stable_wealth, label='Stable Stock')
58 plt.plot(days, volatile_wealth, label='Volatile Stock')
59 plt.title('Simulated Wealth Curves Over One Year')
60 plt.xlabel('Day')
61 plt.ylabel('Wealth')
62 plt.legend()
63 plt.grid(True)
64 plt.tight_layout()
65 plt.show()

```

Listing 2-1 S&P 500 price curve

Note that the parameters used in this simulation constitute two Gaussian (also called normal) distributions. Each normal distribution is specified with two parameters, mean (also called drift) and variance, which roughly correspond to the asset return and volatility. In this simulation, we used the same mean but different variances for the two distributions. Figure 2-2 shows the Probability Density Function (PDF) of stable and volatile stocks in terms of their daily returns. A PDF specifies the different outcomes and the corresponding probabilities. Clearly, the volatile stock shows wider coverage and, therefore, a higher degree of uncertainty.

Listing 2-2 is used to generate the graph. In general, we would observe that a higher standard deviation σ leads to a wider range of possible outcomes in the end, even if the mean parameter μ remains the same. In addition, the PDF with higher volatility is flatter and wider, indicating a higher uncertainty in day-to-day returns.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.stats import norm
4
5 # Parameters for Stable Stock
6 stable_mean = 0.0005 # 0.05% mean daily return
7 stable_vol = 0.005 # 0.5% std. dev.
8
9 # Parameters for Volatile Stock
10 volatile_mean = 0.0005 # same mean for direct comparison

```

```

11 volatile_vol = 0.015      # 1.5% std. dev.
12
13 # Range of possible daily returns for plotting PDF
14 x = np.linspace(-0.05, 0.05, 1000)
15
16 # Calculate the PDF values
17 stable_pdf = norm.pdf(x, loc=stable_mean, scale=stable_vol)
18 volatile_pdf = norm.pdf(x, loc=volatile_mean, scale=volatile_vol)
19
20 # Plot
21 plt.figure(figsize=(8, 5))
22 plt.plot(x, stable_pdf, label='Stable Stock PDF')
23 plt.plot(x, volatile_pdf, label='Volatile Stock PDF')
24 plt.title('PDF of Stable vs. Volatile Stock Daily Returns')
25 plt.xlabel('Daily Return')
26 plt.ylabel('Probability Density')
27 plt.legend()
28 plt.grid(True)
29 plt.tight_layout()
30 plt.show()

```

Listing 2-2 Comparing PDFs of stable and volatile stocks

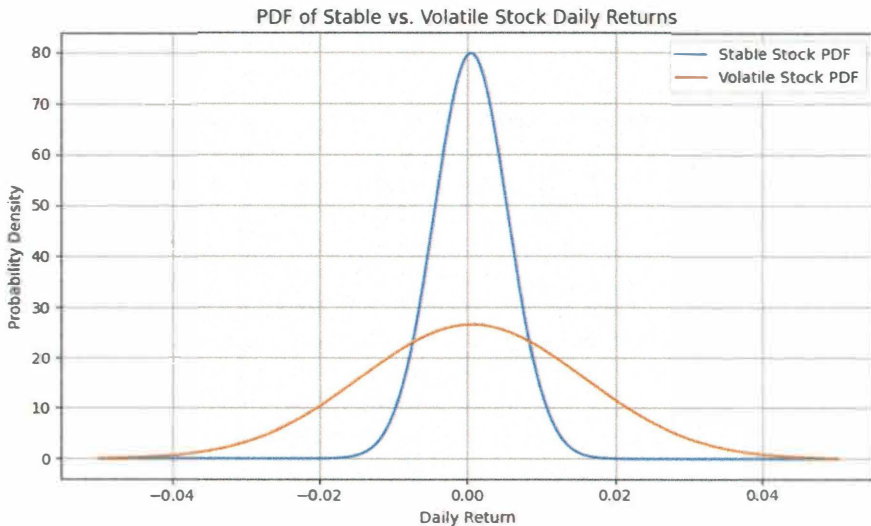


Figure 2-2 Probability density function of stable vs. volatile stock daily returns

2.3 Risk-Return Trade-Off

We have learned that higher potential returns come with higher risks. This relationship is known as the risk-return trade-off. This means that it is very difficult to find a financial product with a higher return and a lower risk in the market. Even if there is, it could be a temporary promotion. This trade-off is often visualized in the mean-variance space of a portfolio, where the mean corresponds to the expected return and the variance (or, equivalently, standard deviation) represents the risk. Picture a two-dimensional plot, with risk on the horizontal axis and return on the vertical axis. The financial instruments we have discussed so far, with different levels of risk, will fill in the graph from the lower left to the upper right, signifying the generally monotone relationship between risk and return.

In this mean-variance graph, the upper-right quadrant contains high-risk and high-return assets, such as stocks, commodities, and derivatives. These investments can offer attractive returns, but come with significant volatility and the potential for loss. The lower-left quadrant contains stable and low-return assets, such as bonds, savings accounts, and other fixed-income securities. These assets tend to provide more stability and lower returns, appealing to risk-averse investors. Also, see Figure 2-3 for an illustration.

This trade-off is the fundamental theme in modern portfolio management, pioneered by Markowitz (1952), who proposed that a diversified portfolio seeks to optimize the balance between risk and return. This balance mixes assets from

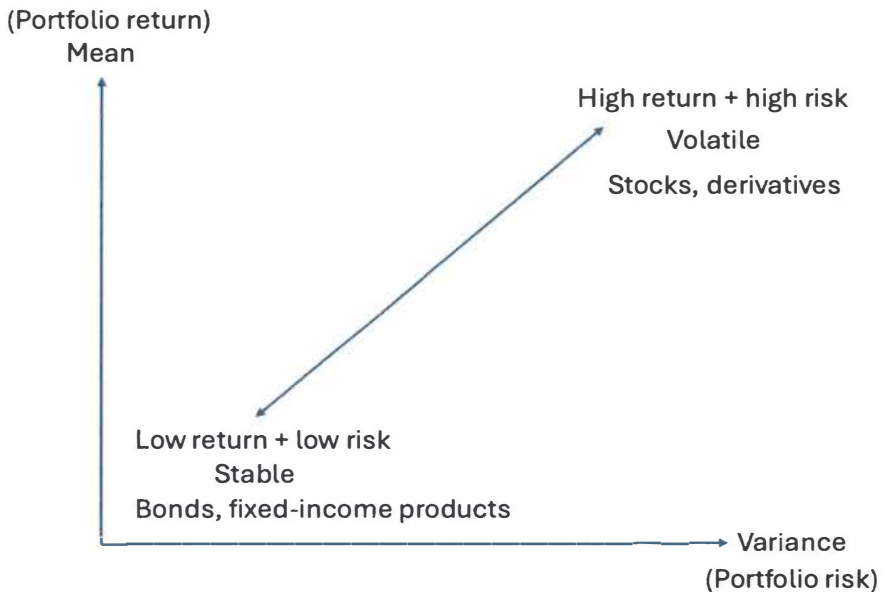


Figure 2-3 Mean-variance space of risk-return trade-off

different parts of the mean-variance space to match the investor's risk tolerance and return objectives. Let us take a closer and more quantitative look at how to measure risk and return.

2.4 Measuring Return

Return is a key measure of financial performance in financial markets. It quantifies how much the asset has appreciated or depreciated over a specific period of time. It also provides a standardized way to assess financial growth in various asset classes. In the following section, we break down the concept of return and explore different ways in which it can be measured and interpreted.

As briefly introduced earlier, return can take several forms, including absolute return and percentage return. We also have logarithmic return, total return, price return, annualized return, single-period return, and multi-period return.

2.4.1 Absolute Return

As the most straightforward form of return, absolute return is the total profit or loss of an investment. For example, if we invested \$1,000 and ended up with \$1,200, the absolute return is simply \$200. Although the calculation is intuitive, the absolute return does not provide a good context about the base, in that a \$200 gain will not be so attractive for a total investment of \$10,000 (which corresponds to a 2% return).

2.4.2 Percentage Return

To create a level playing field for comparing investments, we can convert absolute returns into a percentage term of the initial investment via the following formula:

$$\text{Percentage Return} = \frac{\text{Terminal Value} - \text{Initial Value}}{\text{Initial Value}}$$

Denoting the percentage return of period $t - 1$ to t as R_t and the asset prices in the period t as S_t , we have the following:

$$R_t = \frac{S_t - S_{t-1}}{S_{t-1}} = \frac{S_t}{S_{t-1}} - 1$$

For example, if the investment grows from \$1,000 to \$1,200, the percentage return is

$$\frac{1200 - 1000}{1000} = 20\%$$

This allows us to compare returns across investments of different sizes and time frames. We may also refer to this percentage return as the simple return.

2.4.3 Logarithmic Return

Logarithmic return, or log return (also known as continuously compounded return), offers an alternative to simple percentage returns, especially when analyzing longer-term investments or volatile assets:

$$\text{Log Return} = \ln \left(\frac{\text{Terminal Value}}{\text{Initial Value}} \right)$$

Log returns are additive over time, which simplifies calculations for returns that involve multiple periods. To see this, let us define the single-period log return at time period t as

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right)$$

and for multiple periods, the cumulative log return is

$$r_{\text{total}} = \ln \left(\frac{S_n}{S_0} \right)$$

Using the property of logarithms, we have the following:

$$r_{\text{total}} = \ln \left(\frac{S_n}{S_{n-1}} \cdot \frac{S_{n-1}}{S_{n-2}} \cdots \frac{S_2}{S_1} \cdot \frac{S_1}{S_0} \right)$$

which simplifies to

$$r_{\text{total}} = r_1 + r_2 + \cdots + r_n$$

Hence, log returns are additive over time, while simple returns require compounding. We can convert the cumulative log return back to a simple return via the following:

$$R_{\text{total}} = e^{r_{\text{total}}} - 1$$

2.4.4 Total Return vs. Price Return

The price return measures only the change in the price of an asset, while the total return incorporates both price changes and any cash flows, like dividends or interest,

received during the holding period. By comparison of the two, the total return is a more comprehensive measure of performance, as it includes all components of investment income.

For a stock that pays dividends, the total return would be

$$\text{Total Return} = \frac{\text{Terminal Value} + \text{Dividends Received} - \text{Initial Value}}{\text{Initial Value}}$$

2.4.5 Annualized Returns

One practical challenge when comparing different investments is the different holding periods. For example, comparing the return on a stock held for three months to the return on a bond held for a full year would be difficult without holding them on the same investment horizon. To make returns comparable, we often convert them to an annualized return, which shows the equivalent yearly return as if the investment had grown at a steady rate for a full year.

To annualize a single-period return, we typically employ compounding, the principle that each period's earnings are reinvested to generate additional returns in subsequent periods. At the start, the initial principal is multiplied by $(1 + R)$ to include both the original amount and the gain for that period. If multiple periods follow, this multiplication is repeated each time, so after n periods, the total growth factor becomes $(1 + R)^n$, which gives an exponential growth.

We can use the following formula to calculate the annualized return for an investment period that is within one year:

$$\text{Annualized Return} = (1 + \text{Single Period Return})^{\text{Number of Periods per Year} - 1}$$

For example, if the monthly return is 1%, the annualized return would be $(1 + 0.01)^{12} - 1 = 12.68\%$. The difference from simply multiplying 1% by 12 is due to the effect of compounding.

When an investment is held over multiple years, we use the geometric mean to find the average annual return and compound it to obtain the final value:

$$\text{Annualized Return} = \left(\frac{\text{Terminal Value}}{\text{Initial Value}} \right)^{\frac{1}{\text{Number of Years}}} - 1$$

For example, if an investment grows from \$1,000 to \$2,000 over five years, the annualized return is

$$\left(\frac{2000}{1000} \right)^{\frac{1}{5}} - 1 = 14.87\%$$

Now, if the return is calculated using a frequency that is different than yearly (such as monthly return) and lasts more than a year, we can still convert it into years by adjusting the exponent to account for the total number of years covered by the investment:

$$\text{Annualized Return} = \left(\frac{\text{Terminal Value}}{\text{Initial Value}} \right)^{\frac{12}{\text{Number of Months}}} - 1$$

For example, if an investment grew from \$1,000 to \$1,500 over 18 months, we can calculate the annualized return as $\left(\frac{1500}{1000} \right)^{\frac{12}{18}} - 1 = 1.5^{\frac{2}{3}} - 1 \approx 0.287$ or 28.7%.

2.4.6 Single-Period vs. Multi-Period Returns

When an investment covers more than one period, we need to account for how returns compound as time passes. Compounding does not mean we add up single-period returns; instead, since interest earned in the interim periods can also be reinvested, the net return from compounding will be slightly higher than the additive approach. That is, when returns compound, the return for each period is based on the value generated in the previous periods. This compounding determines the real growth (or decline) of an investment over multiple periods.

To calculate the multi-period return, we resort to the $1 + R$ formatted return, which is in contrast to the simple return R . Thus, to calculate the terminal return R_{0T} (in a simple return format), we can use

$$R_{0T} = (1 + R_{01})(1 + R_{12}) \cdots (1 + R_{T-1,T}) - 1$$

In this formula, we multiply all the single-period $1 + R$ formatted returns for each period, which essentially captures how returns build on each other. We then subtract 1 at the end to bring the result back into a simple return format.

Let's look at an example in which an investment experiences the following returns over three months: +5%, -2%, and +3%. To calculate the multi-period return, we multiply the individual period returns $R_{03} = (1 + 0.05) \times (1 - 0.02) \times (1 + 0.03) - 1 = 0.05954$ or 5.95%.

There is another quicker way to get the same result, which is using the terminal price and the initial price alone. Specifically, the cumulative return over multiple periods can also be calculated as

$$R_{0T} = \frac{S_T}{S_0} - 1$$

where S_0 is the initial price of the asset, and S_T is the price in the final period T . In the previous example, if we know the initial price and the final price, we can simply

calculate $R_{03} = \frac{S_T}{S_0} - 1$. Both approaches give the same result, but compounding makes the step-by-step growth more visible.

2.5 Measuring Risk

So far, we have been discussing the risk of an asset and how it correlates with return. The risk is due to randomness in the outcome (terminal price of the asset), and thus uncertainty in the final result gives the risk. Generally, a more uncertain outcome corresponds to a higher risk, which means that the terminal return could be very high or low. This level of uncertainty is also known as the volatility of the asset. The higher the volatility, the more ups and downs we will see in the asset price curve.

To understand and quantify volatility in a more concrete way, we often use two core measures: variance and standard deviation. These are fundamental to how we measure risk in finance. Assume that we have the simple return R_i on an asset over multiple periods, where $i = 1, 2, \dots, T$. The mean return can be calculated as

$$R_P = \frac{\sum_{i=1}^T R_i}{T}$$

This mean return R_P gives us a sense of where the asset's returns generally sit, on average, over time. In practice, the returns fluctuate, sometimes above and sometimes below. Thus, we need to capture such deviations from the mean and get a sense of its volatility as an indicator of risk.

The first step to quantifying risk is to measure how much each return deviates from the mean. For each single-period return R_i , the deviation from the mean is $R_i - R_P$. Now, deviations can be positive (if a return is above the mean) or negative (if it is below), and simply summing them up could potentially cancel out the ups and downs. To avoid such a cancellation, we often square each deviation to make all the deviations positive.

$$\sigma_P^2 = \frac{\sum_{i=1}^T (R_i - R_P)^2}{T}$$

This gives us the variance σ_P^2 , which says how spread out the returns are around the mean. The larger this number, the more the returns are deviating, and hence the riskier the asset. However, the unit of variance is the square of the returns, which makes it hard to interpret directly, so we could take an additional step to square-root the variance to obtain the standard deviation.

Specifically, to bring variance back to the same scale as the original returns, we can take the square root:

$$\sigma_P = \sqrt{\frac{\sum_{i=1}^T (R_i - R_P)^2}{T}}$$

This square root of the variance is known as the standard deviation, which is often just called volatility. Now we are back to the same units as the returns themselves, making it easier to understand and compare. Volatility essentially tells us how much we could expect the asset return to deviate from its average. The higher the volatility, the more erratic the asset's return, and thus the greater the risk involved. Therefore, volatility can be considered the standardized version of the spread of asset returns.

Note that squaring each deviation is more convenient mathematically than taking the absolute value of each single-period return. Squaring preserves the sign but makes all terms positive, just like taking the absolute value, but it also allows us to use powerful tools from calculus and algebra. For example, when minimizing or maximizing functions (such as variance), the squared deviations are differentiable, whereas absolute value functions are not smooth (they have a “kink” at zero), which makes mathematical analysis more difficult.

By squaring the deviations, we penalize larger deviations more heavily than smaller ones. This makes variance (and hence standard deviation) particularly sensitive to extreme values or outliers. In finance, large deviations from the mean are often of greater concern (i.e., higher risk), so squaring helps capture this idea that bigger deviations should have a disproportionately bigger impact on our risk measure.

In addition, many financial models assume that returns are normally distributed, which fits well with the properties of variance. The normal distribution has a natural relationship with variance and standard deviation, allowing us to easily calculate probabilities and make predictions about return behavior. The use of variance is aligned with these statistical properties, making it a natural fit for risk measurement in contexts where the normal distribution is a good approximation.

2.5.1 Annualization of Risk Measures

Variance and standard deviation give us a gauge of the level of risk for a particular asset. For example, if we look at two assets with the same average return but different volatilities, the one with the higher volatility is considered riskier because its returns deviate more widely from the mean. However, as with asset returns, when assessing the risk of an investment, it's important to compare assets on the same time scale. Risk metrics such as volatility also need to be annualized to make them comparable across different time periods.

The goal of annualizing volatility is thus to estimate what the risk of an asset would be on an annual basis, based on historically observed returns that last either shorter or longer than a full year. To do this, we can use the following formula:

$$\sigma_{P,T} = \sqrt{T} \cdot \sigma_P$$

where $\sigma_{P,T}$ is the annualized volatility, σ_P is the single-period volatility (daily, monthly, or quarterly), and T is the number of periods in a year (252 for trading days, 12 for months, or 4 for quarters). The key insight here is that volatility scales

with the square root of time. This means that we can estimate the volatility for a full year simply by multiplying the square root of the number of periods in a year.

Notice that as the time period T increases, the annualized volatility $\sigma_{P,T}$ grows at a decreasing rate due to the square root operation. This means that doubling the number of periods does not double the annualized volatility; it only increases it by a factor of $\sqrt{2}$.

Correspondingly, the annualized variance $\sigma_{P,T}^2$ grows linearly with time:

$$\sigma_{P,T}^2 = T \cdot \sigma_P^2$$

Although volatility increases with $\sqrt{T}$, variance increases directly with T . This linear relationship with variance makes it straightforward to aggregate risk over multiple periods.

As an example, suppose that the standard deviation (volatility) of a stock's daily returns is 0.1%. To annualize this daily volatility, we can use the following formula:

$$\sigma_{P,T} = 0.001 \cdot \sqrt{252} \approx 0.0159 \text{ or } 1.59\%$$

This means that the annualized volatility would be around 1.59%, assuming 252 trading days in a year.

2.5.2 Difference in Volatility Calculated Using Daily vs. Monthly Data

When calculating stock volatility, the frequency of data (daily vs. monthly) significantly impacts the outcome. For example, daily volatility captures short-term fluctuations in the stock price. This means that it reflects all minor market movements, such as those caused by day-to-day news, investor sentiment, and market microstructure noise. Daily volatility is more sensitive to sudden spikes or drops in the stock price, providing a more granular view of risk. On the other hand, monthly volatility smooths out the day-to-day fluctuations by averaging them over the month. It primarily captures broader trends in the stock price rather than the small movements seen in daily data, which are already smoothed out when aggregating at a monthly level. Monthly volatility is, therefore, less reactive to short-term noise and is influenced by longer-term forces such as economic cycles or company fundamentals. Therefore, in general, daily volatility is higher than monthly volatility because it encompasses the fine details of price movements, while monthly volatility averages out these short-term variations.

Also, when it comes to compounding risk, different data granularity will lead to different results. When calculating volatility using daily returns, we get a compounded picture of risk because we are summing up many individual daily movements (up to 252 trading days in a year). Multiplying by $\sqrt{252}$ further amplifies the impact of these frequent price changes at a yearly level. However,

when using monthly returns, we have only 12 data points in a year. These monthly returns already aggregate the effects of the daily movements within each month. Thus, the annualization multiplier is $\sqrt{12}$, which is much smaller, reflecting less compounding of risk over the same period. Therefore, the volatility derived from daily data will tend to be higher because it captures the compounded effect of frequent price changes, whereas monthly data smooth out this noise.

Daily volatility also offers a more immediate reflection of market conditions. It is sensitive to short-term market shocks like corporate earnings announcements, geopolitical events, or unexpected macroeconomic data releases. This makes daily volatility a good measure for traders who are concerned with short-term risk and potential rapid market movement. It offers a more real-time gauge of risk. On the other hand, monthly volatility provides a more stable view of the risk of a stock, as it does not react as quickly to short-term market shocks. It is more appropriate for long-term investors who are less concerned with daily noise and more focused on general market trends and the company's fundamentals. Choosing which type of volatility to use thus depends on the investment horizon and frequency.

From a statistical estimation perspective, the more data points we have, the more reliable the statistical measures (such as mean and standard deviation) are likely to be. With daily data, we typically have 252 observations per year, compared to just 12 with monthly data. This larger sample size reduces the estimation errors, which leads to a more precise calculation of volatility. More data points mean that estimates are less sensitive to outliers or random noise. With monthly data, a single large return can significantly skew the volatility measure, while daily data spread such impacts across multiple observations.

For example, higher-frequency data often improve the modeling and forecasting of volatility. Many financial models, like GARCH (Generalized Autoregressive Conditional Heteroskedasticity), benefit from high-frequency data to predict volatility more accurately. With more data points, the model can better capture patterns, correlations, and the persistence of volatility over time, leading to more accurate risk forecasts.

It also allows for more effective detection of market anomalies. For example, in volatility clustering, periods of high volatility often follow each other. Daily data allows for the detection of such clusters quickly, while monthly data might obscure these patterns. In mean reversion or momentum, certain stocks exhibit short-term momentum or mean reversion, where prices either continue to move in the same direction or revert to a historical mean. These patterns are more easily detected and taken into account using daily data.

In summary, while monthly data provide a smoother, more aggregated view of stock volatility suitable for long-term trends, daily data offer more granularity, accuracy, and timeliness. This granularity allows better risk management and improved forecasting and is particularly valuable for those who need a real-time assessment of market risks or engage in short-term trading strategies. Therefore, higher-frequency data are generally preferred for calculating volatility and managing financial risks effectively.

2.6 Measuring Risk-Adjusted Return

Now that we have covered both asset return and risk, how can we combine them to obtain a unified view? This combination means merging the risk and return into a single number, which will make it easier to compare different assets on a risk-adjusted basis. In the following, we introduce three of the most widely used metrics: the Sharpe ratio, the Sortino ratio, and the Treynor ratio.

2.6.1 Sharpe Ratio

The Sharpe ratio is one of the most popular risk-adjusted performance metrics and is proposed by Sharpe (1966). It measures how much excess return an asset (or, more generally, a portfolio, if it is a single asset) generates per unit of risk taken. It is defined as

$$\text{Sharpe Ratio} = \frac{R_P - R_f}{\sigma_P}$$

where R_P denotes the average return of the portfolio, R_f is the risk-free rate (such as the return on a Treasury bond), and σ_P is the volatility of the portfolio returns.

The Sharpe ratio essentially tells us how much return is being earned per unit of risk. A higher Sharpe ratio indicates a better risk-adjusted return, suggesting that the investment is generating more return for each unit of risk taken. A Sharpe ratio above one is often considered acceptable or decent, while a Sharpe ratio above two is often deemed excellent. A negative return could also lead to a negative Sharpe ratio.

Let us look at an example of calculating the Sharpe ratio. Consider a portfolio with an annualized return of 5% and an annualized volatility of 20%, and another portfolio with an annualized return of 10% and an annualized volatility of 50%. Given a risk-free rate of 3%, the Sharpe ratios would be calculated as $\frac{0.05-0.03}{0.20} = 0.10$ and $\frac{0.10-0.03}{0.50} = 0.14$, respectively. Here, the second portfolio has a higher Sharpe ratio, making it more attractive on a risk-adjusted basis, even though it has a higher absolute volatility.

However, there is no differentiation between the upside risk and the downside risk in the overall volatility term in the Sharpe ratio. Since most of us are risk averse (taking \$100 dollars away from us inflicts more aversion than the joy brought by giving us \$100), most would prefer to minimize the downside risk more. This leads to the following risk-adjusted return that focuses only on the risk below the target return.

2.6.2 Sortino Ratio

The Sortino ratio is a refinement of the Sharpe ratio that focuses only on downside risk (negative deviations from a target return), rather than overall volatility. This treatment allows us to focus only on the downside risk rather than the upside volatility. The Sortino ratio is defined as

$$\text{Sortino Ratio} = \frac{R_P - R_f}{\sigma_D}$$

where σ_D is the downside deviation (standard deviation of negative returns).

By focusing on returns that fall below a specified threshold (often the risk-free rate), the Sortino ratio does not penalize a portfolio for its upside volatility (positive returns). Thus, a higher Sortino ratio indicates that an investment generates higher returns per unit of downside risk. This means that either the excess return is high or the downside volatility is low. Therefore, the Sortino ratio is especially valuable for investors who are primarily concerned with avoiding losses, such as risk-averse individuals or those near retirement.

2.6.3 Treynor Ratio

The Treynor ratio is another measure of risk-adjusted return, similar to the Sharpe ratio but focusing on the systematic risk (market risk) instead of the total risk. As mentioned earlier, systematic risk is a major type of financial risk and impacts all assets in the market. Based on the Capital Asset Pricing Model (CAPM), it is measured by the beta of a portfolio β_M , which represents its sensitivity to the overall market. It is defined as

$$\text{Treynor Ratio} = \frac{R_P - R_f}{\beta_M}$$

Based on CAPM, β_M is defined as

$$\beta_M = \frac{\text{Cov}(R_P, R_M)}{\text{Var}(R_M)}$$

where the covariance term $\text{Cov}(R_P, R_M)$ measures the strength of comovement between the returns of the portfolio and the overall market. A high positive covariance indicates that portfolio returns generally move in the same direction as the market. $\text{Var}(R_M)$ measures the variance of the market returns.

The Treynor ratio essentially measures how much excess return is generated per unit of market risk. Unlike the Sharpe ratio, which uses total volatility σ_P in the denominator, the Treynor ratio considers only the portion of risk that cannot be diversified (since the market factor is a systematic risk). A higher Treynor ratio

Table 2-1 Comparison of Sharpe, Sortino, and Treynor ratios

Metric	Definition	When to Use
Sharpe ratio	$\frac{R_P - R_f}{\sigma_P}$	Use when you want to assess overall risk-adjusted performance, accounting for all sources of risk (both systematic and idiosyncratic).
Sortino ratio	$\frac{R_P - R_f}{\sigma_D}$	Use when you are more concerned with downside risk and want to avoid penalizing upside volatility. Ideal for conservative or risk-averse investors.
Treynor ratio	$\frac{R_P - R_f}{\beta_M}$	Use when focusing on the market (systematic) risk of a well-diversified portfolio, and you wish to assess performance relative to that systematic risk only.

implies a better risk-adjusted performance, relative to the market risk undertaken. Since it uses β_M , it is particularly relevant when evaluating portfolios against a benchmark index (e.g., the S&P 500). It is useful for comparing well-diversified portfolios.

Table 2-1 provides a summary of the different measures of risk-adjusted return and specific situations on when to use them.

The next section explores the use of these different performance measures in a portfolio optimization context.

2.6.4 Evaluating Performance Measures in Portfolio Optimization

In this section, we will assess these performance metrics in the context of different portfolio selection strategies, each set to optimize a particular performance metric. We will focus on three major US stocks as risky assets and apply various optimization strategies to maximize returns, minimize risk, and optimize risk-adjusted returns. This exercise allows us to better appreciate the roles of different performance metrics in optimization management.

2.6.4.1 Data Preparation

As shown in Listing 2-3, we start by importing the necessary libraries and downloading the necessary financial data for three major US stocks: Apple Inc. (AAPL), Microsoft Corporation (MSFT), and Amazon.com, Inc. (AMZN). We will also use the S&P 500 Index (SPY) as our market benchmark to calculate the Treynor ratio. For all assets, we retrieve five years of daily adjusted closing prices between 2019 and 2024. We also calculate daily and annualized returns and risk measures for the stocks. Note that the first row contains NaN values because there is no previous day to compare. Thus, we remove the first row using the `dropna()` function.

```
1 import numpy as np
2 import pandas as pd
3 import yfinance as yf
4 import matplotlib.pyplot as plt
5 from scipy.optimize import minimize
6
```

```

7 # Define the tickers
8 tickers = ['AAPL', 'MSFT', 'AMZN', 'SPY']
9
10 # Fetch data from Yahoo Finance
11 data = yf.download(tickers, start='2019-01-01', end='2024-01-01')
12     ['Adj Close']
13
14 # Calculate daily returns
15 returns = data.pct_change().dropna()

```

Listing 2-3 Downloading asset data

As shown in Listing 2-4, to make our metrics comparable on a yearly basis, we will annualize the daily returns and risk measures. For simplicity, we will multiply the average daily return by 252 to convert to an annualized scale (and similarly for annualized volatility), assuming independence and stationarity of returns. Note that such arithmetic scaling does not take into account the effect of compounding, and the geometric mean is often used when dealing with real-world financial returns.

```

1 # Define the number of trading days in a year
2 trading_days = 252
3
4 # Calculate annualized mean returns
5 annual_returns = returns.mean() * trading_days
6
7 # Calculate annualized volatility (standard deviation)
8 annual_volatility = returns.std() * np.sqrt(trading_days)
9
10 # Display the annualized returns and volatility
11 print("Annualized Returns:\n", annual_returns)
12 print("\nAnnualized Volatility:\n", annual_volatility)
13
14 # Output
15 Annualized Returns:
16 Ticker
17 AAPL      0.378031
18 AMZN      0.198360
19 MSFT      0.320295
20 SPY       0.167224
21 dtype: float64
22
23 Annualized Volatility:
24 Ticker
25 AAPL      0.322345
26 AMZN      0.352214
27 MSFT      0.304909
28 SPY       0.209957
29 dtype: float64

```

Listing 2-4 Downloading asset data

2.6.4.2 Defining the Portfolios

We'll evaluate five different portfolios based on the following criteria:

- Maximizing Portfolio Return
- Minimizing Portfolio Risk
- Maximizing Sharpe Ratio
- Maximizing Sortino Ratio
- Maximizing Treynor Ratio

We assume a risk-free rate of 3% when calculating the Sharpe ratio and the Sortino ratio.

In addition, to calculate the Treynor ratio, we also need the beta of each stock relative to the market (S&P 500 in this case). As discussed, beta measures the volatility of a stock relative to the market portfolio. See Listing 2-5 for detailed processing.

```

1 risk_free_rate = 0.03
2
3 # Calculate covariance matrix
4 cov_matrix = returns.cov() * trading_days
5
6 # Calculate market variance
7 market_variance = returns['SPY'].var() * trading_days
8
9 # Calculate beta for each stock
10 betas = cov_matrix.loc[['AAPL', 'AMZN', 'MSFT'], 'SPY'] /
    market_variance
11
12 print("Betas:\n", betas)
13
14 # Output
15 Betas:
16 Ticker
17 AAPL      1.235198
18 AMZN      1.087888
19 MSFT      1.197498
20 Name: SPY, dtype: float64

```

Listing 2-5 Calculating asset betas

Note that in the code above, we have used the `.var()` function to calculate the sample variance, which assumes a default setting of `ddof=1` in the input arguments. Given limited data, it is more often to calculate sample variance as a better (consistent) approximation to the theoretical population variance, if we were to be able to access all the data available.

As shown in Listing 2-6, for all portfolios, we also enforce the following common constraints:

- **Fully Invested:** The sum of all portfolio weights is equal to 1. This is also referred to as the budget constraint, in that all available capital should be allocated among the assets in the portfolio and should not be kept in the pocket.
- **No Short Selling:** All weights are between 0 and 1. This is often an optional constraint to reflect the fact that, in some markets such as China, shorting is not allowed.

```

1 # Number of assets
2 num_assets = len(tickers) - 1
3
4 # Initial guess for weights
5 init_guess = num_assets * [1. / num_assets]
6
7 # Bounds for weights: no short selling
8 bounds = tuple((0, 1) for _ in range(num_assets))
9
10 # Constraint: sum of weights is 1
11 constraints = ({'type': 'eq', 'fun': lambda x: np.sum(x) - 1})

```

Listing 2-6 Defining portfolio constraints

We will also define different objective functions for each portfolio optimization criterion. For the first portfolio that aims at maximizing return, we can convert it to a minimization problem (a common practice in optimization) by minimizing the negative of the portfolio's expected return.

```

1 def portfolio_return(weights, mean_returns):
2     return np.dot(weights, mean_returns)
3
4 def neg_portfolio_return(weights, mean_returns):
5     return -portfolio_return(weights, mean_returns)

```

Listing 2-7 Defining optimization objective for portfolio 1

For the second portfolio, our aim is to minimize the portfolio's variance.

```

1 def portfolio_variance(weights, cov_matrix):
2     return np.dot(weights.T, np.dot(cov_matrix, weights))

```

Listing 2-8 Defining optimization objective for portfolio 2

For the third portfolio that aims at maximizing the portfolio Sharpe ratio, we define a function that calculates the portfolio excess return over the risk-free interest rate (the numerator of the Sharpe ratio) and the portfolio volatility (the denominator of Sharpe ratio).

```

1 def sharpe_ratio(weights, mean_returns, cov_matrix,
2     risk_free_rate):
3     p_return = portfolio_return(weights, mean_returns)
4     p_variance = portfolio_variance(weights, cov_matrix)
5     p_volatility = np.sqrt(p_variance)
6     return (p_return - risk_free_rate) / p_volatility
7
8 def neg_sharpe_ratio(weights, mean_returns, cov_matrix,
9     risk_free_rate):
10    return -sharpe_ratio(weights, mean_returns, cov_matrix,
11        risk_free_rate)

```

Listing 2-9 Defining optimization objective for portfolio 3

The fourth portfolio aims to maximize the Sortino ratio, which focuses on downside risk instead of total volatility. Note that the target return can be set to the risk-free rate or any desired threshold.

```

1 def sortino_ratio(weights, returns, risk_free_rate, target_return
  =0):
2     portfolio_returns = returns[['AAPL', 'MSFT', 'AMZN']].dot(
  weights)
3     downside = portfolio_returns[portfolio_returns <
  target_return]
4     expected_return = portfolio_returns.mean() * trading_days
5     downside_deviation = np.sqrt((np.square(downside -
  target_return)).mean()) * np.sqrt(trading_days)
6     return (expected_return - risk_free_rate) /
  downside_deviation
7
8 def neg_sortino_ratio(weights, returns, risk_free_rate,
  target_return=0):
9     return -sortino_ratio(weights, returns, risk_free_rate,
  target_return)

```

Listing 2-10 Defining optimization objective for portfolio 4

Finally, the last portfolio aims at maximizing the Treynor ratio, which measures excess return per unit of systematic risk (the beta).

```

1 def treynor_ratio(weights, mean_returns, betas, risk_free_rate):
2     p_return = portfolio_return(weights, mean_returns)
3     p_beta = np.dot(weights, betas)
4     return (p_return - risk_free_rate) / p_beta
5
6 def neg_treynor_ratio(weights, mean_returns, betas,
  risk_free_rate):
7     return -treynor_ratio(weights, mean_returns, betas,
  risk_free_rate)

```

Listing 2-11 Defining optimization objective for portfolio 5

2.6.4.3 Optimizing the Portfolios

Now that we've defined our objective functions and constraints, let us perform the optimizations for each portfolio, starting with the first portfolio that maximizes return.

```

1 # Optimize Portfolio 1: Maximizing Return
2 def optimize_max_return(mean_returns, constraints, bounds,
  init_guess):
3     result = minimize(neg_portfolio_return, init_guess, args=(
  mean_returns, ),
4                       method='SLSQP', bounds=bounds, constraints=
  constraints)
5     return result
6
7 result_max_return = optimize_max_return(annual_returns[['AAPL',
  'AMZN', 'MSFT']], constraints, bounds, init_guess)
8
9 print("Portfolio 1: Maximizing Return")
10 print("Weights:", result_max_return.x)

```

```

11 print("Expected Annual Return:", portfolio_return(
    result_max_return.x, annual_returns[['AAPL', 'AMZN', 'MSFT']
    ]))
12 print("Annual Volatility:", np.sqrt(portfolio_variance(
    result_max_return.x, cov_matrix.loc[['AAPL', 'AMZN', 'MSFT'],
    ['AAPL', 'AMZN', 'MSFT']]))))
13 print("\n")
14
15 # Output
16 Portfolio 1: Maximizing Return
17 Weights: [1.0000000000e+00  6.31439345e-16  0.0000000000e+00]
18 Expected Annual Return: 0.37803071152250534
19 Annual Volatility: 0.3223453947637612

```

Listing 2-12 Optimizing portfolio 1

The optimal allocation result is not surprising. Since Apple has the highest annualized return, the maximum return portfolio will allocate all weight to this asset.

Now, let us turn to portfolio 2 that minimizes risk.

```

1 # Optimize Portfolio 2: Minimizing Risk
2 def optimize_min_risk(cov_matrix, constraints, bounds, init_guess
    ):
3     result = minimize(portfolio_variance, init_guess, args=(
        cov_matrix,),
4                       method='SLSQP', bounds=bounds, constraints=
        constraints)
5     return result
6
7 result_min_risk = optimize_min_risk(cov_matrix.loc[['AAPL', 'AMZN',
    'MSFT'], ['AAPL', 'AMZN', 'MSFT']], constraints, bounds,
    init_guess)
8
9 print("Portfolio 2: Minimizing Risk")
10 print("Weights:", result_min_risk.x)
11 print("Expected Annual Return:", portfolio_return(result_min_risk
    .x, annual_returns[['AAPL', 'AMZN', 'MSFT']]))
12 print("Annual Volatility:", np.sqrt(portfolio_variance(
    result_min_risk.x, cov_matrix.loc[['AAPL', 'AMZN', 'MSFT'], [
    'AAPL', 'AMZN', 'MSFT']]))))
13 print("\n")
14
15 # Output
16 Portfolio 2: Minimizing Risk
17 Weights: [0.3180772  0.2094853  0.4724375]
18 Expected Annual Return: 0.313115822392309
19 Annual Volatility: 0.288101952001344

```

Listing 2-13 Optimizing portfolio 2

Now, each individual weight allocation is closer to each other, with MSFT getting close to half of the total allocation, possibly due to the lowest volatility out of the three stocks.

As shown in Listing 2-14, now we turn to portfolio 3 that maximizes the Sharpe ratio.

```

1 # Optimize Portfolio 3: Maximizing Sharpe Ratio
2 def optimize_max_sharpe(mean_returns, cov_matrix, risk_free_rate,
3     constraints, bounds, init_guess):
4     result = minimize(neg_sharpe_ratio, init_guess, args=(
5         mean_returns, cov_matrix, risk_free_rate),
6         method='SLSQP', bounds=bounds, constraints=
7         constraints)
8     return result
9
10 result_max_sharpe = optimize_max_sharpe(annual_returns[['AAPL', '
11     AMZN', 'MSFT']],
12     cov_matrix.loc[['AAPL', '
13     AMZN', 'MSFT'], ['AAPL', 'AMZN', 'MSFT']],
14     risk_free_rate,
15     constraints, bounds, init_guess)
16
17 print("Portfolio 3: Maximizing Sharpe Ratio")
18 print("Weights:", result_max_sharpe.x)
19 print("Expected Annual Return:", portfolio_return(
20     result_max_sharpe.x, annual_returns[['AAPL', 'AMZN', 'MSFT'
21     ]]))
22 print("Annual Volatility:", np.sqrt(portfolio_variance(
23     result_max_sharpe.x, cov_matrix.loc[['AAPL', 'AMZN', 'MSFT'],
24     ['AAPL', 'AMZN', 'MSFT']]))))
25 print("Sharpe Ratio:", sharpe_ratio(result_max_sharpe.x,
26     annual_returns[['AAPL', 'AMZN', 'MSFT']],
27     cov_matrix.loc[['AAPL', 'AMZN',
28     'MSFT'], ['AAPL', 'AMZN', 'MSFT']], risk_free_rate))
29 print("\n")
30
31 # Output
32 Portfolio 3: Maximizing Sharpe Ratio
33 Weights: [7.17495581e-01 5.81132364e-17 2.82504419e-01]
34 Expected Annual Return: 0.3617201069615052
35 Annual Volatility: 0.3018898876667841
36 Sharpe Ratio: 1.0988115883088032

```

Listing 2-14 Optimizing portfolio 3

Indeed, the third portfolio gives the highest Sharpe ratio out of the three portfolios analyzed so far (whose Sharpe ratios are 1.079 for the first portfolio and 0.982 for the second portfolio).

Now we turn to portfolio 4 that maximizes the Sortino ratio. We also set the target return to zero in Listing 2-15.

```

1 # Optimize Portfolio 4: Maximizing Sortino Ratio
2 def optimize_max_sortino(weights, returns, risk_free_rate,
3     constraints, bounds, init_guess, target_return=0):
4     return minimize(neg_sortino_ratio, init_guess, args=(returns,
5         risk_free_rate, target_return),
6         method='SLSQP', bounds=bounds, constraints=
7         constraints)
8

```

```

6 result_max_sortino = optimize_max_sortino(init_guess, returns,
    risk_free_rate, constraints, bounds, init_guess)
7
8 # Extract weights from optimization result
9 weights_sortino = result_max_sortino.x
10
11 # Calculate expected return and downside deviation
12 expected_return_sortino = portfolio_return(weights_sortino,
    annual_returns[['AAPL', 'AMZN', 'MSFT']])
13 portfolio_returns_sortino = returns[['AAPL', 'MSFT', 'AMZN']].dot
    (weights_sortino)
14 downside_sortino = portfolio_returns_sortino[
    portfolio_returns_sortino < 0]
15 downside_deviation_sortino = np.sqrt((np.square(downside_sortino)
    ).mean()) * np.sqrt(trading_days)
16 sortino = (expected_return_sortino - risk_free_rate) /
    downside_deviation_sortino
17
18 print("Portfolio 4: Maximizing Sortino Ratio")
19 print("Weights:", weights_sortino)
20 print("Expected Annual Return:", expected_return_sortino)
21 print("Downside Deviation:", downside_deviation_sortino)
22 print("Sortino Ratio:", sortino)
23 print("\n")
24
25 # Output
26 Portfolio 4: Maximizing Sortino Ratio
27 Weights: [7.51230682e-01 2.48769318e-01 4.48859699e-18]
28 Expected Annual Return: 0.3333341715302924
29 Downside Deviation: 0.30636490028556385
30 Sortino Ratio: 0.9901074543707634

```

Listing 2-15 Optimizing portfolio 4

Finally, we optimize the fifth portfolio by maximizing the Treynor ratio as shown in Listing 2-16.

```

1 # Optimize Portfolio 5: Maximizing Treynor Ratio
2 def optimize_max_treynor(mean_returns, betas, risk_free_rate,
    constraints, bounds, init_guess):
3     result = minimize(neg_treynor_ratio, init_guess, args=(
        mean_returns, betas, risk_free_rate),
4                       method='SLSQP', bounds=bounds, constraints=
        constraints)
5     return result
6
7 result_max_treynor = optimize_max_treynor(annual_returns[['AAPL',
    'AMZN', 'MSFT']],
8                                           betas[['AAPL', 'AMZN', '
    MSFT']],
9                                           risk_free_rate,
10                                          constraints, bounds, init_guess)
11
12 print("Portfolio 5: Maximizing Treynor Ratio")
13 print("Weights:", result_max_treynor.x)

```

```

15 print("Expected Annual Return:", portfolio_return(
    result_max_treynor.x, annual_returns[['AAPL', 'AMZN', 'MSFT']
    ]))
16 print("Beta:", np.dot(result_max_treynor.x, betas[['AAPL', 'AMZN',
    'MSFT']]))
17 print("Treynor Ratio:", treynor_ratio(result_max_treynor.x,
    annual_returns[['AAPL', 'AMZN', 'MSFT']], betas[['AAPL', '
    AMZN', 'MSFT']], risk_free_rate))
18 print("\n")
19
20 # Output
21 Portfolio 5: Maximizing Treynor Ratio
22 Weights: [1.00000000e+00 0.00000000e+00 3.88578059e-16]
23 Expected Annual Return: 0.3780307115225055
24 Beta: 1.2351976772098658
25 Treynor Ratio: 0.2817611447494436

```

Listing 2-16 Optimizing portfolio 5

Let us summarize the optimized portfolios' weights, expected returns, volatility, and risk-adjusted ratios.

```

1 #####
2 # Additional code snippet for generating the final summary table
3 #####
4
5 # 1) Helper to compute the same Sortino ratio
6 # but directly from daily returns (annually scaled).
7 def compute_annual_sortino_ratio(weights, daily_returns, rf=0.03,
    target_return=0.0):
8     """
9     Computes the annualized Sortino ratio for a given portfolio (
10     weights) and daily returns.
11     The logic here mirrors the approach in 'sortino_ratio()', but
12     uses daily returns
13     and scales to annual figures, ensuring consistency with your
14     optimization code.
15     """
16
17     # 1) Construct daily portfolio returns from the chosen assets
18     pf_daily_returns = daily_returns[['AAPL', 'MSFT', 'AMZN']].
19     dot(weights)
20
21     # 2) Annualized portfolio return = mean(daily) * 252
22     pf_annual_return = pf_daily_returns.mean() * 252
23
24     # 3) Extract returns below the target threshold
25     negative_returns = pf_daily_returns[pf_daily_returns <
26     target_return]
27     if len(negative_returns) == 0:
28         # Means no negative returns at all -> "infinite" Sortino
29         return np.nan
30
31     # 4) Annualize the downside deviation

```

```

26 # (mean of squared deviations) -> multiply by #days ->
    sqrt
27 downside_var = np.mean((negative_returns - target_return)**2)
28 downside_dev_annual = np.sqrt(downside_var * 252)
29 if downside_dev_annual == 0:
30     return float('inf')
31
32 # 5) Compute the actual Sortino ratio
33 sortino_ratio_val = (pf_annual_return - rf) /
    downside_dev_annual
34 return sortino_ratio_val
35
36 # 2) Define the summary table as a dictionary with performance
    metrics for each portfolio
37 portfolios = {
38     'Max Return': {
39         'Weights': result_max_return.x,
40         'Return': portfolio_return(result_max_return.x,
    annual_returns[['AAPL', 'AMZN', 'MSFT']]),
41         'Volatility': np.sqrt(portfolio_variance(
    result_max_return.x, cov_matrix.loc[['AAPL', 'AMZN', 'MSFT'],
    ['AAPL', 'AMZN', 'MSFT']])),
42         'Beta': np.dot(result_max_return.x, betas[['AAPL', 'AMZN',
    'MSFT']].values.flatten()),
43         'Downside Deviation': downside_deviation_sortino, #
    Replace with actual downside deviation calc
44         'Sharpe Ratio': (portfolio_return(result_max_return.x,
    annual_returns[['AAPL', 'AMZN', 'MSFT']]) - risk_free_rate) /
    np.sqrt(portfolio_variance(result_max_return.x, cov_matrix.
    loc[['AAPL', 'AMZN', 'MSFT'], ['AAPL', 'AMZN', 'MSFT']])),
45         'Sortino Ratio': (portfolio_return(result_max_return.x,
    annual_returns[['AAPL', 'AMZN', 'MSFT']]) - risk_free_rate) /
    downside_deviation_sortino, # Replace with actual
    calculation
46         'Treyner Ratio': treynor_ratio(result_max_return.x,
    annual_returns[['AAPL', 'AMZN', 'MSFT']], betas[['AAPL', '
    AMZN', 'MSFT']].values.flatten(), risk_free_rate)
47     },
48     'Min Risk': {
49         'Weights': result_min_risk.x,
50         'Return': portfolio_return(result_min_risk.x,
    annual_returns[['AAPL', 'AMZN', 'MSFT']]),
51         'Volatility': np.sqrt(portfolio_variance(result_min_risk.
    x, cov_matrix.loc[['AAPL', 'AMZN', 'MSFT'], ['AAPL', 'AMZN',
    'MSFT']])),
52         'Beta': np.dot(result_min_risk.x, betas[['AAPL', 'AMZN',
    'MSFT']].values.flatten()),
53         'Downside Deviation': downside_deviation_sortino, #
    Replace with actual downside deviation calc
54         'Sharpe Ratio': (portfolio_return(result_min_risk.x,
    annual_returns[['AAPL', 'AMZN', 'MSFT']]) - risk_free_rate) /
    np.sqrt(portfolio_variance(result_min_risk.x, cov_matrix.loc
    [['AAPL', 'AMZN', 'MSFT'], ['AAPL', 'AMZN', 'MSFT']])),

```

```

55     'Sortino Ratio': (portfolio_return(result_min_risk.x,
annual_returns[['AAPL', 'AMZN', 'MSFT']]) - risk_free_rate) /
    downside_deviation_sortino, # Replace with actual
calculation
56     'Treynor Ratio': treynor_ratio(result_min_risk.x,
annual_returns[['AAPL', 'AMZN', 'MSFT']], betas[['AAPL', '
AMZN', 'MSFT']].values.flatten(), risk_free_rate)
57 },
58 'Max Sharpe': {
59     'Weights': result_max_sharpe.x,
60     'Return': portfolio_return(result_max_sharpe.x,
annual_returns[['AAPL', 'AMZN', 'MSFT']]),
61     'Volatility': np.sqrt(portfolio_variance(
result_max_sharpe.x, cov_matrix.loc[['AAPL', 'AMZN', 'MSFT'],
['AAPL', 'AMZN', 'MSFT']])),
62     'Beta': np.dot(result_max_sharpe.x, betas[['AAPL', 'AMZN'
, 'MSFT']].values.flatten()),
63     'Downside Deviation': downside_deviation_sortino, #
Replace with actual downside deviation calc
64     'Sharpe Ratio': sharpe_ratio(result_max_sharpe.x,
annual_returns[['AAPL', 'AMZN', 'MSFT']],
65         cov_matrix.loc[['AAPL', '
AMZN', 'MSFT'], ['AAPL', 'AMZN', 'MSFT']], risk_free_rate),
66     'Sortino Ratio': (portfolio_return(result_max_sharpe.x,
annual_returns[['AAPL', 'AMZN', 'MSFT']]) - risk_free_rate) /
    downside_deviation_sortino, # Replace with actual
calculation
67     'Treynor Ratio': treynor_ratio(result_max_sharpe.x,
annual_returns[['AAPL', 'AMZN', 'MSFT']], betas[['AAPL', '
AMZN', 'MSFT']].values.flatten(), risk_free_rate)
68 },
69 'Max Sortino': {
70     'Weights': weights_sortino,
71     'Return': expected_return_sortino,
72     'Downside Deviation': downside_deviation_sortino,
73     'Sortino Ratio': sortino
74 },
75 'Max Treynor': {
76     'Weights': result_max_treynor.x,
77     'Return': portfolio_return(result_max_treynor.x,
annual_returns[['AAPL', 'AMZN', 'MSFT']]),
78     'Beta': np.dot(result_max_treynor.x, betas[['AAPL', 'AMZN
', 'MSFT']].values.flatten()),
79     'Treynor Ratio': treynor_ratio(result_max_treynor.x,
annual_returns[['AAPL', 'AMZN', 'MSFT']], betas[['AAPL', '
AMZN', 'MSFT']].values.flatten(), risk_free_rate)
80 }
81 }
82
83 # 3) Populate missing metrics for each portfolio in the summary
84 for pf in portfolios:
85     w = portfolios[pf]['Weights']
86     p_ret = portfolios[pf]['Return']

```

```

87
88 # -- If needed, compute portfolio volatility
89 if 'Volatility' not in portfolios[pf]:
90     portfolios[pf]['Volatility'] = np.sqrt(
91         portfolio_variance(
92             w,
93             cov_matrix.loc[['AAPL', 'AMZN', 'MSFT'], ['AAPL', '
AMZN', 'MSFT']]
94         )
95     )
96
97 # -- Compute Sharpe Ratio (if you want it in the summary)
98 if portfolios[pf]['Volatility'] != 0:
99     portfolios[pf]['Sharpe Ratio'] = (
100         (p_ret - risk_free_rate) / portfolios[pf]['Volatility
']
101     )
102 else:
103     portfolios[pf]['Sharpe Ratio'] = np.nan
104
105 # -- Compute the annualized Sortino Ratio (using the new
helper)
106 sortino_val = compute_annual_sortino_ratio(
107     w,          # weights
108     returns,    # daily returns DataFrame
109     rf=risk_free_rate,
110     target_return=0.0 # or set to risk_free_rate if you
prefer
111 )
112 portfolios[pf]['Sortino Ratio'] = sortino_val
113
114 # -- Optionally store the raw Downside Deviation if you'd
like
115 # (just for the summary display)
116 pf_daily_returns = returns[['AAPL', 'MSFT', 'AMZN']].dot(w)
117 negative_rets = pf_daily_returns[pf_daily_returns < 0]
118 if len(negative_rets) == 0:
119     dd_annual = 0.0
120 else:
121     dd_annual = np.sqrt(np.mean((negative_rets - 0.0)**2) *
252)
122 portfolios[pf]['Downside Deviation'] = dd_annual
123
124 # -- Compute Treynor Ratio if Beta is known
125 if 'Beta' in portfolios[pf]:
126     b = portfolios[pf]['Beta']
127     if b != 0:
128         portfolios[pf]['Treynor Ratio'] = (p_ret -
risk_free_rate) / b
129     else:
130         portfolios[pf]['Treynor Ratio'] = np.nan
131 else:
132     portfolios[pf]['Treynor Ratio'] = np.nan

```

```
134 # 4) Convert the dictionary to a DataFrame and format
135 summary = pd.DataFrame(portfolios).T
136
137 # Round & format columns
138 summary['Weights'] = summary['Weights'].apply(lambda arr: [round(
139     x, 2) for x in arr])
140 summary['Return'] = summary['Return'].round(4)
141 summary['Volatility'] = pd.to_numeric(summary.get('Volatility',
142     pd.Series()), errors='coerce').round(4)
143 summary['Beta'] = pd.to_numeric(summary.get('Beta', pd.Series()),
144     errors='coerce').round(4)
145 summary['Downside Deviation'] = pd.to_numeric(summary.get('
146     Downside Deviation', pd.Series()), errors='coerce').round(4)
147 summary['Sharpe Ratio'] = pd.to_numeric(summary.get('Sharpe Ratio
148     ', pd.Series()), errors='coerce').round(4)
149 summary['Sortino Ratio'] = pd.to_numeric(summary.get('Sortino
150     Ratio', pd.Series()), errors='coerce').round(4)
151 summary['Treyrnor Ratio'] = pd.to_numeric(summary.get('Treyrnor
152     Ratio', pd.Series()), errors='coerce').round(4)
153
154 pd.set_option('display.max_columns', None)
155 pd.set_option('display.width', 1000)
156
157 # Finally, display the summary table
158 summary
```

Listing 2-17 Analyzing portfolio results

Running the code generates Figure 2-4. The result suggests that the Max Return portfolio and the Max Treynor ratio portfolio achieve the highest return, but also have high volatility and market sensitivity, leading to moderate Sharpe and Treynor ratios. The Min Risk portfolio is more diversified with the lowest volatility, but has lower returns and risk-adjusted performance, making it suitable for conservative investors. The Max Sharpe portfolio balances return and risk efficiently, achieving a high Sharpe ratio and suggesting an optimal risk-adjusted return. The Max Sortino portfolio, with its emphasis on downside risk, yields a maximum Sortino ratio and moderate return. Finally, the Max Treynor portfolio, identical to Max Return in weights, offers a strong Treynor ratio, excelling in return per unit of market risk.

	Weights	Return	Volatility	Beta	Downside Deviation	Sharpe Ratio	Sortino Ratio	Treynor Ratio
Max Return	[1.0, 0.0, 0.0]	0.378031	0.3223	1.2352	0.3210	1.0797	1.0842	0.2818
Min Risk	[0.32, 0.21, 0.47]	0.313116	0.2881	1.1865	0.3059	0.9827	0.8207	0.2386
Max Sharpe	[0.72, 0.0, 0.28]	0.36172	0.3019	1.2245	0.3057	1.0988	0.9725	0.2709
Max Sortino	[0.75, 0.25, 0.0]	0.333334	0.3043	NaN	0.3064	0.9969	1.0891	NaN
Max Treynor	[1.0, 0.0, 0.0]	0.378031	0.3223	1.2352	0.3213	1.0797	1.0832	0.2818

Figure 2-4 Summary results of each portfolio

Finally, let us visualize the efficient frontier in the mean-variance space and plot our optimized portfolios. As shown in Listing 2-18, we generate 5,000 random portfolios (random set of weights) to visualize the efficient frontier, using a scatter plot to represent the trade-off between risk (volatility) and return. We also highlight the five optimized portfolios with different markers and colors.

```

1 # Function to calculate portfolio performance
2 def portfolio_performance(weights, mean_returns, cov_matrix):
3     returns = np.dot(weights, mean_returns)
4     volatility = np.sqrt(np.dot(weights.T, np.dot(cov_matrix,
5         weights)))
6     return returns, volatility
7
8 # Generate random portfolios
9 num_portfolios = 5000
10 results = np.zeros((3, num_portfolios))
11 weight_records = []
12
13 for i in range(num_portfolios):
14     weights = np.random.dirichlet(np.ones(num_assets), size=1)[0]
15     weight_records.append(weights)
16     portfolio_ret, portfolio_vol = portfolio_performance(weights,
17         annual_returns[['AAPL', 'AMZN', 'MSFT']], cov_matrix.loc[['
18         AAPL', 'AMZN', 'MSFT'], ['AAPL', 'AMZN', 'MSFT']])
19     results[0,i] = portfolio_vol
20     results[1,i] = portfolio_ret
21     results[2,i] = (portfolio_ret - risk_free_rate) /
22         portfolio_vol
23
24 # Convert results to DataFrame
25 results_df = pd.DataFrame(results.T, columns=['Volatility', '
26     Return', 'Sharpe Ratio'])
27
28 # Plot the efficient frontier
29 plt.figure(figsize=(10, 7))
30 plt.scatter(results_df.Volatility, results_df.Return, c=
31     results_df['Sharpe Ratio'], cmap='viridis', alpha=0.5)
32 plt.colorbar(label='Sharpe Ratio')
33 plt.xlabel('Annualized Volatility')
34 plt.ylabel('Annualized Return')
35 plt.title('Efficient Frontier')
36
37 # Plot optimized portfolios
38 portfolios_to_plot = {
39     'Max Return': result_max_return.x,
40     'Min Risk': result_min_risk.x,
41     'Max Sharpe': result_max_sharpe.x,
42     'Max Sortino': weights_sortino,
43     'Max Treynor': result_max_treynor.x
44 }
45
46 colors = ['r', 'b', 'g', 'c', 'm']

```

```

41 for (name, weights), color in zip(portfolios_to_plot.items(),
    colors):
42     p_ret, p_vol = portfolio_performance(weights, annual_returns
        [['AAPL', 'AMZN', 'MSFT']], cov_matrix.loc[['AAPL', 'AMZN', '
        MSFT']], ['AAPL', 'AMZN', 'MSFT']))
44     plt.scatter(p_vol, p_ret, marker='*', color=color, s=200,
        label=name)
44
45 plt.legend()
46 plt.show()

```

Listing 2-18 Optimizing portfolio 4

Running the code generates Figure 2-5, where all portfolios with the Max Sortino portfolio are on the efficient frontier, by definition. Each point on the graph is a portfolio, and those lying along the upper boundary are called efficient, in the sense that they have the highest return for a given risk or the lowest risk for a given return. Portfolios whose return lies below risk-free rate are considered inferior and thus not optimal; thus, the actual efficient frontier is only cut to the top half.

In addition, we also observe how optimal asset weights shift in response to changes in key inputs such as expected returns, risk (volatility), and correlations. Correctly estimating these parameters thus greatly impacts the quality of the resulting solutions (portfolio weights), and it is important to perform such a

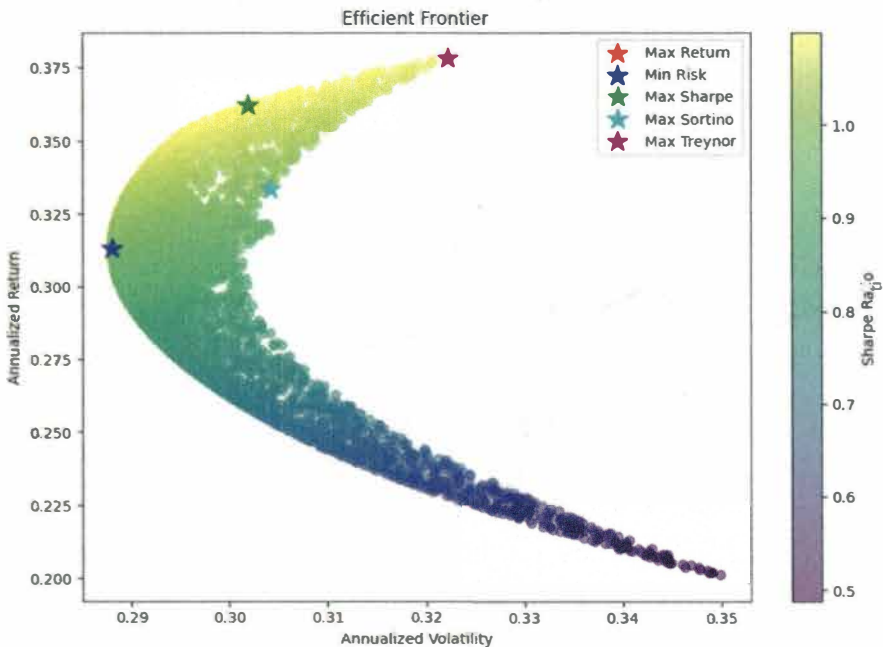


Figure 2-5 Efficient frontier with the five portfolios

sensitivity analysis on the impact of model misspecification. Such an analysis helps identify which assets are most sensitive to estimation errors, in the sense that a small change in the estimated parameter leads to a large change in the corresponding allocation weight.

2.7 Summary

In this chapter, we reviewed different approaches in the definition and calculation of risk and return, two fundamental concepts in finance, and illustrated their use in constructing and managing portfolios effectively. Returns can arise from capital appreciation, dividends, and interest and are often annualized to standardize performance across time frames. Risk comes from the uncertainty of returns and can be quantified using performance metrics such as variance (or standard deviation) or other downside risk measures (we will cover more on this in a later chapter). The risk-return trade-off highlights that higher potential returns generally require accepting greater risk.

Measuring returns involves various forms, including absolute return, percentage return, and logarithmic return. The performance of investments is often evaluated on a risk-adjusted basis, using metrics such as the Sharpe ratio, which evaluates return per unit of total volatility; the Sortino ratio, which focuses only on downside risk; and the Treynor ratio, which assesses excess return per unit of market (systematic) risk. These metrics enable investors to compare the effectiveness with which different portfolios manage risk in pursuit of returns.

Portfolio optimization strategies can vary, focusing on maximizing return, minimizing risk, or maximizing risk-adjusted returns based on the above metrics. For instance, the Max Return portfolio aims for the highest return but carries higher volatility, while the Min Risk portfolio seeks to minimize volatility at the expense of some return. The Max Sharpe portfolio balances risk and return efficiently, delivering the best risk-adjusted return when accounting for total volatility. The Max Sortino portfolio is designed to minimize downside risk, while the Max Treynor portfolio optimizes returns relative to market risk. Understanding these metrics and optimization strategies helps to build portfolios that are aligned with specific risk tolerance and return objectives.

Credit risk is a primary source of concern for institutions engaged in the lending business. When the borrower does not honor the contractual obligation to pay back the interest or the principal, the lending institution will suffer financial loss. Thus, proper management of credit risk plays an important role in sustainable financial operations and stability in the lending industry, including banks and other financial institutions. It is also a primary concern for regulatory authorities. For example, during the 2008 financial crisis, the underestimation of credit risk in subprime mortgage lending was a major contributing factor. Consequently, many modern regulatory frameworks have been proposed and are still being revised to actively detect vulnerabilities before they escalate into crises.

Credit risk refers to the potential that a borrower (also referred to as a debtor) will fail to fulfill their obligation to repay the loan or satisfy other contractual debt terms (such as paying back interest), thus resulting in a financial loss for the lender (also called the creditor). This risk materializes when the borrower defaults by not making timely payments of principal or interest, such as paying back credit card spending or home mortgage. Credit card lending typically involves a higher credit risk because it is unsecured, which means that there is no collateral to support the loan. Thus, credit card issuers typically face challenges in recovering the borrowed amount when the borrower defaults.

However, home ownership loans generally present a lower credit risk because they are secured by the property being financed. In case the borrower defaults, the lender can recover some or all of their losses through the foreclosure process, where the property is seized and sold at the prevailing market price. Despite differences in risk levels, properly and accurately assessing a borrower's creditworthiness is an important first-stage screening task. Such an assessment often comes with two decision outcomes: a binary output that indicates whether the credit card (or home loan application) is approved or not and a continuous output that indicates the credit line (or home loan amount) that limits the maximum amount to be borrowed.

Such an assessment process often involves developing credit scoring models (also called scorecards), setting appropriate interest rates for the loan based on risk levels, or asking for additional support from collateral or guarantees to reduce the lender's exposure to potential losses. This risk-based pricing principle also applies to asset financing for firms that seek loans from banks.

Credit risk management is also a key consideration for the broader financial system. For example, the 2008 global financial crisis was the result of poor credit risk management that led to significant financial loss. Why did it occur? It turned out that many financial institutions were lending money to risky borrowers with a high probability of default, particularly through subprime mortgages (a type of home loan offered to individuals who have low credit scores or limited credit history). These unmanaged credit exposures then accumulated in the banking sector and other financial entities until they reached a financial meltdown. As borrowers defaulted on their loans, a cascade of losses was triggered, resulting in a global financial crisis. Thus, proper credit risk management for the entire financial system is crucial to ensure financial robustness and stability.

The credit risk management life cycle encompasses the end-to-end process of identifying, assessing, and managing the risk of borrower default. It begins with the origination, where the borrower's creditworthiness is evaluated (typically by a scorecard) and the loan is structured. After approval and disbursement, the credit exposure enters the monitoring phase, where the performance, financial health, and external risk factors of obligors are assessed on a regular basis, mostly yearly or even shorter. If signs of stress appear, mitigation strategies, such as restructuring or additional collateral, are used to reduce potential losses. In the event of default, a recovery process would be triggered, which aims to maximize asset realization and minimize credit exposure (more on this later). Throughout the life cycle, regular portfolio-level review and reporting would be performed to ensure that credit risk exposure remains aligned with the institution's appetite and regulatory requirements. See Figure 3-1 for an illustration of the full credit management life cycle.



Figure 3-1 Illustrating the full credit management life cycle. The last two components are drawn with dashed line since not all loans are going to default

This chapter introduces common quantitative measures for credit risk management, focusing on the fundamental components that underpin this field, including probability of default (PD), loss given default (LGD), and exposure at default (EAD). We will also explore how these elements converge to form the expected credit loss (ECL), a critical metric for gauging anticipated losses. Finally, we will implement a PD model that is used to predict the PD of individual obligors.

3.1 Expected and Unexpected Credit Loss

The expected credit loss, or ECL, refers to the anticipated financial loss that a lender (such as a bank or specialized lending institution) may incur due to a borrower's failure to meet their financial obligations, such as paying back the principal or interest. Unlike traditional loss estimation methods that predominantly rely on historical default data, ECL adopts a forward-looking perspective to predict future likely outcomes and sets the expected loss accordingly. It uses both historical performance and predictive indicators to forecast potential future losses, where both statistical and machine learning tools can be used. When future potential losses are forecast, with some degree of accuracy, institutions can take proactive measures to mitigate these risks, such as only selecting borrowers with low ECL. These forecasts can help optimize credit portfolios and thus improve overall financial stability. Furthermore, ECL plays a pivotal role in regulatory compliance frameworks, notably under International Financial Reporting Standards (IFRS 9) and the Current Expected Credit Loss (CECL) model in the United States, which mandate the recognition of expected losses over the entire life of a financial instrument. Similar regulations in Singapore are also in place and set by the Monetary Authority of Singapore.

ECL is systematically calculated by integrating three fundamental components: PD, LGD, and EAD. Each component captures a distinct facet of credit risk and is connected by the following equation:

$$\text{ECL} = \text{PD} \times \text{LGD} \times \text{EAD}$$

This equation characterizes the interaction between the likelihood of default, the severity of the loss on default, and the magnitude of exposure, providing a detailed view of the anticipated losses under normal economic conditions. The expected loss is the product of the probability of a loss event (default), the expected loss per event (LGD), and the size of the exposure at the time of the event (EAD). This multiplicative relationship ensures that the impact of each component is reflected proportionately in the overall expected loss. For example, even if the PD is low, a high LGD or EAD can result in a significant ECL. In practice, PD is often subject to greater scrutiny in financial institutions, as it serves as the initial gatekeeper for identifying and filtering potential high-quality customers.

Let us look at an example of how to calculate the ECL. To calculate the lifetime ECL for a two-year term loan under the assumptions of IFRS 9, consider a loan with

a principal (EAD) of \$500,000, an LGD of 50%, annual marginal PDs of 3% for the first year and 5% for the second year, and a constant discount rate of 5%.

First, note that the maximum possible loss in any default scenario is

$$\$500,000 \times 50\% = \$250,000.$$

In year 1, the probability of default is 3%, giving an undiscounted expected loss of

$$\$250,000 \times 0.03 = \$7,500.$$

For year 2, we incorporate survival from year 1 (97% survival) and then apply the 5% marginal PD, leading to a year 2 unconditional default probability of

$$0.97 \times 0.05 = 0.0485,$$

which yields an undiscounted expected loss of

$$\$250,000 \times 0.0485 = \$12,125.$$

Next, we discount these amounts to the present. The year 1 expected loss is discounted by one year at 5%, giving

$$\frac{\$7,500}{1.05} \approx \$7,142.86,$$

and the year 2 expected loss is discounted for two years, giving

$$\frac{\$12,125}{(1.05)^2} \approx \$10,999.55.$$

Summing these discounted figures leads to a total lifetime ECL of approximately

$$\$7,142.86 + \$10,999.55 = \$18,142.41.$$

Although the ECL measures expected losses under normal circumstances, many financial institutions also consider unforeseen and extreme loss scenarios to ensure complete risk management. This is where unexpected loss (UL) and stress loss (SL) come into play, extending the ECL framework to consider a broader spectrum of potential credit risks. Extending ECL to cover both UL and SL also helps us assess the variability and extreme quantities of credit risk, ensuring that financial institutions are prepared not only for routine credit events but also for unforeseen and severe adverse conditions.

3.1.1 Unexpected Loss

Unexpected loss (UL) measures potential losses that exceed expected loss, reflecting the extent of tail risk across the entire distribution of credit exposures. It is mainly used to quantify and assess the capital reserves necessary to cover losses from unexpected adverse events that are beyond the usual circumstances, such as rare disasters like COVID-19, along with the ensuing economic downturns and market turbulence. These infrequent but potentially disastrous events could lead to considerable financial losses.

We can calculate UL using the Value at Risk (VaR) at a certain confidence level (α):

$$UL = \text{VaR}_{\alpha} - \text{ECL}$$

where VaR_{α} denotes the loss level that is exceeded with a probability of $1 - \alpha$. For example, at a confidence level 99%, $\text{VaR}_{0.99}$ represents the loss that is not expected to be exceeded in cases 99%, thus capturing the extreme tail threshold of the loss distribution (while ignoring those outside 99%). For example, assume that VaR at the 99% confidence level is \$15,000 and ECL is \$10,000. The UL is then calculated as $UL = 15,000 - 10,000 = \$5,000$. This means that there is an additional \$5,000 excess loss beyond the ECL value at the 99% confidence level. Thus, the bank (or financial institution) needs to prepare additional capital reserves to cover such extreme events.

Note that we can also use alternative downside risk measures, such as Conditional Value at Risk (CVaR), to calculate the unexpected loss. We will discuss different ways of VaR calculation and its shortcomings compared to CVaR in a later chapter.

3.1.2 Stress Loss

Stress loss (SL) refers to losses in stress scenarios, which are typically simulated adverse conditions to assess their impact on credit portfolios. This is a periodic exercise in banks and financial institutions when developing credit scoring models, as these models need to be able to withstand significant economic shocks without compromising their financial stability.

Stress testing involves simulating adverse conditions to assess their impact on credit portfolios, ensuring that financial institutions can withstand significant economic shocks without compromising their financial stability. SL is crucial for strategic planning and regulatory compliance, as it provides insight into the resilience of credit portfolios under extreme but plausible conditions.

Specifically, stress testing involves adjusting key credit risk parameters, such as PD, LGD, and EAD, based on predefined or preidentified adverse scenarios.

The resulting SL can be quantified by recalculating the ECL under these stressed conditions:

$$SL = \text{Stress PD} \times \text{Stress LGD} \times \text{Stress EAD}$$

Consider a recession scenario where economic conditions deteriorate significantly, leading to an increase in PD by 100% from 2% to 4% and an increase in LGD by 50% from 60% to 90%. Using an EAD of \$1,100,000 (which is an absolute amount that represents the loss instead of probability), the stressed loss is

$$SL = 0.04 \times 0.90 \times 1,100,000 = 39,600$$

This represents a substantial increase (tripled) in the baseline EL of \$13,200, highlighting the potential impact of adverse economic conditions on credit losses. The multiplicative nature of the ECL formula implies that the influence of each component is captured proportionately, providing a comprehensive measure of expected losses. In essence, it accounts for the frequency (PD) and severity (LGD) of potential credit losses and adjusts for the size of exposures (EAD).

In general, the ECL framework, through its integration of PD, LGD, and EAD, offers a good foundation for estimating credit risk. However, this also means that we need to obtain proper prediction for PD, LGD, and EAD to obtain a good estimate of ECL. For PD, we often develop statistical and econometric models to estimate the probability of default. There could be multiple considerations going into the estimation, such as sparsity (the model should not admit too many input features) and direction of travel (certain features should only have positive or negative signs in the corresponding coefficient), etc. Additional data, such as borrower-specific profile or transactional data, as well as macroeconomic indicators, are available to refine risk assessments.

On the other hand, LGD is influenced by collateral quality and recovery processes, which require quantitative asset valuations and qualitative legal recovery pathways. To evaluate EAD, we also need to have a good understanding of common credit utilization patterns and the potential for future loss.

We will dive into these three components of the ECL calculation in the following sections.

3.2 Probability of Default

Probability of default (PD) quantifies the likelihood that a borrower will fail to meet their financial obligations within a specified time horizon, typically using a one-year horizon, and could be extended to multiple years using conditional probabilities. PD serves as a critical indicator of the creditworthiness of a borrower. A PD model captures both the characteristics of the individual borrower and the broader economic conditions as its predictors. For example, individual profile factors of the borrower could cover aspects related to financial health and credit history, while

broader factors include industry stability, GDP growth, unemployment rates, or other macroeconomic variables.

Additional restrictions on the direction of travel for these factors may also come into play. For example, during an economic downturn, the PD of borrowers can increase due to reduced income streams and increased financial stress, both reflected in the relevant variables. In this case, we would expect the PD to be negatively correlated with the income stream variable (higher income is associated with lower PD), and the resulting coefficient for the income stream must be negative. Similarly, if we expect the obligor PD to decrease if China's GDP increases, then we would expect the China GDP variable to be negatively correlated with PD and the corresponding coefficient to be a negative sign.

PD can be empirically calculated as the percentage between the number of defaults out of the total number of loans:

$$PD = \frac{\text{Number of Defaults}}{\text{Total Number of Loans}}$$

This empirical estimation provides a basic measure of the probability of default based on historical data. For example, consider a financial institution that has issued 10,000 loans in the past year. Of these, 200 loans have been terminated due to flagged defaults. The PD is then calculated as

$$PD = \frac{200}{10,000} = 0.02 \text{ or } 2\%$$

This implies that, based on historical data, there is a 2% probability that any given loan will default within the year. However, the past does not represent future performance; thus, practical PD estimation often uses more sophisticated models that incorporate various borrower-specific and macroeconomic variables, as discussed earlier. These advanced models aim to enhance the predictive accuracy by capturing complex, nonlinear relationships and interactions between predictors. In the following sections, we introduce some of the common methodologies used in PD estimation.

3.2.1 Logistic Regression

Logistic regression is a widely used statistical method for binary classification problems, making it suitable for modeling default events (default vs. non-default). The logistic regression model estimates the probability that a borrower i will default based on a total of k explanatory variables $\{X_j\}_{j=1}^k$.

$$PD_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki})}}$$

where PD_i is the probability of default for the borrower i . $X_{1i}, X_{2i}, \dots, X_{ki}$ are the explanatory (also called independent) variables of the borrower i (such as individual credit score, income, debt-to-income ratio, or other macroeconomic variables). $\beta_0, \beta_1, \dots, \beta_k$ are the coefficients to be estimated from the data, so that the estimated PD is close to the true label (either 1 or 0). The logistic function is designed to ensure that the estimated PD is between 0 and 1, thus providing a probabilistic interpretation (i.e., how likely the underlying obligor will default). The coefficients β_j represent the log-odds change in PD for a one-unit change in the corresponding predictor X_j , which corresponds to a similar marginal interpretation in linear regression.

For example, suppose that we have a logistic regression model with two predictors: third-party credit score X_1 and debt-to-income ratio X_2 , with intercept β_0 and coefficients β_1, β_2 , respectively. The logistic regression model assumes the following form:

$$PD_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i})}}$$

Let us assume that the estimated coefficients are $\beta_0 = -1$, $\beta_1 = -0.5$, and $\beta_2 = 2$. For a borrower with a credit score of 75 and a debt-to-income ratio of 20, the PD is calculated as

$$PD = \frac{1}{1 + e^{-(-1 - 0.5 \times 75 + 2 \times 20)}} \approx 0.818$$

This relatively high PD of 81.8% indicates that this borrower has a high risk of default.

3.2.2 Decision Trees and Random Forests

Decision trees and ensemble methods such as random forests belong to the non-parametric family of models, which can be more powerful in making accurate predictions of PD. The power lies in the approximation capability that can learn complex interactions and nonlinearities automatically, without the need for explicit specification of model form.

A decision tree learns a decision rule by recursively partitioning the feature space based on a particular measure of marginal utility to classify the borrowers into default or non-default classes. Each split aims to maximize the separation between classes, as measured by the level of impurity in the resulting node, often using criteria like Gini impurity or information gain.

A random forest is an enhanced tree model that aggregates multiple decision trees, each trained on a bootstrap sample of the data and a random subset of features. The randomness of both samples and features significantly improves the model's learning capability and generalization performance. The final PD estimate

is obtained by averaging the predictions of individual trees, thus reducing variance and improving robustness.

These models can automatically handle high-dimensional data (where the number of features exceeds the number of observations) and interactions between variables (multicollinearity is a major issue in linear models), providing flexible and accurate PD estimates due to their ability to capture nonlinear relationships. However, compared to simpler models like logistic regression, they typically require larger datasets and computational resources.

3.2.3 Other Machine Learning Classifiers

Advanced machine learning classifiers, such as support vector machines (SVM), gradient boosting machines (GBM), and neural networks, can further enhance PD estimation by leveraging their ability to learn highly complex and possibly nonlinear patterns in the data.

SVM works by constructing hyperplanes in high-dimensional feature spaces to separate default and non-default borrowers. These hyperplanes are learned classifiers to best classify the observations with maximum margin in between supporting hyperplanes, subject to a pre-specified tolerance threshold. It also uses a kernel function to capture nonlinear decision boundaries.

GBMs work by learning an ensemble of weak learners (which are simple decision trees) in a sequential manner, where each subsequent tree is learned to minimize the errors of the previous ones. This iterative process turns out to be particularly effective in minimizing training error.

Neural networks consist of interconnected layers of nodes that digest input features through weight sum and nonlinear transformations. Such information extraction and nonlinear activation enable the modeling of highly complex relationships between predictors and PD. In particular, we can design very powerful and complex neural networks to learn arbitrary PD functions.

These classifiers excel at capturing intricate dependencies and interactions within the data and often outperform traditional statistical models in predictive accuracy. However, complex models often have many hyperparameters, and tuning a good set of hyperparameters could be quite time-consuming. In addition, regularization techniques are often applied to avoid overfitting during the training process.

3.3 Loss Given Default

Loss given default (LGD) represents the proportion (a ratio) of the exposure (an amount) that a lender is likely to lose due to a default event. Unlike PD, which measures the likelihood that a default occurs, LGD quantifies the severity of the loss (in terms of the percentage loss out of total exposure) once a default event has occurred. PD tells us how likely defaults may occur, while LGD measures the potential magnitude of losses associated with those defaults.

The level of LGD provides valuable insight into the potential severity of the losses. For example, a high LGD indicates a relatively significant loss, often associated with illiquid collateral that is difficult to recover. In contrast, a low LGD suggests a smaller loss, which may reflect high liquidity in the underlying collateral. Loans backed by easily recoverable and liquid collateral typically exhibit lower LGD values, whereas loans with illiquid and hard-to-recover collateral are more likely to result in higher LGD. For example, loans secured by tangible assets, such as real estate, typically exhibit lower LGDs compared to unsecured loans such as credit cards, as lenders can liquidate these collaterals to offset losses. The legal system that enforces efficient debt recovery also tends to lead to a lower LGD.

LGD is defined as

$$\text{LGD} = 1 - \text{Recovery Rate}$$

where the recovery rate denotes the percentage of the total exposure that can be recovered if the default event occurs. As an alternative definition, if a lender provides a loan with an outstanding exposure (unpaid total sum) of E and recovers a total amount of R after the default event, the LGD can be expressed as

$$\text{LGD} = \frac{E - R}{E} = 1 - \frac{R}{E}$$

Thus, LGD depends on two quantities: the outstanding exposure E and the recovery amount R . Since E cannot be changed if a default event occurs, LGD essentially depends on R , which also relates to the lender's ability to recover the full or partial outstanding exposure as efficiently as possible. In general, high-quality collateral and efficient recovery mechanisms lead to a high R and a low LGD, improving the resilience of the lender against defaults. In contrast, poor collateral quality or inefficient recovery processes result in low R and high LGD, thus increasing the likelihood of further loss.

There are multiple methods that can be used to estimate LGD. We outline a few commonly used methodologies here:

- **Using Historical Recovery:** This approach involves analyzing historical default data and calculating average recovery rates for similar asset classes, loan types, and economic conditions. In other words, this approach assumes average past performance as a prediction for future LGD prediction. As an example, suppose that a lender reviews ten defaulted loans for the past three years and finds that, on average, only 40% of the outstanding amounts were recovered. Then the estimated LGD is $1 - 0.40 = 0.60$, meaning a 60% loss is expected in the event of default.
- **Using Collateral Valuation:** For secured loans, the quality and liquidity of collateral are critical determinants as they represent the recoverable amount R in the definition of LGD. Valuation models are developed to assess the current market value of collateral assets, considering factors such as depreciation, market

volatility, and liquidation costs. This is different from the previous approach that focuses on estimating the LGD alone; instead, we focus on the collateral assets and try to predict their future value, which is further assumed to be recoverable should a default event happen. For example, imagine a secured loan of \$100,000 backed by collateral currently valued at \$90,000. This means that we will be able to recover at most 90% of the total risk exposure. If liquidating the collateral (such as selling the property collateral) requires 10% of the collateral value (i.e., \$9,000) as an additional cost, the recoverable amount becomes $\$90,000 - \$9,000 = \$81,000$. Thus, our estimate of LGD is $\frac{100,000 - 81,000}{100,000} = 0.19$.

- **Using Expert Judgment and Heuristics:** When historical data is sparse, insufficient, or unreliable, expert judgment often plays a crucial role in supplementing LGD estimation. Subject matter experts leverage their domain knowledge and experience to provide informed LGD estimates, which are particularly valuable when data is inadequate to train a reliable model. In addition, experts can refine their estimates by incorporating insights into current and evolving economic conditions, as well as prevailing market sentiment, ensuring that the predictions remain relevant and contextually accurate. For example, in cases with limited historical data, an expert might assess that, given current economic conditions, similar loans will have an expected recovery rate of 50%, which translates to an implied LGD of $1 - 0.50 = 0.50$, or 50%.
- **Using Statistical Models:** Statistical models, including linear regression and logistic regression, can be used to predict LGD based on various predictor variables such as borrower demographics, loan attributes, collateral information, and macroeconomic indicators. These models are trained to minimize the discrepancy between predicted and actual losses by optimizing the parameters based on historical data. These models, once properly developed, can potentially capture the underlying pattern in the data and make reliable predictions on LGD. In this regard, ensuring the proper training data and modeling assumption is the key to building a reliable model. As an example, consider a linear regression model that estimates LGD as $\hat{LGD} = 0.20 + 0.5 \times (LTV - 0.60)$, where LTV is the loan-to-value ratio expressed as a decimal. For a loan with an LTV of 0.80, the predicted LGD is $0.20 + 0.5 \times (0.80 - 0.60) = 0.30$, or 30%.
- **Using Machine Learning Models:** Advanced machine learning algorithms, such as random forests and neural networks, offer more complex and powerful models for LGD estimation. These techniques can capture complex nonlinear relationships and interactions between predictors. However, it is important to prevent these complex approximators from overfitting the data, which corresponds to good training set performance but poor out-of-sample performance. In addition, the choice of predictors and the modeling approach significantly influence the accuracy and reliability of LGD estimates. Suppose a random forest model, trained on historical loan and borrower data, predicts an LGD of 15% for a new loan of \$100,000. This implies an expected loss of $0.15 \times 100,000 = \$15,000$.

It is also important to note the interaction between LGD and other components of credit risk, such as PD and EAD. For example, a high PD coupled with a high LGD signifies a particularly risky exposure, as this means that the borrower is likely to default, and if such a default happens, most of the exposure will not be recoverable. Thus, such a type of obligor requires more stringent risk mitigation measures and higher capital reserves. In contrast, a high EAD (meaning a large amount of exposure at risk) with a low LGD (meaning that the majority of the exposure can be recovered, likely due to high-quality and liquid collateral) might require careful monitoring of exposure levels to manage potential losses effectively.

3.4 Exposure at Default

Exposure at default (EAD) represents the total value to which a lender is exposed when the borrower defaults (note that the definition of default may vary in different financial institutions). Unlike PD and LGD, which quantify the probability of default and the severity of potential loss, respectively, EAD quantifies the magnitude of exposure that is at risk, thus being an absolute amount instead of a percentage. Accurately estimating EAD is a key component not only for calculating ECL but also for determining the necessary capital reserves required to absorb potential losses.

To obtain EAD, we need to consider both the outstanding principal and accrued interest on a loan, as well as any additional exposures, such as undrawn credit lines or commitments that may be utilized by the borrower before defaulting. Thus, the exact value of EAD is inherently dynamic and may fluctuate in time, similar to both PD and LGD. All three components are, therefore, point-in-time estimates.

EAD is calculated by summing the current outstanding balance of a loan with the expected utilization of any undrawn credit facilities:

$$\text{EAD}_t = E_t + \theta U_t$$

where the outstanding balance E_t denotes the current principal and the interest accrued on the loan at time t . The undrawn credit line U_t represents the portion of a revolving credit facility that remains unused by the borrower at time t . The credit conversion factor (CCF) θ_t is the time-dependent CCF reflecting the expected utilization at time t . It is a factor that estimates the proportion of the undrawn credit line that will be used at the time of default. This equation captures the total exposure at default by accounting for both the utilized and potential future drawdowns on credit facilities. The CCF θ is typically determined based on historical utilization rates and can vary depending on the type of credit facility, borrower characteristics, and prevailing economic conditions.

EAD can be estimated using the following approaches:

- **Using a Standardized Approach Based on Regulations:** Under regulatory frameworks such as Basel II and Basel III, the standardized approach prescribes

fixed CCFs for different types of credit facilities. For example, an undrawn portion of a commercial loan might have a CCF of 50%, implying that 50% of the undrawn amount is expected to be used at default.

- **Using Behavioral Models:** Behavioral models can be trained to estimate EAD by forecasting the future utilization of undrawn credit lines based on borrower-specific behavior and macroeconomic indicators. Similar to PD and LGD estimation, these models often employ statistical models such as time series analysis to predict utilization patterns.
- **Using Machine Learning Models:** Machine learning algorithms, including random forests, XGBoost, and neural networks, are increasingly utilized to model EAD due to their good learning capacity without the need for feature engineering. These models can also incorporate a wide range of features, such as borrower demographics, credit history, loan characteristics, and macroeconomic variables, to enhance predictive accuracy.

As an example, consider the following scenario with an outstanding balance E of \$1,000,000, an undrawn credit line U of \$200,000, and a credit conversion factor θ of 50%. The EAD can be calculated as

$$\text{EAD} = 1,000,000 + (200,000 \times 0.50) = 1,100,000$$

which indicates that the total exposure at default is \$1,100,000, comprising the outstanding balance of \$1,000,000 and an additional \$100,000 expected from the undrawn credit line.

For a portfolio with multiple undrawn credit lines, each with different CCFs based on borrower profiles and economic scenarios, the corresponding EAD for the entire portfolio ($\text{EAD}_{\text{portfolio}}$) can be aggregated as

$$\text{EAD}_{\text{portfolio}} = \sum_{i=1}^n (E_i + U_i \times \theta_i)$$

where n is the total number of credit facilities in the portfolio.

Furthermore, for revolving credit facilities where borrowers can make multiple withdrawals over the loan life, EAD estimation may require integrating over the expected utilization path. This involves predicting not just a single drawdown, but a sequence of potential utilizations at different points in time, leading to a more dynamic and time-sensitive EAD estimation.

$$\text{EAD} = E_0 + \sum_{t=1}^T (U_t \times \theta_t)$$

where E_0 is the initial outstanding balance, U_t is the undrawn credit line at time t , θ_t is the CCF at time t , and T is the total time horizon.

The definition of EAD suggests that it is quite sensitive to the CCF parameter θ . This sensitivity can be characterized using sensitivity analysis, which involves assessing how changes in key parameters, such as the CCF or the undrawn credit line, impact the EAD estimates. For example, an increase in the CCF from 50% to 60% would proportionally increase the EAD by \$20,000 in the aforementioned example:

$$\Delta \text{EAD} = U \times (\theta_{\text{new}} - \theta_{\text{old}}) = 200,000 \times (0.60 - 0.50) = 20,000$$

3.5 Expected Credit Loss

Let us look at a concrete example of calculating the ECL in a housing mortgage loan. Imagine a bank that provides a mortgage loan to a borrower to purchase a residential property. The key parameters of this loan are as follows: The borrower secures a loan amount of \$400,000 to finance a property valued at \$500,000, resulting in a loan-to-value ratio (LTV) of 80%. Over time, the borrower makes repayments that add up to \$40,000, thus reducing the outstanding loan balance to \$360,000. This outstanding balance is calculated by subtracting the repayments made from the initial loan amount. It is the EAD for the bank, representing the total Value at Risk should the borrower default on the loan. This calculation reveals that the lender's exposure at the current time point is \$360,000, which consists of both the principal and any accrued interest that remains unpaid.

Next, we look at PD, which measures the likelihood that the borrower will default on the mortgage within a specified time horizon. Based on historical data, suppose that it is observed that one out of four homeowners default on their mortgage loans within a year. This empirical observation serves as an estimate for future PD:

$$\text{PD} = \frac{\text{Number of Defaults}}{\text{Total Number of Loans}} = \frac{1}{4} = 0.25 \text{ or } 25\%$$

This implies a 25% probability that the borrower will default on the loan within the year.

We also look at LGD, which quantifies the proportion of the exposure that is lost when a borrower defaults, after accounting for recoveries from collateral or other credit enhancements. In this example, assume that if the borrower defaults, the lender is able to immediately sell the property for \$342,000. The remaining loss incurred by the default event is the difference between the total outstanding loan balance and the recovery amount.

$$L = \text{EAD} - R = 360000 - 342000 = 18000$$

LGD is then calculated as the ratio of this loss to the exposure at default:

$$\text{LGD} = \frac{L}{\text{EAD}} = \frac{\$18,000}{\$360,000} = 0.05 \text{ or } 5\%$$

This calculation indicates that, in the event of default, the lender expects to lose 5% of the exposure at default, primarily due to the shortfall between the outstanding balance and the amount of recovery from the property sale.

Finally, let us calculate the ECL by integrating the three components, PD, LGD, and EAD, into a single measure that quantifies the expected average loss from credit exposures via the following:

$$\text{ECL} = \text{PD} \times \text{LGD} \times \text{EAD}$$

Substituting the values from our example:

$$\text{ECL} = 0.25 \times 0.05 \times 360000 = 4500$$

Thus, the bank's ECL is \$4,500. This figure represents the expected average loss from the loan, considering both the likelihood of default and the severity of the loss upon default.

Note that the ECL formula is based on the principle of conditional expectation, where the expected loss is the product of the probability of a loss event (PD), the percentage loss given a default (LGD), and the size of the exposure at the time of the event (EAD). This multiplicative relationship ensures that the impact of each component is reflected proportionately in the overall expected loss. In this example, a PD of 25% combined with an LGD of 5% and an EAD of \$360,000 yields an ECL of \$4,500, highlighting how even relatively low LGD values can result in significant expected losses when combined with substantial exposure and non-negligible default probabilities.

Note that we can further perform sensitivity analysis to analyze the impact of individual components, while assuming the rest is fixed, on the resulting ECL. All of these components have a positive correlation with ECL, meaning that a higher level of PD, LGD, or EAD will tend to increase ECL.

In the following section, we look at a related concept: the risk-weighted asset.

3.5.1 Capital Regulation Using Risk-Weighted Asset

In modern banking, managing credit risk is a crucial function within the risk management space. If not properly managed, credit risk can significantly affect a bank's financial health. To mitigate this risk, many regulatory frameworks have been proposed, such as Basel III, which requires banks to have sufficient capital relative to the level of risk and exposure. This is where the concept of risk-weighted assets (RWAs) comes in.

RWAs provide a mechanism to adjust the total value of bank assets based on the level of risk they carry. Simply put, it is a weighted combination across the value of all available assets. For instance, a government bond may be considered less risky than a corporate loan, and, therefore, it should receive a lower risk weight. By assigning risk weights to different asset classes, RWAs ensure that banks maintain an appropriate capital buffer proportional to the total exposure.

In addition to RWA, we also have a regulatory requirement on the bank's Capital Adequacy Ratio (CAR), a key measure of financial stability used to derive the required amount of capital reserve. Regulatory authorities, through the Basel III framework, have established minimum capital requirements (often set at 8%) to ensure that banks remain resilient in times of economic stress. CAR helps ensure that a bank has enough capital to cover its risks, thereby promoting stability and solvency in the financial system.

The required capital reserve is thus defined as the product of RWA and CAR:

$$\text{Capital} = \text{CAR} \times \text{RWA}$$

where Capital refers to the regulatory capital of a bank, which consists of Tier 1 (core capital, including common equity and retained earnings) and Tier 2 (supplementary capital, such as subordinated debt). Given the regulated CAR and the calculated RWA, we can then derive the amount of capital that the bank needs to reserve.

There are two main methods outlined in the Basel III framework for calculating RWAs for credit risk: the standardized approach (SA) and the internal ratings-based (IRB) approach. Both methods involve assigning risk weights to assets. SA is the simpler method of calculating RWA and is commonly used by smaller banks and financial institutions. This approach uses a fixed set of risk weights predefined by regulators, typically based on the external credit ratings of the counterparty or asset class involved. These credit ratings are then converted to a set of risk weights and further used to calculate RWA through the following:

$$\text{RWA} = \text{EAD} \times \text{Risk Weight}$$

Thus, to calculate the RWA, we would first determine the exposure value of each asset (along with additional adjustments for collateral or guarantees), multiply the exposure value of each asset by its assigned risk weight, and then sum the results across all assets. Consider a bank with the following assets: \$200 million in AAA-rated government bonds with a risk weight of 0%, \$80 million in BBB-rated corporate loans with a risk weight of 100%, and \$50 million in residential mortgages with a risk weight of 50%. The RWA is calculated as $(200 \times 0\%) + (80 \times 100\%) + (50 \times 50\%) = 105$ million. Therefore, the total RWA for credit risk amounts to \$105 million.

There are multiple types of asset classes when it comes to risk exposure, including sovereign exposures, corporate loans, residential mortgages, retail exposures, and off-balance sheet items. Each asset class is assigned a specific risk weight based

on the credit rating of the counterparty (if available) or the asset type. For example, government bonds, which are AAA rated, can get 0% risk weight, while corporate loans (BBB rated) get 100% risk weight (due to their high risk of default), residential mortgages get 50% risk weight (since the property itself can serve as collateral), and retail loans get 75% risk weight (relatively risky especially for unsecured loans).

Compared to SA, the IRB approach is more advanced and allows banks and financial institutions to use their own risk models to calculate the three risk components (PD, LGD, and EAD). Due to its complexity and a considerable amount of development effort, this method is typically used by larger banks with more complex portfolios and requires regulatory approval from local and/or foreign regulators.

There are two variants of the IRB approach: Foundation IRB (F-IRB), where banks estimate PD with LGD and EAD provided by regulators, and Advanced IRB (A-IRB), where banks estimate all three key components. These parameters, once estimated, are then used in the calculation of the RWA.

Specifically, we first calculate the risk weight parameter K , which represents the capital requirement per unit of EAD. A common calculation is based on the PD, LGD, and asset correlation parameter ρ using the following expression:

$$K = \text{LGD} \times \Phi \left(\frac{\Phi^{-1}(\text{PD}) + \sqrt{\rho} \Phi^{-1}(0.999)}{\sqrt{1 - \rho}} \right)$$

This quantity reflects the amount (in percentage) of capital that a bank needs to hold for each unit exposure to cover the unexpected loss (tail risk or extreme loss scenarios beyond the average anticipated loss) with a specified confidence level (99.9%). Φ is the cumulative distribution function of the standard normal distribution (used to map probabilities to the corresponding Z scores), and $\Phi^{-1}(p)$ is the inverse cumulative distribution function (quantile function) of the standard normal distribution, representing the Z score corresponding to a probability p . Note that more advanced models may also consider the maturity of the exposure in the calculation.

Once K is calculated, we can then calculate RWA via $\text{RWA} = K \times \text{EAD}$. This process reflects the risk sensitivity of the bank's exposures and helps determine how much capital the bank needs to hold to cover potential losses.

Consider a scenario with the following parameters: an EAD of \$120 million, a PD of 5%, an LGD of 55%, an asset correlation (ρ) of 0.2, a CAR of 8%, and a target confidence level 99.9%. To calculate the capital requirement, we first determine K :

$$K = 0.55 \times \Phi \left(\frac{\Phi^{-1}(0.05) + \sqrt{0.2} \times \Phi^{-1}(0.999)}{\sqrt{1 - 0.2}} \right) \approx 21.12\%$$

Next, the RWA is calculated as follows:

$$\text{RWA} = 0.2112 \times 120,000,000 = \$25,344,000$$

Finally, the required capital is computed as follows:

$$\text{Capital} = 0.08 \times 25,344,000 \approx \$2.03 \text{ million}$$

This implies that a bank with an RWA value of \$25.344 million must maintain \$2.03 million in capital to meet the regulatory CAR of 8%. This calculation highlights the interaction between credit risk parameters and regulatory requirements, ensuring sufficient capital buffers to absorb potential losses.

In the next section, we will look at how to build a PD model using the logistic regression model. We focus on this component due to its important role as a first-level screening between high-risk and low-risk borrowers.

3.6 Building a PD Model

A PD model takes relevant borrower information and outputs the probability of default within the next period of time, which could be one year, two years, etc. There are typically two types of obligor data: one-time application data and recurrent behavioral data. By digesting both application and behavioral data, credit risk models aim to predict the probability of default, which helps to set the interest rate and other risk management guidelines throughout the borrowing period.

Application data refers to the information provided by the borrower at the time of the credit application. It typically includes key demographic and loan-specific characteristics that give an initial indication of the borrower's risk profile. The key attributes in the application data, once approved, include interest rate (which positively correlates with credit risk), credit grade (an internal grade assigned to potential obligors as an indicator of creditworthiness), and loan amount (the total amount of money requested by the borrower).

Behavioral data reflect the borrower's financial status and historical background information. This information is often collected and assessed on a periodic basis, often yearly, after the borrower has taken out a loan. This periodic assessment ensures that the borrower is in good financial condition on a regular basis. Example behavior attributes include employment history (such as length of service and income level), historical default (past default history, which is a key predictor of future default risk), and other demographics such as spending or transaction history.

When developing a PD model, the default definition, or good-bad definition, is typically based on the delinquency status of the borrower in terms of the number of days past the due date. A common definition is that a borrower would be considered as default if it has been more than 90 days past the due date. However, a borrower who has committed fraud may also be considered a default. The default status is often stored in a binary variable where 1 indicates default and 0 indicates non-default. When modeling the default outcome in logistic regression, the log odds of the default probability are modeled as a linear combination of the features.

In the following sections, we first look at common best practices in data processing and exploration, including dealing with outliers and missing data. We then move on to model development using logistic regression and its evaluation.

3.6.1 Data Processing and Exploration

We will look at a credit bureau dataset that contains both application and behavior features. The dataset contains 32,581 rows and 12 columns, as shown in Listing 3-1.

```

1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 from sklearn.model_selection import train_test_split
5 from sklearn.linear_model import LogisticRegression
6 from sklearn.metrics import (
7     accuracy_score,
8     precision_score,
9     recall_score,
10    f1_score,
11    confusion_matrix,
12    roc_curve,
13    auc,
14    classification_report
15 )
16 import seaborn as sns
17
18 cr_loan = pd.read_csv("data/cr_loan2.csv")
19 cr_loan.shape
20
21 # Output
22 (32581, 12)

```

Listing 3-1 Importing data

In this dataset, the *loan_status* variable indicates the default status. As mentioned earlier, banks and financial institutions demand a higher interest for risky obligors whose probability of default is higher than less risky borrowers. We can verify this via Listing 3-2, which shows the average interest rate for each type of *loan_status* and *person_home_ownership*. The resulting table, as shown in Table 3-1, suggests that the average interest rate is the lowest for non-default cases but rises up for default cases.

```

1 # Crosstab between 'person_home_ownership' and 'loan_status',
2 # calculating the mean of 'loan_int_rate' and rounding it to 2
   decimal places
3 pd.crosstab(
4     cr_loan['person_home_ownership'], # Index: Homeownership
      status
5     cr_loan['loan_status'],           # Columns: Loan status
      values=cr_loan['loan_int_rate'], # Values to aggregate:
6     Loan interest rate

```

Table 3-1 Comparing the interest rates across different home ownerships and default status

Loan status	0	1
Person home ownership		
MORTGAGE	10.06	13.43
OTHER	11.41	13.56
OWN	10.75	12.24
RENT	10.75	12.97

```
aggfunc='mean' # Aggregation function:
Mean
).round(2) # Round the results to 2
decimal places
```

Listing 3-2 Analyzing interest rate across different home ownerships and default status

3.6.2 Dealing with Outliers

When developing credit risk models, outliers can arise from various sources, such as incorrect data entry, rare but extreme financial behaviors, or unique economic events that lead to significant deviation from the norm. These outliers are typically data points that deviate considerably from the majority of observations and can distort statistical analyses and predictions. For example, exceptionally long or short employment terms, income levels, or loan amounts may not represent typical borrower behavior and can skew the model’s understanding of default risk. Thus, it is important to detect and handle these outliers because they can influence the estimation process and lead to biased PD predictions. Once detected, outliers can be treated by either removing them, transforming them, or using robust modeling techniques that reduce their influence.

Let us start by analyzing *person_emp_length* and *loan_int_rate*, where we show the original scatterplot in the left panel of Figure 3-2. Since it is unlikely that a person will have more than 60 years of employment duration, we can identify these rows as outliers and remove them from the original dataframe. This generates the right panel of Figure 3-2.

See Listing 3-3 for details on removing outlier observations.

```
# Copy original data to preserve it for "before" plot
df_before = cr_loan.copy()

# Identify outliers: employment length >= 60 months
outlier_indices = df_before[df_before['person_emp_length'] >=
60].index

# Remove those outliers to create 'after' set
df_after = df_before.drop(outlier_indices)

# Create a 1x2 subplot
```

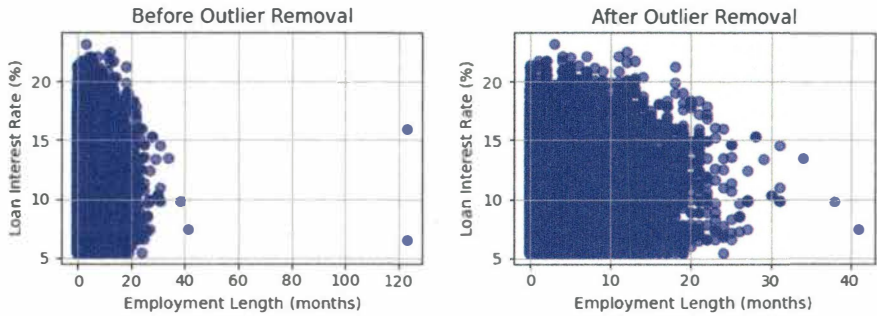


Figure 3-2 Scatter plot of Employment Length vs. Loan Interest Rate. Left: before removing outliers. Right: after removing outliers

```

11 fig, axes = plt.subplots(nrows=1, ncols=2, figsize=(8, 3))
12
13 # -- Plot 1: Before Outlier Removal
14 axes[0].scatter(df_before['person_emp_length'],
15                df_before['loan_int_rate'],
16                alpha=0.5,
17                color='blue')
18 axes[0].set_title('Before Outlier Removal')
19 axes[0].set_xlabel('Employment Length (months)')
20 axes[0].set_ylabel('Loan Interest Rate (%)')
21 axes[0].grid(True)
22
23 # -- Plot 2: After Outlier Removal
24 axes[1].scatter(df_after['person_emp_length'],
25                df_after['loan_int_rate'],
26                alpha=0.5,
27                color='blue')
28 axes[1].set_title('After Outlier Removal')
29 axes[1].set_xlabel('Employment Length (months)')
30 axes[1].set_ylabel('Loan Interest Rate (%)')
31 axes[1].grid(True)
32
33 plt.tight_layout()
34 plt.show()

```

Listing 3-3 Removing outliers

In addition to removing outliers, we can also transform them into meaningful observations so that they can still be used for modeling purposes. For example, applying a log transformation compresses the range of the data by converting large values into a logarithmic scale, which reduces the disproportionate influence of extreme observations and stabilizes variance. Standardization, on the other hand, rescales the data so that each feature has a mean of 0 and a standard deviation of 1, thereby facilitating comparisons across different variables and improving the convergence of many algorithms. Additionally, winsorization involves capping

extreme values at specific percentile thresholds, which limits the impact of outliers while preserving the overall distribution of the data. Together, these methods enable us to retain valuable information from outliers while ensuring that they do not unduly distort the modeling process.

3.6.3 Dealing with Missing Data

The handling of missing data is also a critical step in ensuring the integrity and predictive power of the model. Missing data can occur for a variety of reasons, such as incomplete customer applications, data entry errors, or limitations in data collection systems. If not addressed properly, missing values can introduce bias, reduce the efficiency of the model, or even render the model invalid. When a feature has most of the observations missing, the common practice is to remove it. However, when only a few observations are missing, we tend to fill these missing values with some estimate.

There are several techniques for treating missing data in credit risk modeling, each depending on the extent and nature of the missing observations. Common methods include mean or median imputation, where missing values are replaced by the average or median of the observed data, which is simple and often effective for continuous variables. Mode imputation is useful for categorical variables. In some cases, the missing data itself may be informative and indicative of increased risk, in which case creating a separate “missing” category can be useful.

In Listing 3-4, we first count the number of rows with missing data in *loan_int_rate* and *person_emp_length*, followed by mean value replacement and row removal for the latter.

```
1 cr_loan = df_after.copy()
2
3 # Fill missing values in the 'loan_int_rate' column with the mean
  # of the non-missing values
4 cr_loan['loan_int_rate'].fillna(cr_loan['loan_int_rate'].mean(),
  inplace=True)
5 # Get the indices of rows where 'person_emp_length' is null (NaN)
6 indices = cr_loan[cr_loan['person_emp_length'].isnull()].index
7 # Drop the rows with missing 'person_emp_length' values from the
  # DataFrame
8 cr_loan.drop(indices, inplace=True)
```

Listing 3-4 Dealing with missing value

3.6.4 Dealing with Categorical Data

Logistic regression requires numerical input, so any categorical variables in the dataset must be converted to a numerical format. One common method for this

conversion is one-hot encoding, which transforms categorical variables into a series of binary columns, as shown in Listing 3-5.

```
1 # Identify categorical variables
2 categorical_vars = ['person_home_ownership', 'loan_intent', '
   loan_grade', 'cb_person_default_on_file']
3
4 # Perform dummy encoding on categorical variables, dropping the
   first category to avoid multicollinearity
5 cr_loan_encoded = pd.get_dummies(cr_loan, columns=
   categorical_vars, drop_first=True)
```

Listing 3-5 One-hot encoding categorical data

By applying one-hot encoding to the identified categorical variables and dropping the first category, we prevent multicollinearity, which can adversely affect the model's performance.

3.6.5 Train-Test Split

As shown in Listing 3-6, after encoding the categorical variables, the next step is to separate the dataset into features and the target variable. The features (X) are all the columns except *loan_status*, which is the target (y) that we aim to predict.

```
1 # Define feature matrix X and target vector y
2 X = cr_loan_encoded.drop('loan_status', axis=1)
3 y = cr_loan_encoded['loan_status']
```

Listing 3-6 Creating features and target variables

To effectively evaluate the performance of the model, it is important to divide the data into training and testing sets. As shown in Listing 3-7, typically, a common split is 70% for training and 30% for testing. This division allows the model to learn from the training data and be evaluated on unseen testing data.

```
1 # Split the data into training and testing sets (e.g., 70% train,
   30% test)
2 X_train, X_test, y_train, y_test = train_test_split(
3     X, y, test_size=0.3, random_state=42, stratify=y
4 )
```

Listing 3-7 Performing train-test split

Here, setting *random_state=42* ensures reproducibility of the results, and *stratify=y* maintains the same distribution of the target variable in both training and testing sets.

Now, let us move on to the model development stage.

3.6.6 Developing Logistic Regression Model

With the data prepared, the next step is to build and train the logistic regression model. As shown in Listing 3-8, we will initialize the model with a higher maximum number of iterations to ensure convergence and use the “lbfgs” solver, which is suitable for smaller datasets and supports multinomial loss.

```
1 # Initialize the Logistic Regression model with increased
   max_iter to ensure convergence
2 logreg = LogisticRegression(max_iter=1000, solver='lbfgs')
3 # Fit the model on the training data
4 logreg.fit(X_train, y_train)
```

Listing 3-8 Building logistic regression model

As shown in Listing 3-9, training the predictive model involves fitting it to the training data, allowing it to learn the relationships between the features and the target variable. After training the model, we use it to make predictions on the test set. This involves predicting both the class labels and the probabilities associated with the positive class, which are useful for evaluating the model’s performance using ROC analysis.

```
1 # Make predictions on the test set
2 y_pred = logreg.predict(X_test)
3 y_pred_proba = logreg.predict_proba(X_test)[: , 1]
```

Listing 3-9 Generating model predictions

Here, *y_pred* contains the predicted class labels, while *y_pred_proba* contains the predicted probabilities for the positive class.

3.6.7 Model Evaluation

Evaluating the model’s performance is crucial to understand how well it generalizes to new data. As shown in Listing 3-10, we will use several metrics, including accuracy, precision, recall, F1 score, confusion matrix, and the ROC-AUC curve, to assess the model’s performance. Specifically, accuracy measures the proportion of correctly predicted instances out of all instances. Precision assesses the correctness of positive predictions, recall evaluates the model’s ability to capture all positive instances, and the F1 score balances precision and recall.

```
1 # Evaluate the model
2 accuracy = accuracy_score(y_test, y_pred)
3 precision = precision_score(y_test, y_pred)
4 recall = recall_score(y_test, y_pred)
5 f1 = f1_score(y_test, y_pred)
6
7 print("Model Evaluation Metrics:")
8 print(f"Accuracy : {accuracy:.4f}")
9 print(f"Precision: {precision:.4f}")
10 print(f"Recall   : {recall:.4f}")
```

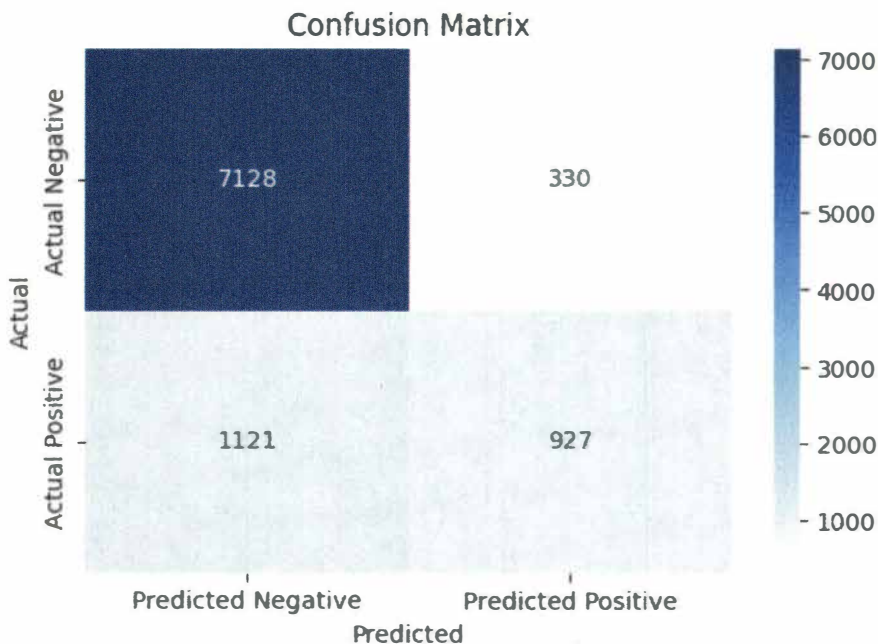


Figure 3-3 Confusion matrix between model predictions and actual targets

```

11 print(f"F1 Score : {f1:.4f}")
12
13 # Output
14 Accuracy : 0.8474
15 Precision: 0.7375
16 Recall   : 0.4526
17 F1 Score : 0.5610

```

Listing 3-10 Model evaluation

The result suggests that there might be a trade-off between precision and recall. Although the model is good at avoiding false positives (as indicated by the high precision), it is poor at capturing actual defaults (low recall). In credit risk modeling, missing actual defaults can be risky, so recall is often more important than precision.

As shown in Listing 3-11, the confusion matrix provides a more detailed breakdown of the model predictions, showing the number of true positives, true negatives, false positives, and false negatives. This visualization, as shown in Figure 3-3, helps to understand where the model is performing well and where it might be making mistakes.

```

1 # Display the confusion matrix
2 conf_matrix = confusion_matrix(y_test, y_pred)
3 plt.figure(figsize=(6,4))
4 sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues',
5             xticklabels=['Predicted Negative', 'Predicted
    Positive'],

```

```

6 yticklabels=['Actual Negative', 'Actual Positive'])
7 plt.title('Confusion Matrix')
8 plt.xlabel('Predicted')
9 plt.ylabel('Actual')
10 plt.show()

```

Listing 3-11 Displaying confusion matrix

As shown in Listing 3-12, we can also use the classification report to obtain a comprehensive overview of the model's performance, including precision, recall, F1 score, and support for each class. This report also helps us to understand how the model performs in different categories.

```

1 # Classification report for detailed metrics
2 print("Classification Report:")
3 print(classification_report(y_test, y_pred))
4
5 # Output
6 # Classification Report:
7
8      precision    recall  f1-score   support
9
10     0.86    0.96    0.91     7458
11     0.74    0.45    0.56     2048
12
13 accuracy          0.85     9506
14 macro avg          0.80     9506
15 weighted avg       0.84     9506

```

Listing 3-12 Classification report

3.6.8 ROC Curve

Note that such a classification report is based on a threshold value of 0.5 by default. Choosing the appropriate classification threshold is a critical step in building an effective logistic regression model, especially in sensitive applications such as credit risk modeling. The threshold determines how predicted probabilities are converted into class labels (e.g., default or no default), directly impacting the balance between true positives and false positives. In this context, selecting the optimal threshold involves not only statistical considerations but also a deep understanding of the business implications.

The classification threshold is the probability value above which a prediction is classified as the positive class (e.g., loan default) and below which it is classified as the negative class (e.g., loan non-default). Adjusting this threshold affects the model's sensitivity (true positive rate) and specificity (true negative rate).

The Receiver Operating Characteristic (ROC) curve is a graphical representation that illustrates the trade-off between the true positive rate (TPR) and the false positive rate (FPR) in different threshold settings. Each point on the ROC curve corresponds to a specific threshold value and the corresponding trade-off between

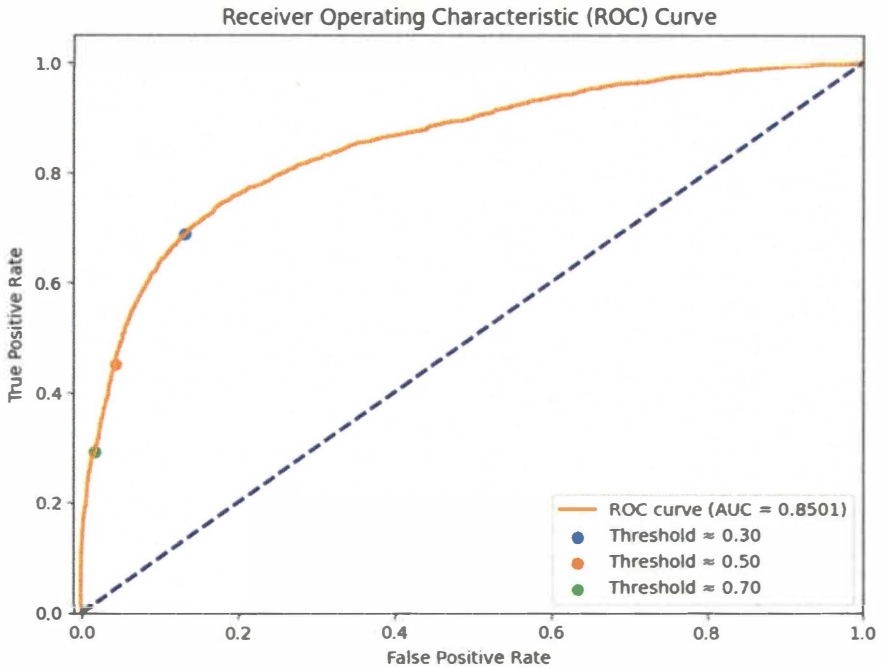


Figure 3-4 ROC curve

precision and recall. By plotting TPR against FPR at various thresholds, the ROC curve provides a comprehensive view of the model's performance. The Area Under the Curve (AUC) then quantifies the overall ability of the model to discriminate between the two classes, with a higher AUC indicating better performance.

However, the ROC curve alone does not specify the best threshold for classification. To determine the optimal threshold, especially in the context of credit risk modeling, it is still essential to consider the specific costs and benefits associated with different types of prediction errors.

To better understand how the threshold impacts model performance and to identify the optimal threshold for credit risk modeling, let us delve deeper into the ROC curve and incorporate additional analyses. In Listing 3-13, we plot the ROC curve and highlight specific threshold points (0.3, 0.5, and 0.7) to visualize how different thresholds affect the true positive and false positive rates, as shown in Figure 3-4. The `find_closest_threshold` function ensures that we select the nearest available threshold value to avoid indexing errors. In addition, we also demonstrate how changing the threshold to 0.7 affects key performance metrics.

```
# ROC Curve and AUC
fpr, tpr, thresholds = roc_curve(y_test, y_pred_proba)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(8,6))
```

```

6 plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (
    AUC = {roc_auc:.4f})')
7 plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
8
9 # Define the thresholds you want to highlight
10 desired_thresholds = [0.3, 0.5, 0.7]
11
12 # Function to find the closest threshold
13 def find_closest_threshold(thresholds, desired_thresh):
14     idx = np.argmin(np.abs(thresholds - desired_thresh))
15     return idx
16
17 # Highlight specific threshold points
18 for thresh in desired_thresholds:
19     idx = find_closest_threshold(thresholds, thresh)
20     plt.scatter(fpr[idx], tpr[idx], label=f'Threshold {
        thresholds[idx]:.2f}', marker='o')
21
22 plt.xlim([-0.01, 1.0])
23 plt.ylim([0.0, 1.05])
24 plt.xlabel('False Positive Rate')
25 plt.ylabel('True Positive Rate')
26 plt.title('Receiver Operating Characteristic (ROC) Curve')
27 plt.legend(loc="lower right")
28 plt.show()
29
30 # Example of threshold impact on predictions
31 threshold = 0.7
32 y_pred_threshold = (y_pred_proba >= threshold).astype(int)
33
34 # Evaluate performance at the chosen threshold
35 accuracy = accuracy_score(y_test, y_pred_threshold)
36 precision = precision_score(y_test, y_pred_threshold)
37 recall = recall_score(y_test, y_pred_threshold)
38 f1 = f1_score(y_test, y_pred_threshold)
39
40 print(f"Performance Metrics at Threshold = {threshold}:")
41 print(f"Accuracy : {accuracy:.4f}")
42 print(f"Precision: {precision:.4f}")
43 print(f"Recall    : {recall:.4f}")
44 print(f"F1 Score  : {f1:.4f}")
45
46 # Output
47 Performance Metrics at Threshold = 0.7:
48 Accuracy : 0.8340
49 Precision: 0.8210
50 Recall    : 0.2935
51 F1 Score  : 0.4324

```

Listing 3-13 ROC curve and AOC

Assuming that there is no preference between precision and recall, we can use the F1 score to balance the two. The F1 score is the harmonic mean of precision and recall, which can be particularly useful when the classes are imbalanced. In

Listing 3-14, we iterate through all thresholds and calculate the F1 score to select the threshold that achieves the highest balance between precision and recall. In this case, the optimal threshold is 0.34.

```

1 # Maximizing the F1-Score
2 f1_scores = []
3 for thresh in thresholds:
4     y_pred_thresh = (y_pred_proba >= thresh).astype(int)
5     f1 = f1_score(y_test, y_pred_thresh)
6     f1_scores.append(f1)
7
8 optimal_idx = np.argmax(f1_scores)
9 optimal_threshold = thresholds[optimal_idx]
10 print(f"Optimal Threshold based on F1-Score: {optimal_threshold
    :.2f}")
11
12 # Output
13 Optimal Threshold based on F1-Score: 0.34

```

Listing 3-14 Maximizing the F1 score

3.7 Summary

Managing credit risk is a fundamental aspect of financial institutions, who need to assess potential losses due to potential default event. This chapter began with an introduction to credit risk, highlighting its importance in maintaining financial stability and preventing substantial financial losses. The discussion then moved deep into the core components of credit risk assessment: PD, LGD, and EAD. These elements collectively form ECL, a pivotal metric that integrates the likelihood, severity, and magnitude of potential defaults to provide a nuanced estimate of future expected losses.

This chapter also examined advanced concepts such as unexpected loss (UL) and stress loss (SL), which extend the ECL framework to account for extreme and unforeseen adverse events. In addition, the discussion transitioned to capital regulation, explaining how RWA and CAR are used under regulatory frameworks like Basel III to ensure that banks have sufficient capital relative to their risk exposures.

In the realm of predictive modeling, we introduced the construction of a PD model using logistic regression, starting with data preprocessing steps such as outlier detection, handling missing data, and categorical variable encoding. The model development process involves training the logistic regression model, making predictions, and evaluating its performance through metrics like accuracy, precision, recall, F1 score, and the ROC curve. We also discussed the importance of selecting an optimal classification threshold, particularly in credit risk contexts where the balance between true positives and false positives has significant financial implications. As an example, we introduced techniques such as maximizing the F1 score to select the most appropriate threshold.

In general, this chapter provided a framework for understanding and managing credit risk, integrating quantitative methodologies with practical considerations to improve financial institutions' ability to predict, assess, and mitigate potential losses. We further enhanced the discussion by introducing a hands-on example of building a PD model using logistic regression, which serves as an important first-level check on the creditworthiness of potential and existing obligors.

Market risk encompasses the uncertainty and potential financial loss generated by broad-based fluctuations in asset prices, driven by factors such as interest rates, inflation, currency movements, and macroeconomic indicators. Unlike idiosyncratic risk, which pertains to specific firms or sectors and can be minimized through diversification, market risk is systemic throughout the global financial landscape. It can affect the performance of equities, bonds, commodities, and other asset classes simultaneously. The inherent challenge lies in managing this pervasive risk by balancing portfolio construction, hedging strategies, and regulatory oversight to mitigate vulnerabilities while still pursuing returns. For example, we can adopt a market-neutral strategy to bet on the relative difference between two assets rather than focusing on the directional change of a single asset, which is often random and difficult to predict.

Market risk is ubiquitous in multiple asset classes, such as equities, interest rates, foreign exchange rates, and commodity prices. For example, fluctuations in stock prices can result in substantial gains or losses, affecting both individual investors and institutional portfolios. Changes in interest rates affect borrowing costs, savings returns, and the valuations of fixed-income securities, which, in turn, influence investment decisions and overall economic activity. Fluctuations in currency exchange rates can also significantly affect multinational corporations, investors with foreign assets, and economies that depend significantly on imports and exports, such as Singapore. Furthermore, changes in the prices of essential commodities such as oil, gold, and agricultural products can have a profound impact on various industries, consumer prices, and geopolitical stability.

Note that market risk is widespread and cannot be completely eliminated through diversification strategies, which are primarily effective against unsystematic and idiosyncratic risk, which is the risk associated with individual assets or specific sectors. To manage market risk, it is essential to have a thorough understanding of the factors that influence market movements and to implement robust risk management frameworks. On this front, we often use various quantitative metrics to

manage and reduce market risk effectively. These metrics offer crucial insights into different aspects of risk, including volatility, the potential scale of tail losses, and the overall exposure of investment portfolios to negative market conditions.

In this chapter, we focus on three fundamental market risk metrics: variance, maximum drawdown, and Value at Risk (VaR). First, variance is a statistical measure that quantifies how asset returns fluctuate from the mean, serving as an indicator of volatility and the level of uncertainty in investment returns. Second, maximum drawdown measures the largest decline from a peak to a trough in the value of an investment portfolio, highlighting the potential for significant losses during unfavorable market conditions. Lastly, VaR is a probabilistic measure that estimates the maximum expected loss of a portfolio over a specified time period at a given confidence level. Each of these metrics offers a distinct perspective on market risk, allowing a comprehensive assessment of potential downside risks for portfolios. These three risk measures jointly provide a benchmark for assessing risk levels and can be used together to guide capital allocation.

We examine each risk measure in detail in the following sections.

4.1 Variance

Variance is an important statistical metric that measures how much variability or dispersion exists within a dataset. In statistics, it is the second moment that is used to describe the distribution of the data. In finance, it is also a crucial tool for evaluating an asset's volatility, showing how much the asset's returns differ from the expected average over a specific timeframe. A high variance indicates greater uncertainty and risk, as the asset's returns fluctuate widely. In contrast, a low variance suggests more stable and, therefore, more predictable returns.

High volatility should not be universally considered a negative characteristic. In fact, a highly volatile asset can offer the potential for large gains, which can be particularly appealing to investors with a strong appetite for risk or those targeting substantial returns. At the same time, this elevated volatility carries an equally heightened possibility of severe losses. Consequently, such assets are generally not suitable for risk-averse individuals or those seeking more stable and predictable returns.

The variance of a series of returns can be calculated as the average of the squared differences between each return and the mean return. Here, we would differentiate between sample variance and population variance, where the former denotes the variance for the current limited sample, and the latter refers to the whole population (which we may not have access to). Specifically, for an asset with a series of returns

$R = \{R_1, R_2, \dots, R_n\}$, the sample variance (s^2) is determined using the following formula:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (R_i - \bar{R})^2$$

where R_i represents the return of the asset in the i -th period, $\bar{R}$ denotes the mean return calculated as $\bar{R} = \frac{1}{n} \sum_{i=1}^n R_i$, and n is the total number of observations.

Note that squaring the deviations from the mean ensures that both positive and negative differences contribute to the total variability, preventing them from canceling each other out. Furthermore, using $n-1$ in the denominator, known as Bessel's correction, adjusts for the bias that occurs when estimating the population variance from a sample. This adjustment ensures that the calculated variance is an unbiased estimator of the true variance of the population.

Let us look at an example of how to calculate the variance of a stock. Suppose a collection of monthly returns valued at 3%, -2%, and 4%. The average return is $\bar{R} = \frac{3+(-2)+4}{3} \approx 1.67\%$. Next, the squared deviations from the mean are computed: $(3 - 1.67)^2 \approx 1.77$, $(-2 - 1.67)^2 \approx 13.44$, and $(4 - 1.67)^2 \approx 5.44$. Summing these squared deviations gives approximately 20.65. Applying Bessel's correction by dividing by $n-1 = 2$ yields a sample variance of $\frac{20.65}{2} \approx 10.33\%$. This indicates that the monthly returns of the stock fluctuate considerably around the average return of 1.67%, implying higher volatility and, therefore, a higher level of risk and the potential for significant gains or losses.

4.1.1 Unbiasedness in Sample Variance

When estimating the variance from a sample of data, it is important to use an unbiased estimator. This means that the expected value of the estimator should equal the true population variance as the number of observations gets bigger. To achieve this, the formula for sample variance uses a denominator of $n-1$ instead of n . This adjustment, known as Bessel's correction, addresses the bias that can occur when estimating the population variance from a finite sample.

Consider a set of independently and identically distributed (i.i.d.) random variables $R_1, R_2, \dots, R_n$ with a mean μ and variance σ^2 . The sample mean $\bar{R}$ serves as an unbiased estimator of the population mean μ , as indicated by the fact that $E[\bar{R}] = \mu$ according to the law of large numbers. Recall the definition of sample variance s^2 :

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (R_i - \bar{R})^2$$

Now we would also like to have $E[s^2] = \sigma^2$, which means that s^2 is an unbiased estimator of σ^2 in expectation. We begin by expanding the squared deviations by adding and subtracting μ .

$$\sum_{i=1}^n (R_i - \bar{R})^2 = \sum_{i=1}^n (R_i - \mu + \mu - \bar{R})^2$$

Now, we expand the squared term inside the summation:

$$\sum_{i=1}^n [(R_i - \mu) + (\mu - \bar{R})]^2 = \sum_{i=1}^n [(R_i - \mu)^2 + 2(R_i - \mu)(\mu - \bar{R}) + (\mu - \bar{R})^2]$$

We can break the expanded expression into three separate sums:

$$\sum_{i=1}^n (R_i - \bar{R})^2 = \sum_{i=1}^n (R_i - \mu)^2 + 2(\mu - \bar{R}) \sum_{i=1}^n (R_i - \mu) + \sum_{i=1}^n (\mu - \bar{R})^2$$

Note that the first term $\sum_{i=1}^n (R_i - \mu)^2$ is the sum of squared deviations from the population mean. For the second term, we know that

$$\sum_{i=1}^n (R_i - \mu) = n\bar{R} - n\mu = n(\bar{R} - \mu)$$

Therefore, the second term becomes

$$2(\mu - \bar{R}) \cdot n(\bar{R} - \mu) = -2n(\bar{R} - \mu)^2$$

The third term is $\sum_{i=1}^n (\mu - \bar{R})^2 = n(\mu - \bar{R})^2$; since $\mu - \bar{R}$ is constant with respect to i , it can be factored out.

Putting it all together, we have

$$\sum_{i=1}^n (R_i - \bar{R})^2 = \sum_{i=1}^n (R_i - \mu)^2 - 2n(\bar{R} - \mu)^2 + n(\bar{R} - \mu)^2$$

Further simplifying, we have the following:

$$\sum_{i=1}^n (R_i - \bar{R})^2 = \sum_{i=1}^n (R_i - \mu)^2 - n(\bar{R} - \mu)^2$$

Now we can take the expected value of both sides:

$$E \left[\sum_{i=1}^n (R_i - \bar{R})^2 \right] = E \left[\sum_{i=1}^n (R_i - \mu)^2 \right] - E \left[n(\bar{R} - \mu)^2 \right]$$

For the first term, since R_i are i.i.d. with variance σ^2 , we have

$$E \left[\sum_{i=1}^n (R_i - \mu)^2 \right] = \sum_{i=1}^n E \left[(R_i - \mu)^2 \right] = n\sigma^2$$

For the second term, since the variance of the sample mean $\bar{R}$ is $\text{Var}(\bar{R}) = \frac{\sigma^2}{n}$, thus

$$E \left[n(\bar{R} - \mu)^2 \right] = n \cdot E \left[(\bar{R} - \mu)^2 \right] = n \cdot \text{Var}(\bar{R}) = n \cdot \frac{\sigma^2}{n} = \sigma^2$$

Substituting the expected values back gives

$$E \left[\sum_{i=1}^n (R_i - \bar{R})^2 \right] = n\sigma^2 - \sigma^2 = (n-1)\sigma^2$$

Recall that

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (R_i - \bar{R})^2$$

Taking the expected value gives

$$E[s^2] = \frac{1}{n-1} E \left[\sum_{i=1}^n (R_i - \bar{R})^2 \right] = \frac{1}{n-1} \cdot (n-1)\sigma^2 = \sigma^2$$

Thus, we have shown that the sample variance s^2 is an unbiased estimator of the population variance σ^2 . This result indicates that using n as the denominator can lead to an underestimation of the true population variance, especially when the sample size n is small. That is

$$E \left[\frac{1}{n} \sum_{i=1}^n (R_i - \bar{R})^2 \right] = \frac{n-1}{n} \sigma^2 < \sigma^2$$

which would underestimate the true population variance. Thus, Bessel's correction addresses this bias by using $n-1$ the denominator, ensuring that the sample variance serves as an unbiased estimator of the population variance.

4.1.2 Variance in Practice

Let us use Python to demonstrate how to compute the variance using real stock price data. In this example, we will retrieve historical stock prices, calculate monthly returns, and determine both sample and population variance. We first download daily adjusted closing prices for Apple (AAPL), covering the period from January 1, 2024, to December 1, 2024. The adjusted closing prices are used because they account for dividends and stock splits, providing a more accurate representation of the stock's true value over time.

We then calculate the returns after obtaining the data. First, we compute the daily returns by measuring the percentage change between consecutive trading days using the “pct_change()” function, which effectively assesses the day-to-day performance of the stock. To gain a broader understanding of the stock's performance, we then aggregate these daily returns into monthly returns. The aggregation is done by selecting the last available price for each month and calculating the percentage change from the previous month (also for the last day). Such aggregation smooths out daily volatility and highlights longer-term trends in the stock's performance.

As shown in Listing 4-1, the next step is to calculate the variance, which includes both sample variance and population variance. The sample variance is calculated with the degree of freedom (known as “ddof”) set to 1. This means that the denominator in the variance formula is $n - 1$. As discussed earlier, this adjustment helps ensure that the sample variance is an unbiased estimator of the population variance. On the other hand, the population variance is calculated with “ddof” set to 0, which is the default setting, using n as the denominator. This approach assumes that the dataset represents the entire population rather than just a sample. Although this results in a biased estimator, it is useful for understanding how variance is computed when considering the entire population.

```

1 import pandas as pd
2 import numpy as np
3 import yfinance as yf
4 import matplotlib.pyplot as plt
5
6 # Suppress warnings for clean output
7 import warnings
8 warnings.filterwarnings('ignore')
9
10 # Step 1: Fetch Historical Stock Data
11 ticker = 'AAPL' # Example: Apple Inc.
12 start_date = '2024-01-01'
13 end_date = '2024-12-01'
14
15 # Download daily adjusted closing prices
16 stock_data = yf.download(ticker, start=start_date, end=end_date,
17                           progress=False)['Adj Close']
18
19 # Display the first few rows
20 print("Historical Adjusted Closing Prices:")
21 print(stock_data.head())

```

```

21
22 # Step 2: Calculate Daily Returns
23 daily_returns = stock_data.pct_change().dropna()
24
25 # Display the first few daily returns
26 print("\nDaily Returns:")
27 print(daily_returns.head())
28
29 # Step 3: Resample to Monthly Returns
30 monthly_returns = stock_data.resample('M').ffill().pct_change().
    dropna()
31
32 # Display the monthly returns
33 print("\nMonthly Returns:")
34 print(monthly_returns)
35
36 # Step 4: Calculate Sample Variance (Unbiased Estimator)
37 sample_variance = monthly_returns.var(ddof=1) # ddof=1 for
    sample variance
38
39 # Step 5: Calculate Population Variance (Biased Estimator)
40 population_variance = monthly_returns.var(ddof=0) # ddof=0 for
    population variance
41
42 print(f"\nSample Variance (Unbiased Estimator): {sample_variance
    :.4%}")
43 print(f"Population Variance (Biased Estimator): {
    population_variance:.4%}")
44
45 # Step 6: Visualization (Optional)
46 plt.figure(figsize=(10, 5))
47 plt.plot(stock_data.index, stock_data, label='Adjusted Close
    Price')
48 plt.title(f'{ticker} Adjusted Close Price in 2024')
49 plt.xlabel('Date')
50 plt.ylabel('Price ($)')
51 plt.legend()
52 plt.show()
53
54 # Output
55 Sample Variance (Unbiased Estimator): 0.3275%
56 Population Variance (Biased Estimator): 0.2947%

```

Listing 4-1 Downloading stock data and calculating variance

Figure 4-1 shows the daily adjusted close price of Apple in 2024.

4.1.3 Limitations of Variance As a Risk Measure

Although it is a commonly used metric for measuring volatility, variance has its own limitations. One major drawback is its increased sensitivity to outliers. Because variance is calculated by squaring the deviations from the mean, it tends

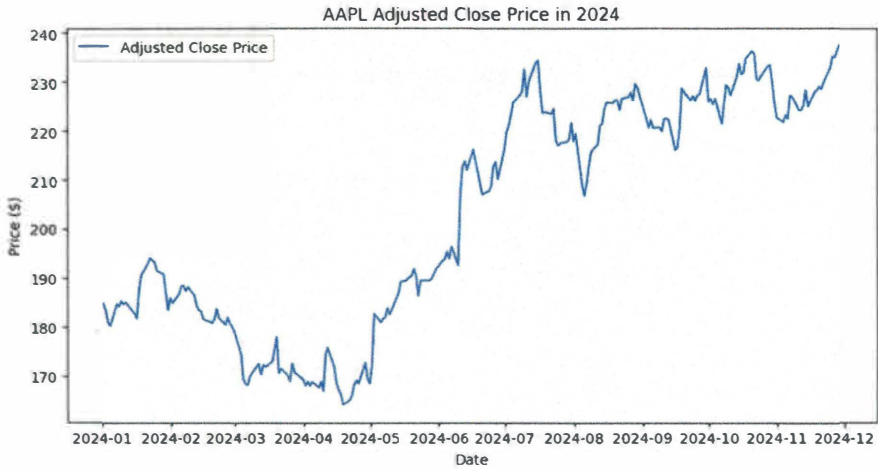


Figure 4-1 Daily adjusted close price of Apple in 2024

to exaggerate the effects of extreme values. This means that outliers—whether exceptionally high or low data points—can distort the variance, possibly resulting in misleading conclusions about the true variability of the dataset.

The Python code in Listing 4-2 demonstrates the impact of outliers on variance. By adding two extreme returns of 20% and -20% to the dataset, the population variance increases from approximately 0.0021 to 0.0028.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Generate a dataset with normal returns
5 np.random.seed(42)
6 returns = np.random.normal(loc=0.02, scale=0.05, size=100)
7
8 # Calculate initial variance
9 initial_variance = np.var(returns, ddof=1)
10 print(f"Initial Variance: {initial_variance:.4f}")
11
12 # Introduce outliers
13 outliers = np.array([0.2, -0.2]) # Extreme positive and negative
    returns
14 returns_with_outliers = np.concatenate((returns, outliers))
15
16 # Calculate variance after adding outliers
17 variance_with_outliers = np.var(returns_with_outliers, ddof=1)
18 print(f"Variance with Outliers: {variance_with_outliers:.4f}")
19
20 # Visualization
21 plt.figure(figsize=(12, 6))
22
23 plt.subplot(1, 2, 1)

```

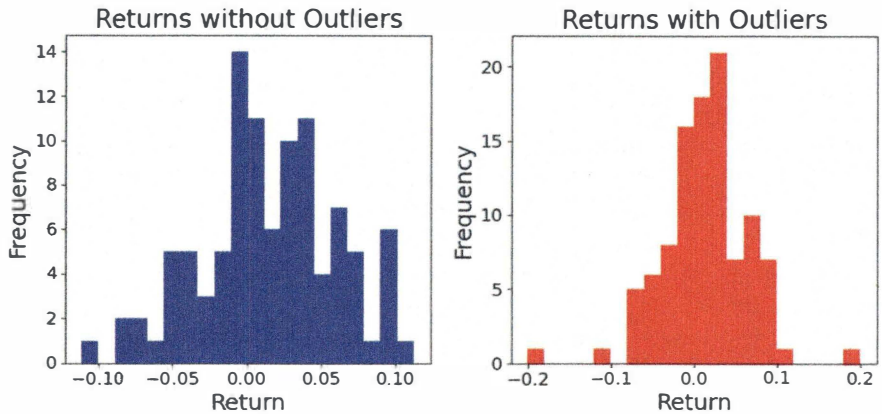


Figure 4-2 Distribution of returns before and after adding outliers

```

24 plt.hist(returns, bins=20, color='blue', alpha=0.7)
25 plt.title('Returns without Outliers')
26 plt.xlabel('Return')
27 plt.ylabel('Frequency')
28
29 plt.subplot(1, 2, 2)
30 plt.hist(returns_with_outliers, bins=20, color='red', alpha=0.7)
31 plt.title('Returns with Outliers')
32 plt.xlabel('Return')
33 plt.ylabel('Frequency')
34
35 plt.tight_layout()
36 plt.show()
37
38 # Output
39 Initial Variance: 0.0021
40 Variance with Outliers: 0.0028

```

Listing 4-2 Assessing the impact of outliers to variance

To further illustrate the distorting impact of outliers on the distribution, we plot the histograms before and after including extreme returns. As shown in Figure 4-2, the scale of the x-axis doubles, highlighting how outliers can skew the variance measure and alter the perceived distribution of returns. Specifically, the left histogram displays the distribution of returns without outliers, showing a concentration around the mean with relatively few extreme values. However, the right histogram includes outliers that extend the range of returns and increase the overall variance. Therefore, when extreme values are present or likely, relying only on variance alone can result in misleading risk assessments. On this front, it is often helpful to use additional risk measures, such as the median absolute deviation or the interquartile range, which are more resilient to the influence of outliers.

Another limitation of using variance as a risk measure is its treatment of non-symmetric returns. Variance treats both positive and negative deviations from the mean equally, which may not align with the specific risk preferences of most investors. In particular, a typical investor would be more concerned with the downside risk, which refers to the negative deviations, rather than fluctuations in the upside. Consequently, the equal weighting of upward and downward movements in variance may not adequately capture the types of risks that are most relevant to investors.

In addition, variance also depends on the assumption that returns follow a normal distribution. In a normal distribution, represented by $N(\mu, \sigma^2)$, variance accurately describes the spread of the data. However, financial returns frequently show skewness and kurtosis, which indicate asymmetry and fat tails, respectively, thus deviating from normality. Skewness (γ_1) measures the asymmetry of the distribution and is defined as follows:

$$\gamma_1 = \frac{E[(R - \mu)^3]}{\sigma^3}$$

Kurtosis (γ_2) measures the heaviness of the distribution's tails and is defined as

$$\gamma_2 = \frac{E[(R - \mu)^4]}{\sigma^4} - 3$$

These higher-order moments provide additional insights that variance alone cannot capture. Since variance does not take into account the asymmetry or the tendency for extreme values in the distribution of returns, using these higher-order moments could better reflect the actual risk.

Besides, another important limitation of variance as a risk measure is that it assumes that asset returns do not depend on each other over time. As mentioned earlier, variance is based on the idea that returns are independent and identically distributed (i.i.d.). This means that the return R_t at any time t does not affect the returns at other times $t' \neq t$. This independence can be formally stated as

$$P(R_t | R_{t-1}, R_{t-2}, \dots, R_1) = P(R_t)$$

This equation implies that past returns do not influence the probability distribution of future returns. In addition, each return R_t is assumed to follow the same probability distribution with constants μ and σ^2 :

$$E[R_t] = \mu \quad \text{and} \quad \text{Var}(R_t) = \sigma^2 \quad \forall t$$

This uniformity ensures that the statistical properties of the returns remain consistent over various time periods. However, empirical observations in financial markets frequently show deviations from these assumptions, leading to time dependence in

return series. Two main forms of time dependence are autocorrelation (or serial correlation) and volatility clustering.

Autocorrelation refers to intercorrelation between observations at different time points in a time series. It could also pose modeling challenges, especially when current returns are correlated with past returns. The autocorrelation function (ACF) at lag k quantifies this relationship and is defined as

$$\rho_k = \frac{\text{Cov}(R_t, R_{t+k})}{\text{Var}(R_t)}$$

Here, $\text{Cov}(R_t, R_{t+k})$ is the covariance between the returns at times t and $t + k$, and ρ_k quantifies the (linear) correlation between R_t and R_{t+k} . When $\rho_k \neq 0$ for any k , it means that the time series data are autocorrelated, suggesting that past returns can be used to predict future returns. Positive autocorrelation ($\rho_k > 0$) means that high returns in the future are likely to be observed by high returns observed now, and vice versa.

When returns have autocorrelation, the assumption of independence is no longer valid. Positive autocorrelation can underestimate true variance because groups of high or low returns make consecutive periods less varied. However, negative autocorrelation can overestimate variance since the alternating high and low returns create a perception of greater variability.

In terms of volatility clustering, it is a commonly observed phenomenon in financial markets. It often occurs when periods of high volatility are followed by more high volatility, and similarly, periods of low volatility tend to follow other low volatility periods. This indicates that the variance of returns is not constant over time; instead, it fluctuates in response to past volatility. To model this behavior, we can use Autoregressive Conditional Heteroskedasticity (ARCH) or Generalized ARCH (GARCH) models. These models allow the variance σ_t^2 at a given time to depend on past squared returns and variances by the following:

$$\sigma_t^2 = \alpha_0 + \alpha_1 R_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Here, α_0 , α_1 , and β_1 are the parameters to be estimated. The term $\alpha_1 R_{t-1}^2$ captures the impact of past squared returns (information about recent volatility), while $\beta_1 \sigma_{t-1}^2$ accounts for the persistence of volatility from previous periods.

As shown in Listing 4-3, to empirically investigate the time dependence of financial returns, we can analyze Apple's daily returns by computing the autocorrelation coefficients to evaluate the presence of serial correlation.

```
1 import matplotlib.pyplot as plt
2 from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
3 from statsmodels.tsa.stattools import acf
4
5 # Create a figure with two subplots side by side
6 fig, axes = plt.subplots(1, 2, figsize=(14, 6))
7
8 # Autocorrelation Function (ACF) Plot
```

```

9 plot_acf(daily_returns, lags=30, alpha=0.05, ax=axes[0])
10 axes[0].set_xlabel('Lag')
11 axes[0].set_ylabel('Autocorrelation')
12
13 # Partial Autocorrelation Function (PACF) Plot
14 plot_pacf(daily_returns, lags=30, alpha=0.05, ax=axes[1])
15 axes[1].set_xlabel('Lag')
16 axes[1].set_ylabel('Partial Autocorrelation')
17
18 # Adjust layout
19 plt.tight_layout()
20 plt.show()
21
22 # Calculate and print autocorrelation coefficients for
    verification
23 autocorr_coeffs = acf(daily_returns, nlags=30, fft=False)
24 print("\nAutocorrelation Coefficients (First 30 Lags):")
25 print(autocorr_coeffs)
26
27 # Output
28 Autocorrelation Coefficients (First 30 Lags):
29 [ 1.00000000e+00  5.37236795e-02 -3.70519869e-02 -8.65535152e-02
30   7.83734376e-02 -1.05379582e-01  1.68068543e-02  3.55114712e-02
31   1.81508577e-02 -2.04299197e-03 -5.55866151e-02 -9.83094485e-03
32   4.56800420e-03  5.31590184e-02 -9.50762751e-02  6.75666460e-02
33  -4.03906304e-02  1.89938306e-02  2.36953557e-02  3.23774660e-02
34   5.57144925e-03  4.98538062e-03  7.48970516e-02  4.48094570e-05
35  -1.34294058e-02 -8.26390596e-02  4.67667504e-02  9.16426999e-02
36   4.78552297e-02 -5.53208738e-02 -1.74259679e-02]

```

Listing 4-3 Assessing autocorrelation

The ACF plot of daily returns, shown in Figure 4-3, illustrates the correlation between returns at various time lags. For Apple's daily returns in 2024, the autocorrelation coefficients for the first 30 lags are nearly zero, indicating minimal serial correlation. This suggests that the returns are approximately independent over time, supporting the assumption of i.i.d. returns. As a result, this validates the use of variance as a risk measure in this context.

However, this observation may not apply universally to all assets or in different time periods. For example, the value at lag 5 is very close to the threshold at the 95% confidence interval, suggesting a potential autocorrelation at this specific lag. The 95% confidence interval indicates that there is a 95% probability that the true autocorrelation coefficient for each lag falls within this range, assuming that the data follow a white noise process (i.e., no autocorrelation).

Financial markets are inherently dynamic, and during times of market stress or significant economic events, characteristics such as volatility clustering can become more pronounced. For example, at the onset of a financial crisis, returns may show high autocorrelation and persistent volatility. This situation requires more advanced risk measures that go beyond just variance. To address these limitations, variance is frequently combined with other risk measures such as Value at Risk (VaR)

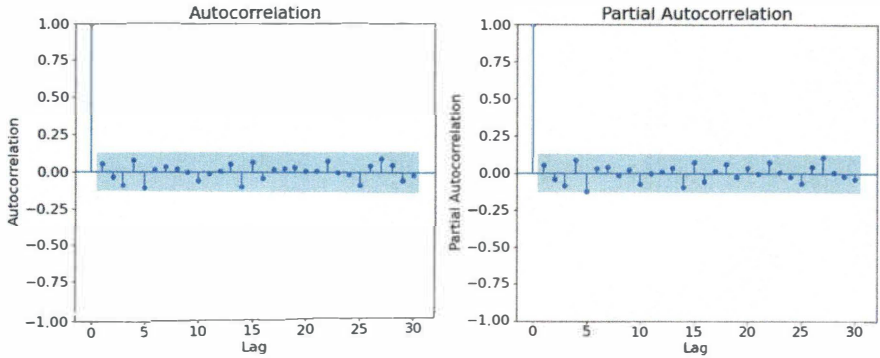


Figure 4-3 ACF and PACF plots of Apple's daily returns

and maximum drawdown. These downside risk measures offer a more thorough assessment of market risk, capturing elements that variance alone may miss. By using measures that consider time dependence and asymmetric risks, investors can gain a clearer and more comprehensive understanding of the risks involved in the investment portfolios.

4.2 Maximum Drawdown (Max Drawdown)

Unlike variance, which places equal weight on both upside and downside risk, the maximum drawdown (or Max Drawdown) measure specifically focuses on the downside risk in an extreme case. It identifies the largest up-to-date decline in a portfolio's value from its peak to its lowest point before recovering. This measure represents the worst potential loss an investor could experience over a specific time frame. By focusing on the most significant drop in value, Max Drawdown highlights the potential severity of losses during challenging market conditions, offering valuable insights into the investment's risk and resilience. Formally, for a series of portfolio values $P = \{P_1, P_2, \dots, P_n\}$, the Max Drawdown (MDD) is defined as

$$MDD = \max_{1 \leq i \leq j \leq n} \left(\frac{P_i - P_j}{P_i} \right) \times 100\%$$

In this equation, P_i represents the portfolio value at its peak time i , and P_j is the value at the trough time j that occurs after the peak i . The fraction $\frac{P_i - P_j}{P_i}$ calculates the percentage decline from the peak P_i to the trough P_j . Essentially, Max Drawdown identifies the most significant loss during the investment period, which is unlike variance that measures average volatility. This focused approach provides a clear view of downside risk, presenting a better view of the potential substantial losses in the portfolios.

To illustrate the calculation of Max Drawdown, consider an investor monitoring the monthly portfolio values over a four-month period, as shown below:

Month Portfolio Value (P)	
1	\$800
2	\$850
3	\$830
4	\$810

To calculate the Max Drawdown, first, we need to identify all the peak-trough pairs within the four-month period. A peak refers to a portfolio value that is followed by a decline, while a trough is the lowest portfolio value that occurs after that peak before a new peak is reached. In this case, we identify the first peak at Month 2 (\$850), with troughs at Month 3 (\$830) and Month 4 (\$810).

Next, for each peak-trough pair, we can calculate the drawdown as follows:

$$\text{Drawdown} = \left(\frac{P_{\text{peak}} - P_{\text{trough}}}{P_{\text{peak}}} \right) \times 100\%$$

Here, P_{peak} is the portfolio (or asset) value at the peak, and P_{trough} is the value at the trough. Specifically, from Peak Month 2 (\$850) to Trough Month 3 (\$830), we have

$$\text{Drawdown} = \left(\frac{850 - 830}{850} \right) \times 100\% = \left(\frac{20}{850} \right) \times 100\% \approx 2.35\%$$

From Peak Month 2 (\$850) to Trough Month 4 (\$810), we have

$$\text{Drawdown} = \left(\frac{850 - 810}{850} \right) \times 100\% = \left(\frac{40}{850} \right) \times 100\% \approx 4.71\%$$

Among these daily drawdowns, the largest is 4.71%, which comes from Peak Month 2 (\$850) to Trough Month 4 (\$810). Thus, this value represents the maximum drawdown for the portfolio over the four-month period, indicating the deepest decline experienced. Since Max Drawdown is typically reported as a negative percentage, it would be -4.71% in this case.

4.2.1 Distinctive Features of Maximum Drawdown

Max Drawdown measures the largest loss from a peak to a trough during an investment period, giving us a clear perspective on potential extreme downside risk. In contrast to variance, which assesses average volatility, Max Drawdown focuses specifically on extreme losses. It allows us to grasp the potential for significant declines in the portfolio. With Max Drawdown, we can evaluate the possible severity

of losses and make informed decisions about risk management and investment strategies. For example, a profiting strategy may not be so attractive if it comes with a big Max Drawdown. This metric is especially valuable for assessing portfolio performance during turbulent market conditions and for comparing the risk profiles of different investment options.

Specifically, Max Drawdown is notable for its focus on the sequence of returns, distinguishing it from variance. Although variance sums all deviations from the mean regardless of their order, Max Drawdown specifically identifies the largest decline from a peak to a trough in a portfolio's history. This means that Max Drawdown indicates not only how much a portfolio can lose but also when those losses occur. For example, a sharp decline following a high peak results in a higher Max Drawdown compared to a gradual decrease, even if the total loss is the same. This ability to take into account the order of returns allows Max Drawdown to provide a clearer understanding of extreme loss events, which is essential for effective risk management.

In addition, Max Drawdown is a non-parametric measure, meaning it does not assume that investment returns follow a specific distribution. This characteristic makes it versatile and applicable across various market conditions and asset types. Unlike variance, which is most effective when returns are normally distributed, Max Drawdown remains relevant even when returns are skewed or exhibit heavy tails. This is especially important in financial markets, where returns often deviate from normal patterns and can display irregular behaviors and extreme values. Therefore, it can provide an accurate assessment of downside risk without being constrained by rigid statistical models, thus being able to cater to a more realistic perspective on investment risks.

Instead of reporting separate measures, Max Drawdown can complement other risk measures, with variance included. Although variance provides an average assessment of volatility by considering all deviations from the mean, Max Drawdown focuses on the most significant losses. This combination gives investors a comprehensive understanding of both regular and extreme risks. For example, a portfolio with a low variance but a high Max Drawdown may indicate that it typically experiences manageable volatility but is at risk of substantial losses during unfavorable market conditions. Conversely, a portfolio with a high variance but a low Max Drawdown suggests frequent fluctuations without severe declines, which might appeal to investors seeking stability.

We can also use Max Drawdown to identify the maximum potential loss an investment could experience, thus assessing how much capital we should retain as reserves for significant downturns. Including Max Drawdown in capital allocation ensures that portfolios maintain sufficient liquidity and buffers during periods of market stress. For example, if a portfolio has a Max Drawdown of 10%, we may decide to hold additional reserves or implement hedging strategies to mitigate the impact of similar future losses. This proactive approach improves the durability of the portfolio and protects against challenging market conditions, which further aligns investment strategies with individual risk tolerance.

However, we should note that the Max Drawdown is highly sensitive to the time period chosen for calculation. The size of Max Drawdown can vary significantly based on the duration of the measurement. Shorter periods may reflect temporary market fluctuations, resulting in greater variability in Max Drawdown. In contrast, longer periods incorporate larger market movements, leading to more stable and accurate Max Drawdown values. This sensitivity is related to the time frame used to identify peaks and troughs. For example, calculating Max Drawdown over six months may reveal different risks compared to a one-year period due to varying exposure to market cycles and volatility. Therefore, selecting the appropriate time horizon is an important decision when assessing the risk associated with a given portfolio strategy.

The time scale we choose for calculating Max Drawdown, whether daily or monthly, can also affect the results. Daily data capture quick market changes, showing sharp declines that occur in just a few trading days. This often leads to higher or more frequent drawdowns, as daily prices react quickly to news and market events. In contrast, using monthly data smooths out these daily ups and downs, focusing on longer-term trends. This typically results in smaller or fewer declines. In practice, short-term traders may favor daily Max Drawdown to manage immediate risks, while long-term investors may prefer monthly Max Drawdown to understand overall performance and how well they can withstand longer market downturns.

4.2.2 Calculating Max Drawdown

To better appreciate how Max Drawdown is calculated and why it is important, let us consider an example that calculates total returns over time, identifies daily declines, and determines the largest drop (i.e., the Max Drawdown).

```

1 # Step 1: Calculate Cumulative Returns
2 cumulative_returns = (1 + daily_returns).cumprod()
3
4 # Step 2: Calculate Running Maximum
5 running_max = cumulative_returns.cummax()
6
7 # Step 3: Calculate Drawdown
8 drawdown = (cumulative_returns - running_max) / running_max
9
10 # Step 4: Calculate Max Drawdown
11 max_drawdown = drawdown.min() * 100 # Convert to percentage
12
13 print(f"\nMaximum Drawdown: {max_drawdown:.2f}%")
14
15 # Step 5: Visualization of Cumulative Returns and Drawdown
16 plt.figure(figsize=(12, 6)) # Increased figsize for better
    readability
17 plt.plot(cumulative_returns, label='Cumulative Returns')
18 plt.plot(running_max, label='Running Maximum', linestyle='--')
19 plt.fill_between(cumulative_returns.index, cumulative_returns,
    running_max,

```

```

20         color='red', alpha=0.3, label='Drawdown')
21
22 # Uncomment and set 'ticker' if needed
23 # plt.title(f'{ticker} Cumulative Returns and Drawdown (Jan-Jun
    2023)')
24
25 plt.xlabel('Date')
26 plt.ylabel('Cumulative Returns')
27 plt.legend()
28
29 # Rotate x-axis labels for better readability
30 plt.xticks(rotation=45)
31
32 # Adjust layout to prevent clipping of tick-labels
33 plt.tight_layout()
34
35 plt.show()
36
37 # Step 6: Plot Drawdown Over Time
38 plt.figure(figsize=(12, 6)) # Increased figsize for consistency
39 plt.plot(drawdown, label='Drawdown', color='red')
40
41 # Uncomment and set 'ticker' if needed
42 # plt.title(f'{ticker} Drawdown Over Time (Jan-Jun 2023)')
43
44 plt.xlabel('Date')
45 plt.ylabel('Drawdown')
46 plt.legend()
47
48 # Rotate x-axis labels for better readability
49 plt.xticks(rotation=45)
50
51 # Adjust layout to prevent clipping of tick-labels
52 plt.tight_layout()
53
54 plt.show()
55
56 # Output
57 Maximum Drawdown: -15.35%

```

Listing 4-4 Calculating max drawdown

In Listing 4-4, we first calculate the cumulative returns by multiplying the daily returns over a specific period. We then monitor the highest value that the returns have reached up to each day. Following this, we assess how much the current returns have decreased from this peak value. The smallest value in the drawdown series is identified and converted into a percentage, which represents a Max Drawdown of -15.35% in this example.

Figure 4-4 shows the total returns alongside the highest value reached. The shaded area indicates the periods when the returns were below the peak, which denotes drawdowns.

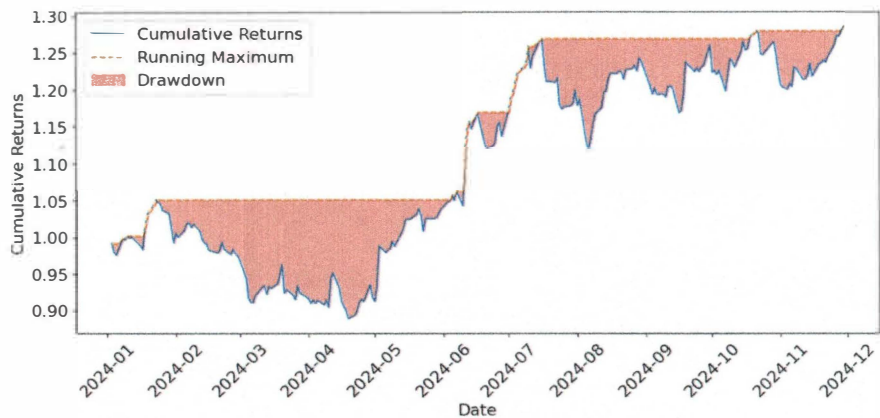


Figure 4-4 Daily cumulative returns and running maximum

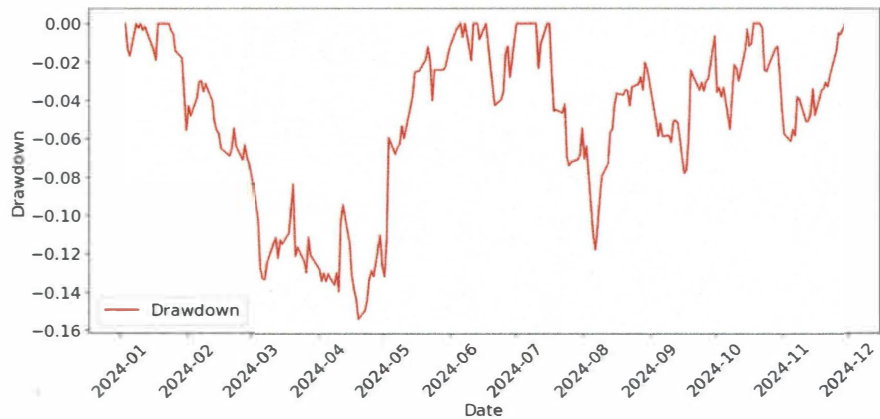


Figure 4-5 Daily drawdown

Figure 4-5 displays the daily drawdowns over time, illustrating the extent and duration of each decline. The Max Drawdown is then identified around mid-April 2024.

Max Drawdown is an important measure of risk, but it is most effective when used alongside other risk metrics to provide a comprehensive view of portfolio risk. For example, combining Max Drawdown with measures like VaR and Expected Shortfall (ES) offers insight into both the probability and magnitude of potential losses. Additionally, examining the rolling Max Drawdown over various time periods can reveal the frequency and duration of drawdowns, which is valuable for managing risk dynamically. Understanding the recovery time from drawdowns is also crucial, as it indicates the portfolio’s ability to rebound from losses and impacts long-term investment performance. Integrating it into Monte Carlo simulations and

stress testing can also enhance risk assessments by evaluating how the portfolio may perform under different adverse conditions.

The next section introduces another commonly used downside risk measure: Value at Risk (VaR).

4.3 Value at Risk

Value at Risk, commonly known as VaR, is a key measure used in finance to assess the risk associated with an investment portfolio. It is an estimate of the maximum potential loss that a portfolio could experience over a specific time period, such as a day, week, or month, at a certain confidence level. For example, a daily VaR of 1% at a 95% confidence level indicates that there is only a 5% chance that the portfolio will lose more than 1% of its value in a single day.

VaR is formally defined as the smallest loss that will not be exceeded with a given probability over a specified time frame. Formally, for a portfolio with returns $R = \{R_1, R_2, \dots, R_n\}$, the VaR at a confidence level α over a time horizon T is expressed as

$$VaR_\alpha = \inf\{x \in \mathbb{R} : P(L \leq x) \geq \alpha\}$$

Here, $L = P_0 - P_T$ represents the loss, where P_0 is the initial value of the portfolio and P_T is its value at time T . This definition ensures that there is at most $(1 - \alpha)$ probability that the loss L exceeds the VaR.

Note that VaR is essentially a quantile measure, meaning it identifies the threshold loss value below which a specific percentage of losses occur. It represents the inverse of the cumulative distribution function (CDF) of the loss distribution at a given confidence level.

$$VaR_\alpha = F_L^{-1}(1 - \alpha)$$

where F_L^{-1} is the quantile function of the loss distribution.

Regulatory bodies, including the Basel Committee on Banking Supervision, recognize and require the use of VaR to assess market risk in financial institutions. VaR can be applied to portfolios of various sizes and complexities, from straightforward stock holdings to complex combinations of assets. This flexibility allows many investors and financial professionals to evaluate and manage risk in a wide range of investment products.

In terms of mathematical properties, although VaR has several desirable properties, such as being monotonic and positively homogeneous, it does not always meet the requirement of subadditivity. Subadditivity means that the VaR of a combined

portfolio should be less than or equal to the sum of the VaRs of the individual portfolios. That is, for two portfolios A and B , we would like to see

$$VaR_{\alpha}(A + B) \leq VaR_{\alpha}(A) + VaR_{\alpha}(B)$$

However, in cases where portfolios exhibit heavy-tailed return distributions or have high levels of dependency, this property may not be valid. As a result, VaR might not always promote diversification, since the VaR of a combined portfolio can sometimes exceed the sum of the individual VaRs. This makes VaR much less appealing compared to improved measures such as conditional VaR, since diversification is the only free lunch in finance.

Besides, VaR only considers extreme risk up to the specified quantile level when it comes to capturing the tail risk while ignoring further extreme losses that occur beyond the VaR threshold. To address this limitation, Expected Shortfall (ES) is often used alongside VaR. ES measures the average loss that occurs when losses exceed the VaR level, offering a more comprehensive understanding of potential extreme losses. It is defined as

$$ES_{\alpha} = E[L \mid L > VaR_{\alpha}]$$

When it comes to calculation, VaR can be calculated using various methods, including Historical Simulation, the Variance-Covariance (or parametric) approach, and Monte Carlo simulation. Each method is based on different mathematical principles and assumptions, providing unique perspectives and varying levels of accuracy depending on the portfolio's characteristics and the data available.

4.3.1 Historical Simulation Approach

The Historical Simulation approach is a simple method for calculating VaR that does not require any specific statistical distribution for returns. Instead, it relies entirely on historical return data, operating under the premise that past returns can help predict future performance.

To start, we first collect a series of past portfolio returns $R = \{R_1, R_2, \dots, R_n\}$ collected over the time period of analysis. We then sort the returns from lowest to highest to create an ordered list, $R_{(1)} \leq R_{(2)} \leq \dots \leq R_{(n)}$. The next step is to identify the return that corresponds to the $(1 - \alpha)$ percentile of this sorted list, which represents the VaR at a confidence level α . Mathematically, VaR at confidence level α is calculated as

$$VaR_{\alpha} = R_{(\lceil (1-\alpha)n \rceil)}$$

Here, $R_{(k)}$ refers to the k -th return in the sorted list. The ceiling function $\lceil \cdot \rceil$ ensures that we select the smallest integer greater than or equal to $(1 - \alpha)n$, which accurately captures the desired percentile in the dataset.

For example, to determine 95% VaR using the Historical Simulation method, we would look for the return at the fifth percentile of the sorted historical returns. This process is illustrated in Listing 4-5.

```

1 # Historical Simulation VaR
2 confidence_level = 0.95
3 VaR_hist = np.percentile(daily_returns, (1 - confidence_level) *
4                           100)
5
6
7 # Visualization
8 plt.figure(figsize=(10, 6))
9 plt.hist(daily_returns, bins=50, alpha=0.7, color='blue',
10          edgecolor='black')
11 plt.axvline(x=VaR_hist, color='red', linestyle='dashed',
12             linewidth=2, label=f'VaR at {confidence_level*100}%')
13 plt.title(f'Histogram of {ticker} Daily Returns with VaR ({
14             confidence_level*100}%)')
15 plt.xlabel('Daily Return')
16 plt.ylabel('Frequency')
17 plt.legend()
18 plt.show()
19
20 # Output
21 Historical Simulation VaR at 95.0% confidence level: -2.18%

```

Listing 4-5 Calculating VaR using the historical simulation approach

Running this code reveals that Apple's VaR at a 95% confidence level for the year 2024 (until the first of December) is -2.18% . This means there is a 95% confidence that the daily loss will not exceed 2.18%. Figure 4-6 displays a histogram of Apple's daily returns for 2024, highlighting the VaR threshold with a red dashed line. The histogram illustrates the distribution of returns and indicates where VaR falls within this distribution. Again, this method does not consider losses that exceed the VaR threshold, and we can use ES to measure the average loss that occurs beyond the VaR level.

In summary, the Historical Simulation approach offers a straightforward and intuitive method for estimating VaR using historical return data. Although this method effectively illustrates potential losses based on past performance, it may not account for extremely rare or severe losses.

The next section introduces the second approach for the VaR calculation: the Variance-Covariance (parametric) approach.

4.3.2 Variance-Covariance (Parametric) Approach

The Variance-Covariance method, also known as the parameter approach, is another technique to calculate VaR. This method assumes that portfolio returns follow a

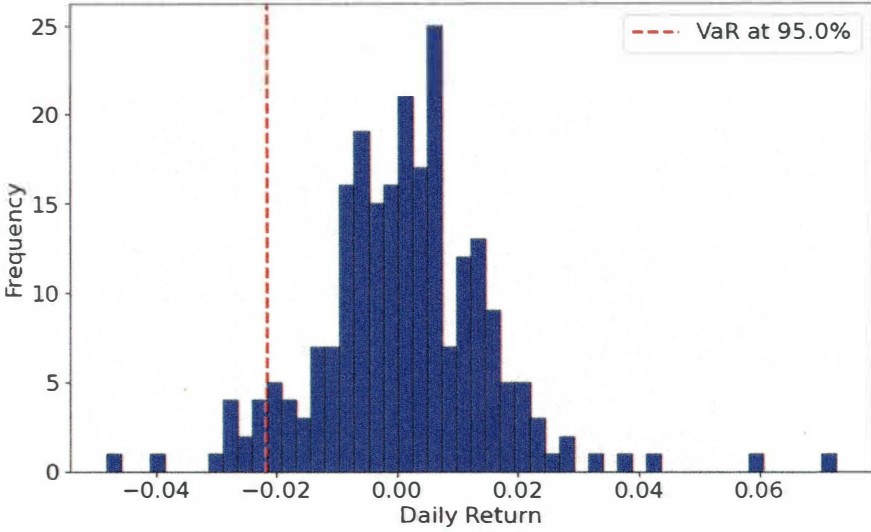


Figure 4-6 Histogram of daily returns with VaR at 95%

normal distribution, which simplifies the analytical computation of VaR by using the mean and standard deviation of the returns.

Specifically, consider portfolio returns denoted by R , which are normally distributed as $R \sim N(\mu, \sigma^2)$, where μ represents the mean return and σ is the standard deviation. The loss over the time horizon can be expressed as $L = -P_0 R$, where P_0 is the initial portfolio value. Our objective is to determine VaR_α , the VaR at a confidence level α .

We begin by writing the probability that the loss exceeds the VaR:

$$P(L > VaR_\alpha) = 1 - \alpha$$

Substituting $L = -P_0 R$ into the equation gives

$$P(-P_0 R > VaR_\alpha) = 1 - \alpha$$

Since P_0 is positive, dividing both sides of the inequality $-P_0 R > VaR_\alpha$ by $-P_0$ reverses the inequality:

$$P\left(R < -\frac{VaR_\alpha}{P_0}\right) = 1 - \alpha$$

Given that R is normally distributed, we can standardize it using the z-score transformation:

$$Z = \frac{R - \mu}{\sigma}$$

This allows the probability statement to be rewritten in terms of the standard normal distribution:

$$P\left(Z < \frac{-\frac{VaR_\alpha}{P_0} - \mu}{\sigma}\right) = 1 - \alpha$$

Let $z_{1-\alpha}$ denote the z-score corresponding to the cumulative probability $1 - \alpha$. Therefore, the equation becomes

$$\frac{-\frac{VaR_\alpha}{P_0} - \mu}{\sigma} = z_{1-\alpha}$$

Recognizing that the standard normal distribution is symmetric, $z_{1-\alpha}$ is equal to $-z_\alpha$, where z_α is the z-score corresponding to the cumulative probability α . Substituting this into the equation yields

$$\frac{-\frac{VaR_\alpha}{P_0} - \mu}{\sigma} = -z_\alpha$$

We proceed with the simplification to solve for VaR_α :

$$-\frac{VaR_\alpha}{P_0} - \mu = -\sigma z_\alpha$$

$$\frac{VaR_\alpha}{P_0} + \mu = \sigma z_\alpha$$

$$VaR_\alpha = P_0(\sigma z_\alpha - \mu)$$

This formula adjusts the mean return μ by multiplying the standard deviation σ and the z-score z_α , thus capturing the risk-adjusted threshold for potential losses.

For instance, consider a portfolio with an initial value $P_0 = \$1,000$, an average daily return $\mu = 0.2\%$, a standard deviation $\sigma = 1.5\%$, and a confidence level $\alpha = 99\%$ (so $z_\alpha = 2.33$). The VaR can be computed as follows:

$$VaR_{0.99} = 1,000 \times (0.015 \times 2.33 - 0.002) = \$32.95$$

This means that there is a 1% chance that the portfolio will lose more than \$32.95 in a single day. In other words, among the worst 1% of trading days, losses could exceed this amount, which is crucial information for risk management and capital allocation decisions.

Listing 4-6 demonstrates how to calculate VaR using the Variance-Covariance (parametric) approach. We normalize the portfolio value by setting $P_0 = 1$ to express VaR as a percentage.

```

1 from scipy.stats import norm
2
3 # Variance-Covariance VaR
4 confidence_level = 0.95
5 mu = daily_returns.mean()
6 sigma = daily_returns.std()
7 z_score = norm.ppf(confidence_level) # z-score for the given
   confidence_level
8
9 # Portfolio Value (normalized)
10 P0 = 1
11
12 # VaR Calculation
13 VaR_var_cov = P0 * (sigma * z_score - mu)
14
15 print(f"Variance-Covariance VaR at {confidence_level*100}%
   confidence level: -{VaR_var_cov:.2%}")
16
17 # Output Variance-Covariance VaR at 95.0% confidence level: -2.26%

```

Listing 4-6 Calculating VaR using the parametric approach

Here, we set *confidence_level* to 0.95 to reflect a 95% confidence level. The variables *mu* and *sigma* represent the mean and standard deviation of the daily returns, respectively. The *z_score* is calculated using the inverse cumulative distribution function of the normal distribution for the specified confidence level. By setting the portfolio value P_0 to 1, the VaR is expressed as a percentage. Running this code yields a VaR of approximately -2.26% , indicating a 5% chance that the portfolio will lose more than 2.26% of its value in a day using the parametric approach. This VaR is slightly higher than the one obtained using the Historical Simulation method, reflecting a greater potential loss based on the assumption of a normal distribution.

Note that the z-score z_α is a key element in this calculation, as it indicates how many standard deviations a data point is from the mean for a given cumulative probability. For a 95% confidence level with $\alpha = 0.95$, we have $z_\alpha \approx 1.645$. This z-score directly connects VaR to the characteristics of the normal distribution, allowing for a straightforward calculation based on the portfolio's mean and standard deviation.

Also, note that the parametric approach depends on the assumption that return distributions are normal. This assumption may not always hold, particularly in financial markets, where returns can show large deviations from a normal distribution. Although this method provides an analytical solution for VaR, making it computationally efficient, its accuracy is based on how well the normality assumption fits the actual return distribution. In situations where the return distribution has long tails or high levels of autocorrelation, the VaR estimates may be inaccurate, and extreme events might not be adequately captured. Therefore, despite its usefulness, the Variance-Covariance approach has limitations and should be applied with an understanding of its underlying assumptions and potential inaccuracies.

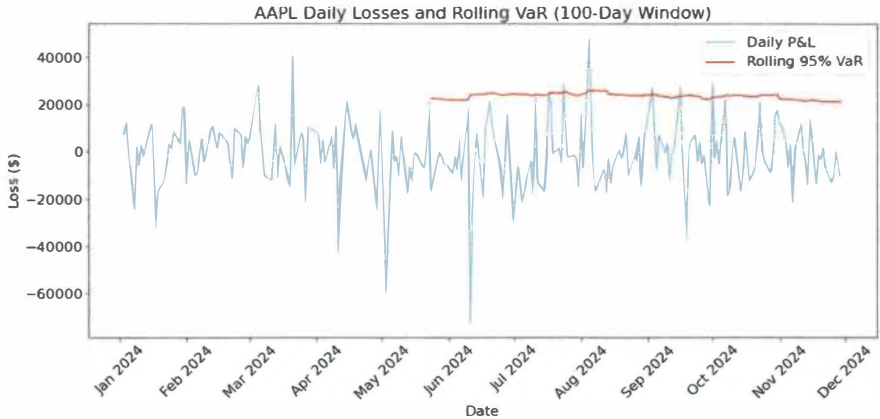


Figure 4-7 Daily P&L and rolling VaR using the Variance-Covariance approach

Figure 4-7 plots the daily P&L and rolling VaR at the confidence level 95% with a window size of 100 days, along with the code snippet shown in Listing 4-7. Plotting the running VaR gives us an idea of the long-term trend of extreme loss metrics.

```

1 from scipy.stats import norm
2 import numpy as np
3 import pandas as pd
4 import matplotlib.pyplot as plt
5 import matplotlib.dates as mdates
6
7 window_size = 100 # Number of days in the rolling window
8
9 # Calculate Rolling VaR using Variance-Covariance Approach
10 mu_rolling = daily_returns.rolling(window=window_size).mean()
11 sigma_rolling = daily_returns.rolling(window=window_size).std()
12 z_score = norm.ppf(0.95) # 95% confidence level
13
14 # Portfolio Value
15 P0 = 1_000_000 # $1,000,000
16
17 # Calculate Rolling VaR
18 rolling_VaR = P0 * (sigma_rolling * z_score - mu_rolling)
19
20 # Calculate Daily Losses in Dollar Terms
21 daily_losses = -P0 * daily_returns
22
23 # Plot Overlay of Daily Losses and Rolling VaR
24 plt.figure(figsize=(14, 7))
25 plt.plot(daily_losses.index, daily_losses, label='Daily P&L',
26         alpha=0.5)
27 plt.plot(rolling_VaR.index, rolling_VaR, label='Rolling 95% VaR',
28         color='red')
29 plt.title(f'{ticker} Daily Losses and Rolling VaR ({window_size}-
30         Day Window)')

```

```

28 plt.xlabel('Date')
29 plt.ylabel('Loss ($)')
30 plt.legend()
31
32 # Format x-axis to show monthly ticks
33 ax = plt.gca()
34 ax.xaxis.set_major_locator(mdates.MonthLocator()) # Set major
    ticks to monthly intervals
35 ax.xaxis.set_major_formatter(mdates.DateFormatter('%b %Y')) #
    Format ticks as 'Month Year'
36 plt.xticks(rotation=45) # Rotate x-axis labels for better
    readability
37
38 plt.tight_layout() # Adjust layout to prevent clipping of tick-
    labels
39 plt.show()

```

Listing 4-7 Plotting daily P&L and rolling VaR

Next, we look at the third approach to calculate VaR.

4.3.3 Monte Carlo Simulation

The Monte Carlo simulation method calculates VaR by generating numerous simulated portfolio returns based on statistical models. This approach is highly flexible, enabling the modeling of complex return distributions and the relationships between assets without relying on strict assumptions of normality.

To calculate VaR using Monte Carlo simulation, we first define the statistical properties of the portfolio returns, including the mean vector and the covariance matrix. Next, we generate a large number N of random return scenarios based on these statistical parameters. For each simulation test, we compute the corresponding loss and then determine the VaR at the $(1 - \alpha)$ percentile of the loss distribution. This identified loss represents the VaR at the specified confidence level.

Formally, if portfolio returns follow a multivariate normal distribution $R \sim N(\mu, \Sigma)$, we simulate N returns $R_1, R_2, \dots, R_N$. For each simulated return R_i , we compute the loss $L_i = -P_0 R_i$, where P_0 is the initial portfolio value. The VaR at the confidence level α is then calculated as

$$VaR_\alpha = \text{Percentile}(L, 100 \times (1 - \alpha))$$

where $L = \{L_1, L_2, \dots, L_N\}$ represents the set of simulated losses.

Listing 4-8 demonstrates how to calculate VaR using the Monte Carlo simulation approach.

```

1 # Monte Carlo Simulation of Returns
2 np.random.seed(42) # For reproducibility
3 num_simulations = 10000
4 simulated_returns = np.random.normal(mu, sigma, num_simulations)
5

```

```

6 # Calculate Simulated Losses
7 simulated_losses = -P0 * simulated_returns
8
9 # Calculate VaR from Simulated Losses
10 VaR_monte_carlo = np.percentile(simulated_losses, 100 * (1 -
    confidence_level))
11
12 print(f"Monte Carlo Simulation VaR at {confidence_level*100}%
    confidence level: {VaR_monte_carlo:.2%}")
13
14 # Output
15 Monte Carlo Simulation VaR at 95.0% confidence level: -2.54%

```

Listing 4-8 Calculating VaR using the Monte Carlo approach

In this code listing, we first set the random seed to ensure that the results can be reproduced. The *simulated_returns* are generated using a normal distribution with a specified mean and standard deviation, simulating a defined number of returns (*num_simulations*). The *simulated_losses* are then calculated by multiplying each simulated return by $-P_0$, where P_0 represents the initial portfolio value. Finally, *VaR_monte_carlo* is computed using the Monte Carlo method by identifying the percentile of the simulated losses that corresponds to $100 \times (1 - \alpha)$.

The result shows that the VaR calculated using the Monte Carlo simulation approach is approximately -2.54% , which is the highest potential loss compared to the Historical Simulation and Variance-Covariance methods. Figure 4-8 shows the simulated daily returns/losses along with the associated VaR at 95%.

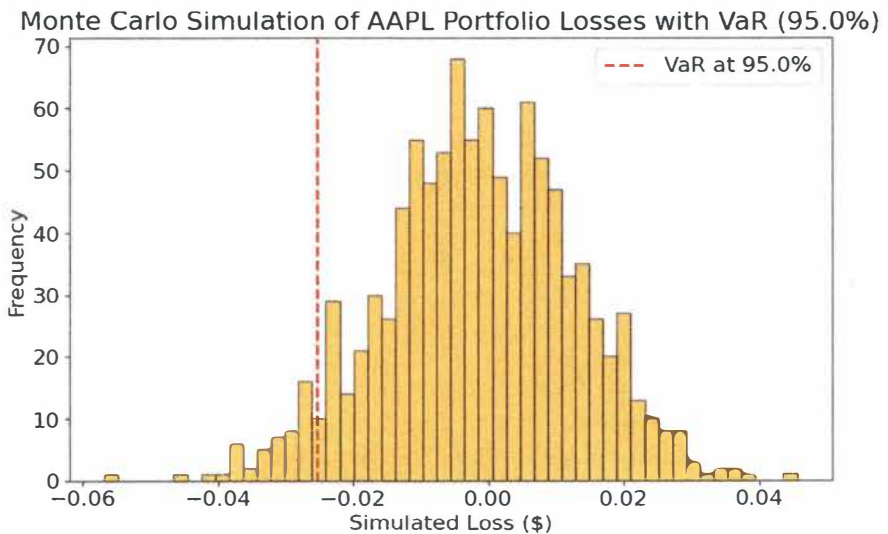


Figure 4-8 Monte Carlo simulation of daily P&L with VaR at 95%

Overall, the Monte Carlo simulation approach provides considerable flexibility by accommodating more complex return distributions and asset dependencies within a portfolio. However, this method can be computationally intensive, as we need to generate a large number of simulated returns. The computation becomes more involved, particularly for large portfolios or when a high number of simulations are required. In practice, financial institutions typically employ a combination of methods, including Historical Simulation, Variance-Covariance, and Monte Carlo simulation, to cross-validate VaR estimates. This approach ensures that estimates are robust under various market scenarios.

4.4 Summary

In this chapter, we explored market risk, which is a crucial type of systematic risk that can result in financial losses due to fluctuations in market factors throughout the entire financial system. We also introduced several metrics to help quantify, manage, and mitigate market risk, including Variance, Max Drawdown, and VaR.

Variance measures the variability of asset returns by calculating the average of the squared deviations from the mean return. This provides insight into the volatility and risk associated with an asset or portfolio. A higher variance indicates a greater spread of returns, which in turn signifies a higher risk. However, variance has several drawbacks. It is sensitive to outliers, treats positive and negative deviations equally, assumes a normal distribution, and assumes that returns are independent and identically distributed. These assumptions may not always hold in real financial markets.

Maximum drawdown measures the largest observed loss from a portfolio's peak value to its lowest point before a new peak is reached. This metric highlights the severity of potential losses, which makes it especially useful for understanding worst-case scenarios that investors may face. Unlike variance, maximum drawdown is path dependent, meaning it does not rely on any specific distribution of returns. This attribute makes it a robust measure under different market conditions. However, it is sensitive to the chosen time horizon.

VaR is a measure that estimates the maximum expected loss over a specified time period at a given confidence level. It provides a probabilistic assessment of potential losses. Essentially, VaR is a quantile measure that indicates the threshold loss value below which a certain percentage of losses is expected to fall. It is versatile and can be applied to different time horizons and portfolio sizes, making it a valuable tool for evaluating risk. However, it does not capture risks that exist beyond the specified confidence level, is sensitive to underlying model assumptions, and lacks the subadditive property. These shortcomings can reduce the advantages of diversification.

VaR can be calculated through various methods, each with different mathematical foundations. The Historical Simulation approach is non-parametric and relies on historical return data, effectively reflecting market behavior but assuming that past returns predict future performance. In contrast, the Variance-Covariance (para-

metric) approach assumes returns are normally distributed, allowing for analytical VaR calculations based on the mean and standard deviation, but it strongly relies on the normality assumption. The Monte Carlo simulation approach generates many hypothetical return scenarios using statistical models, providing flexibility for complex distributions; however, it is computationally intensive and requires accurate specification. Each method has its own strengths and weaknesses, making it important to choose the right one based on the context and available data.

Integrating these risk metrics provides a comprehensive view of market risk. For example, variance measures the average volatility, while Max Drawdown focuses on the most extreme losses. VaR estimates potential losses within a specified confidence interval. Considering additional measures such as Expected Shortfall, stress testing, and dynamic models such as GARCH can further enhance risk assessment by addressing tail risks and accounting for time-varying volatility. All methods are frequently used to cross-validate VaR estimates and ensure their robustness against various market scenarios.